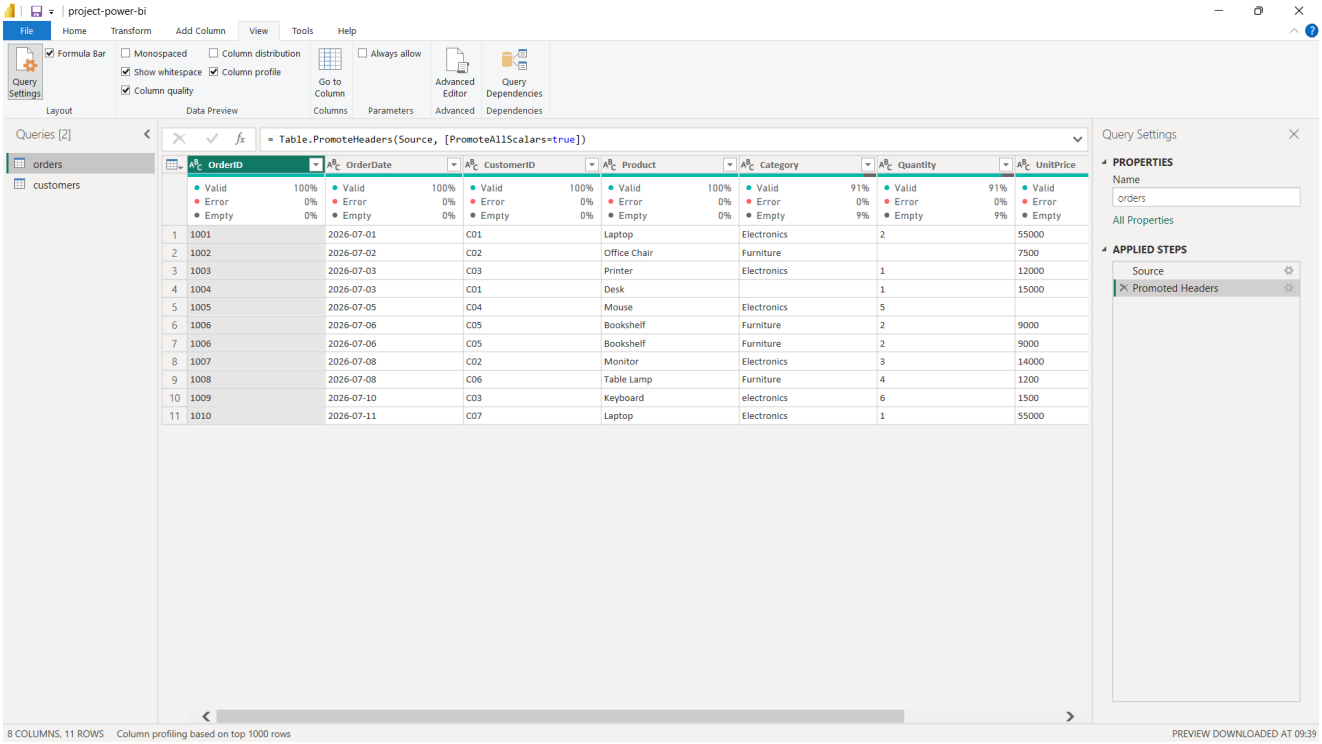
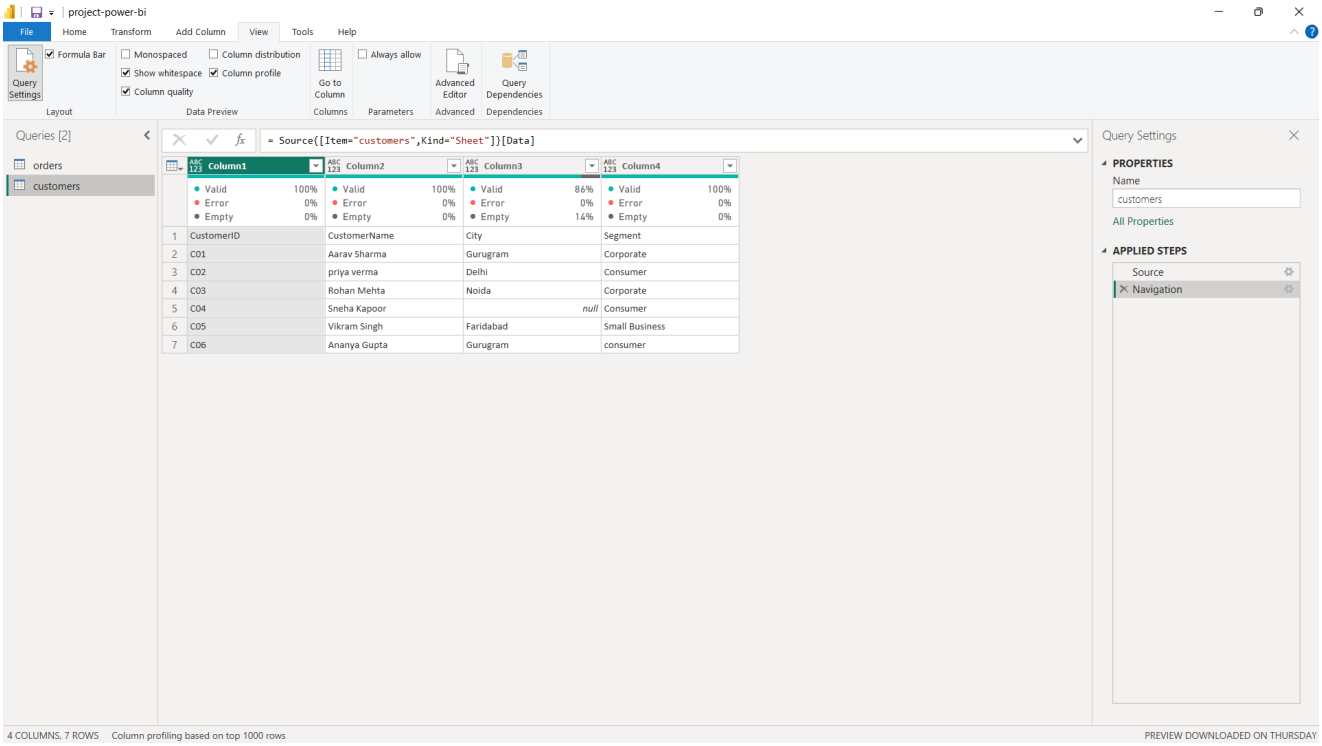
Data Cleaning and Preparation using Power BI

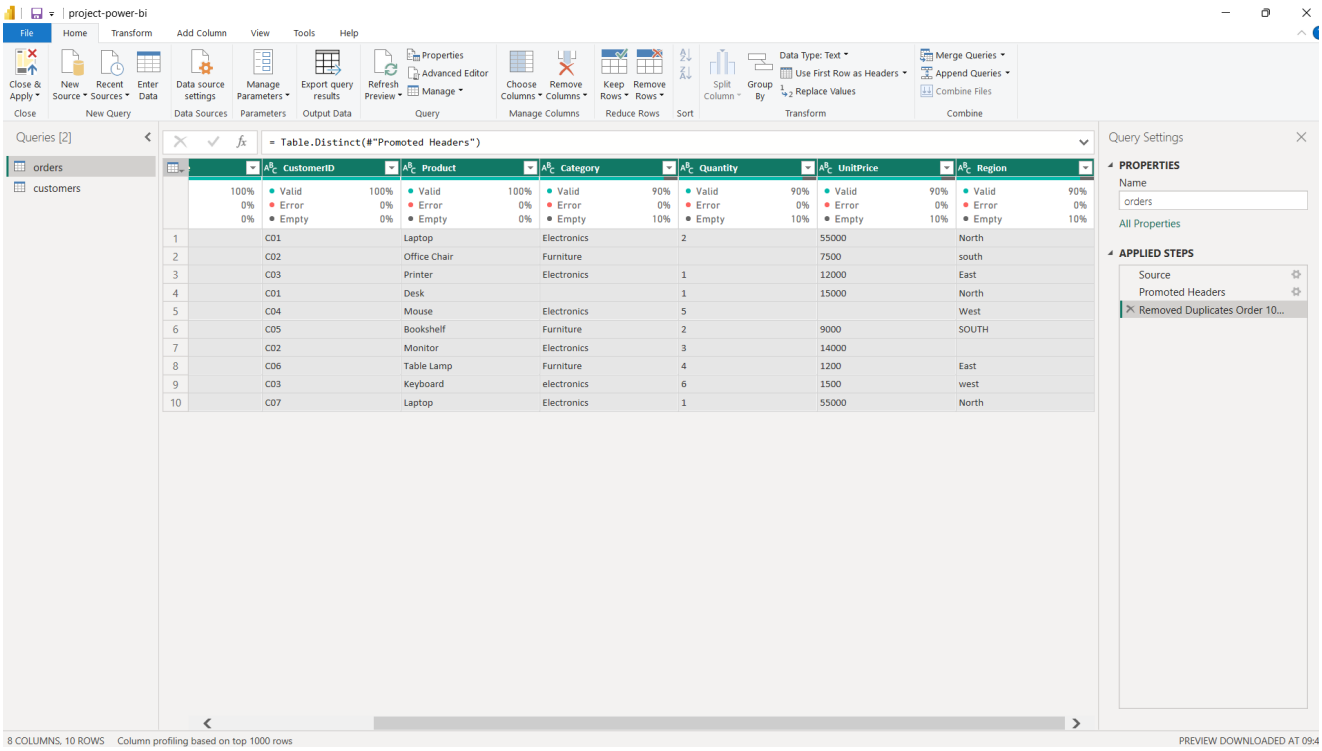
Step 1: Imported orders.csv into Power Query and verified the orders table and its column-quality information.



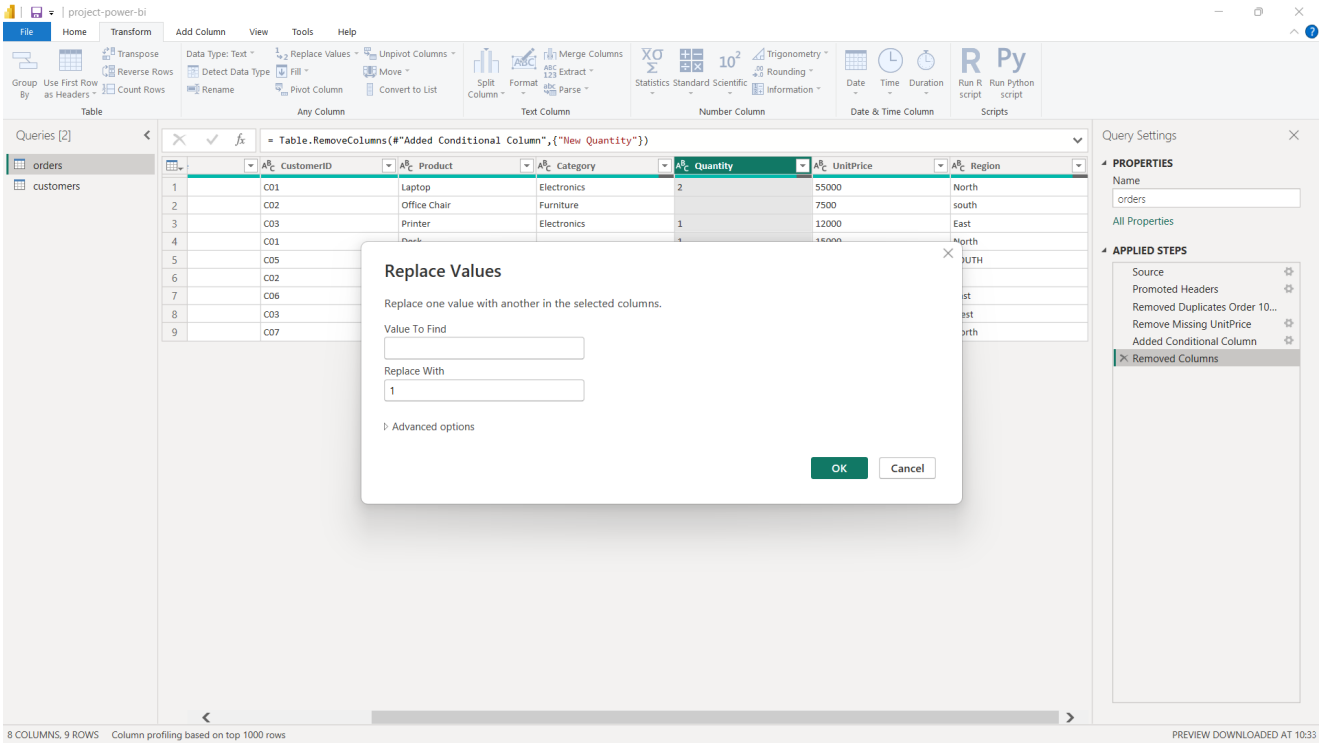
Step 2: Imported customers.xlsx into Power Query and verified the customer table and its column-quality information.



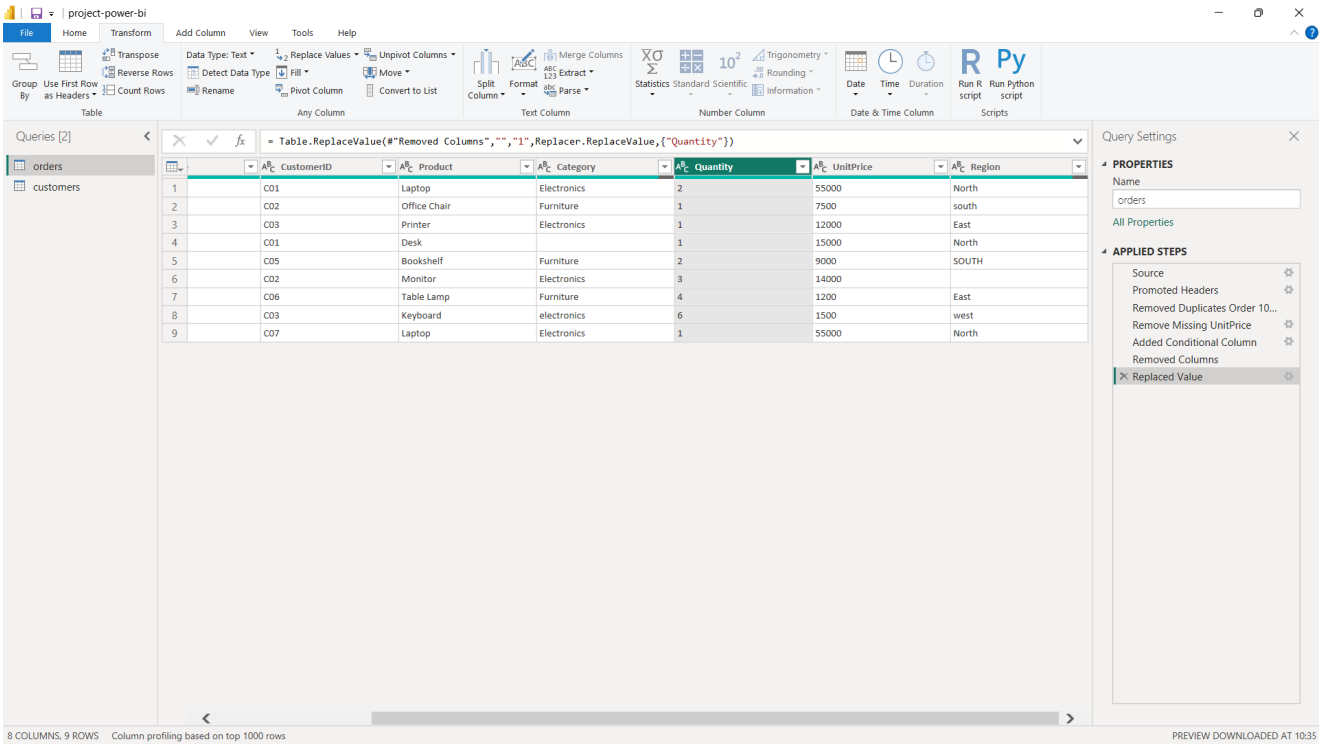
Step 3: Removed the duplicate OrderID 1006 from the orders table.



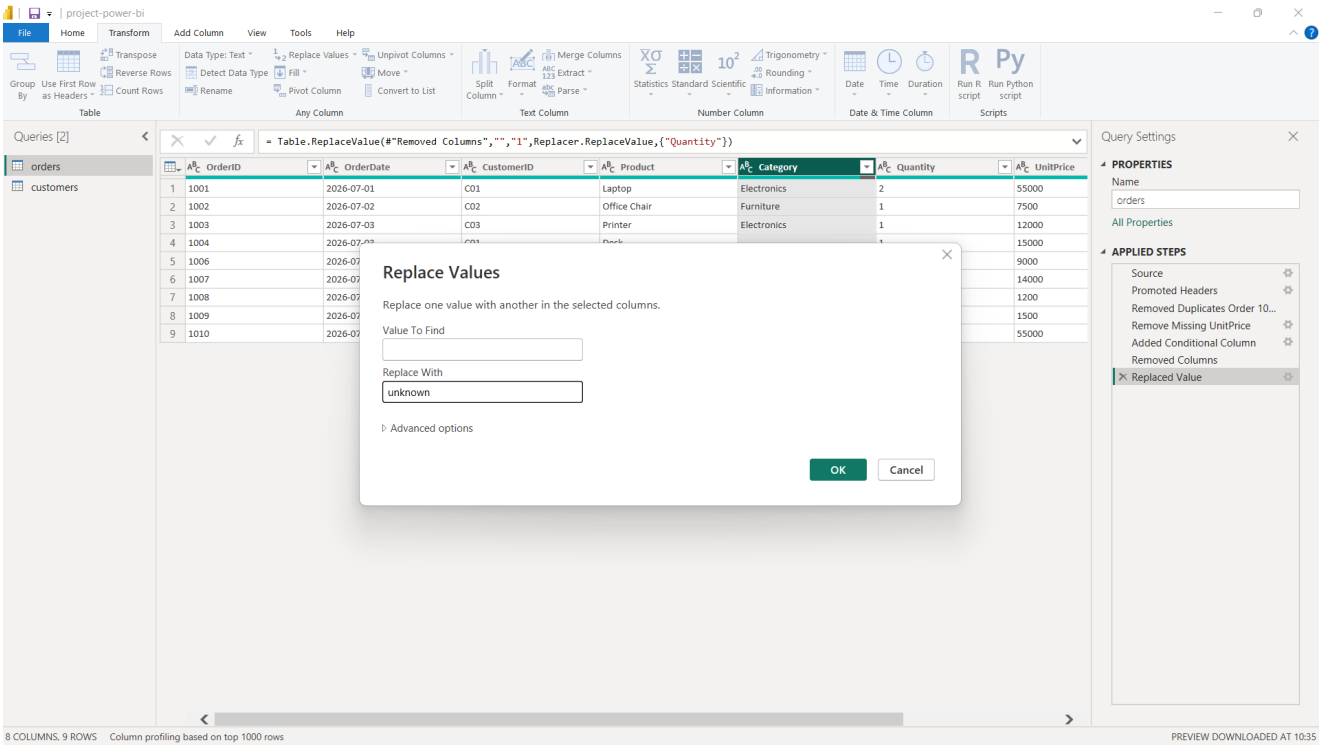
Step 4: Prepared the missing-Quantity transformation and removed the temporary helper column used during the cleaning process.



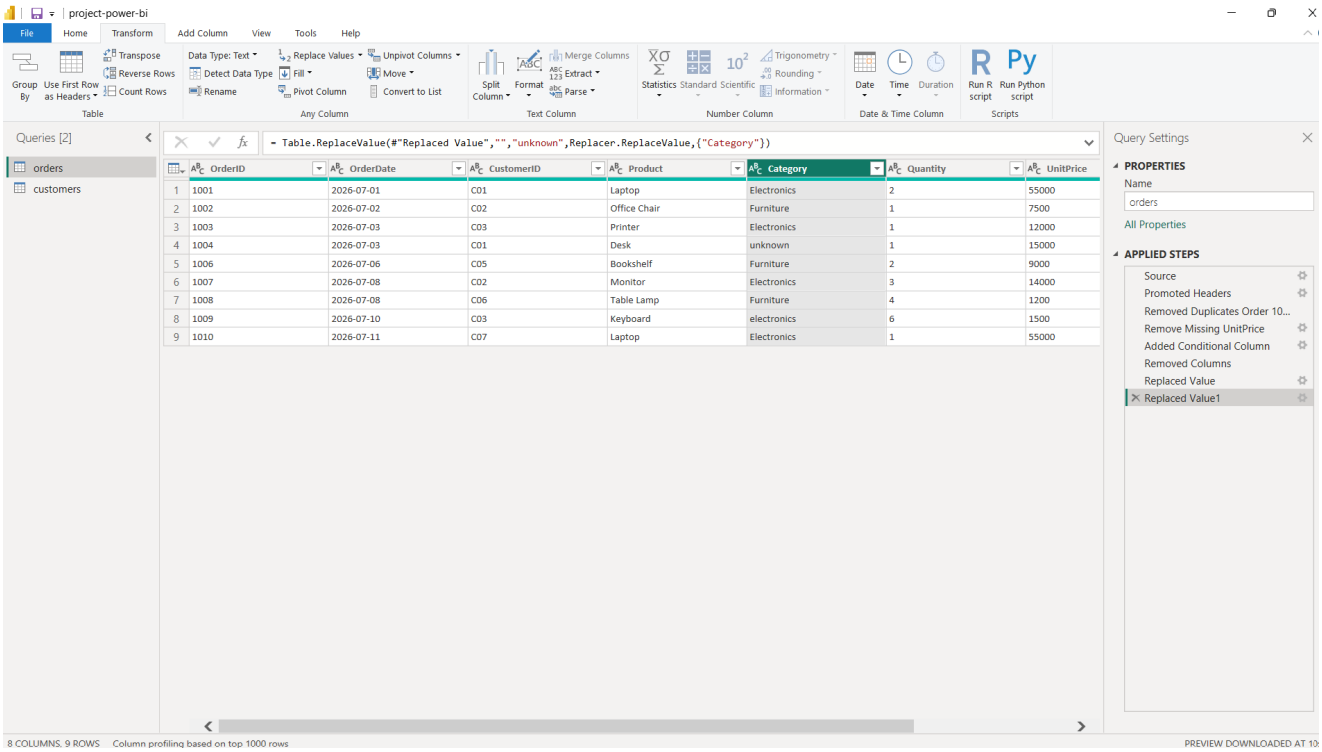
Step 5: Replaced the blank Quantity value with 1.



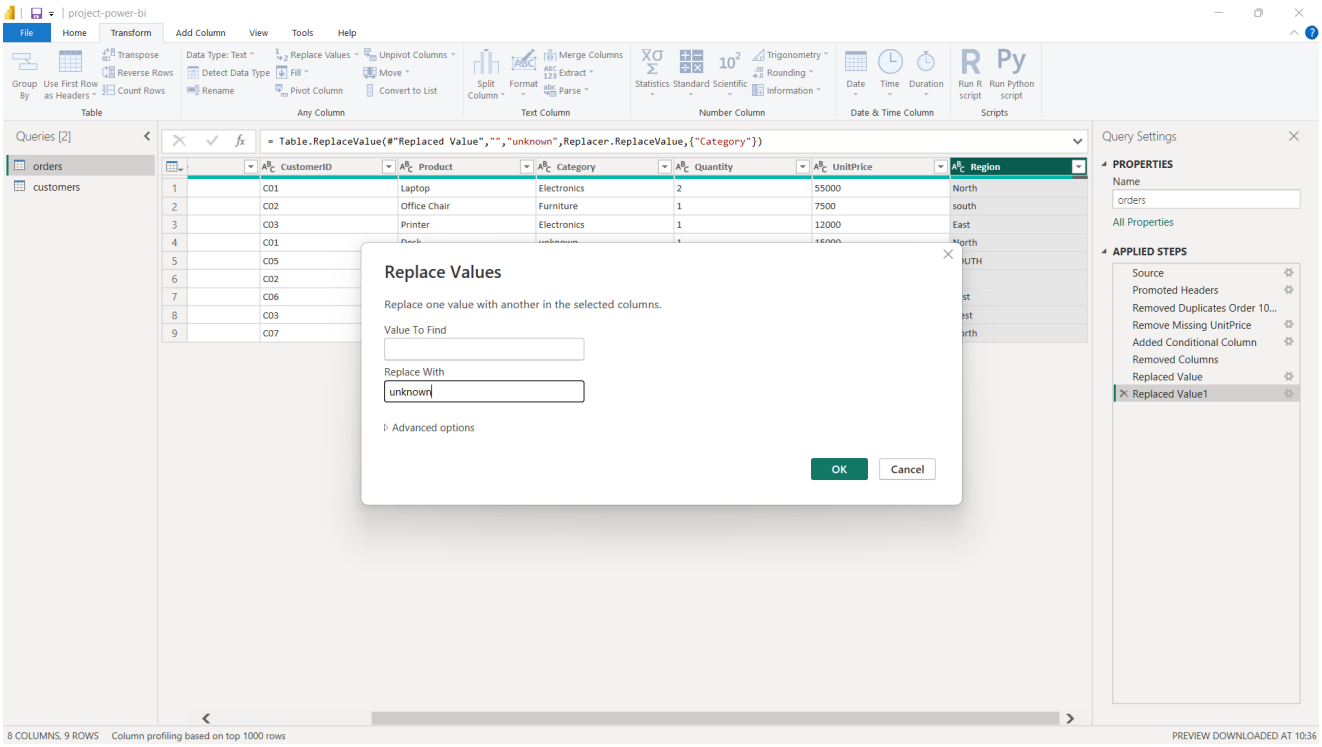
Step 6: Verified the Quantity replacement and the resulting cleaned orders data.



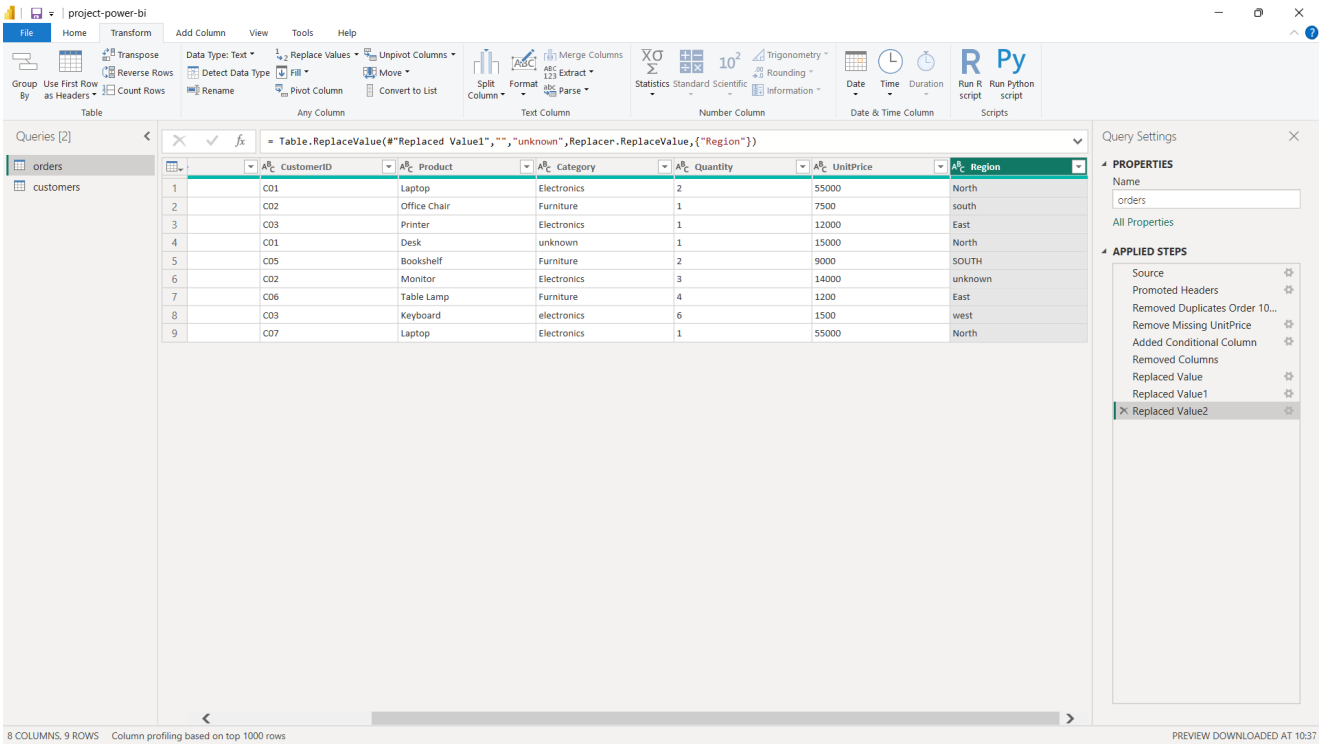
Step 7: Replaced the blank Category value with Unknown.



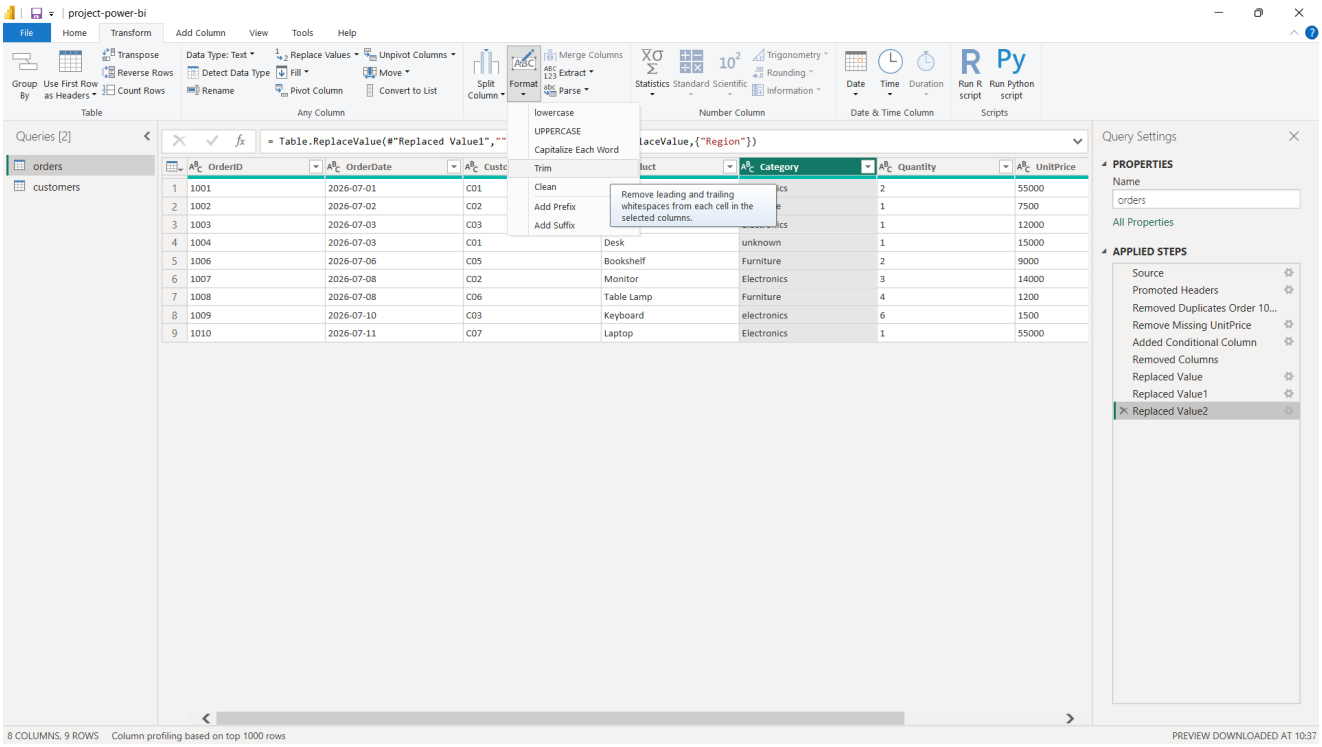
Step 8: Verified the Category replacement in the cleaned orders table.



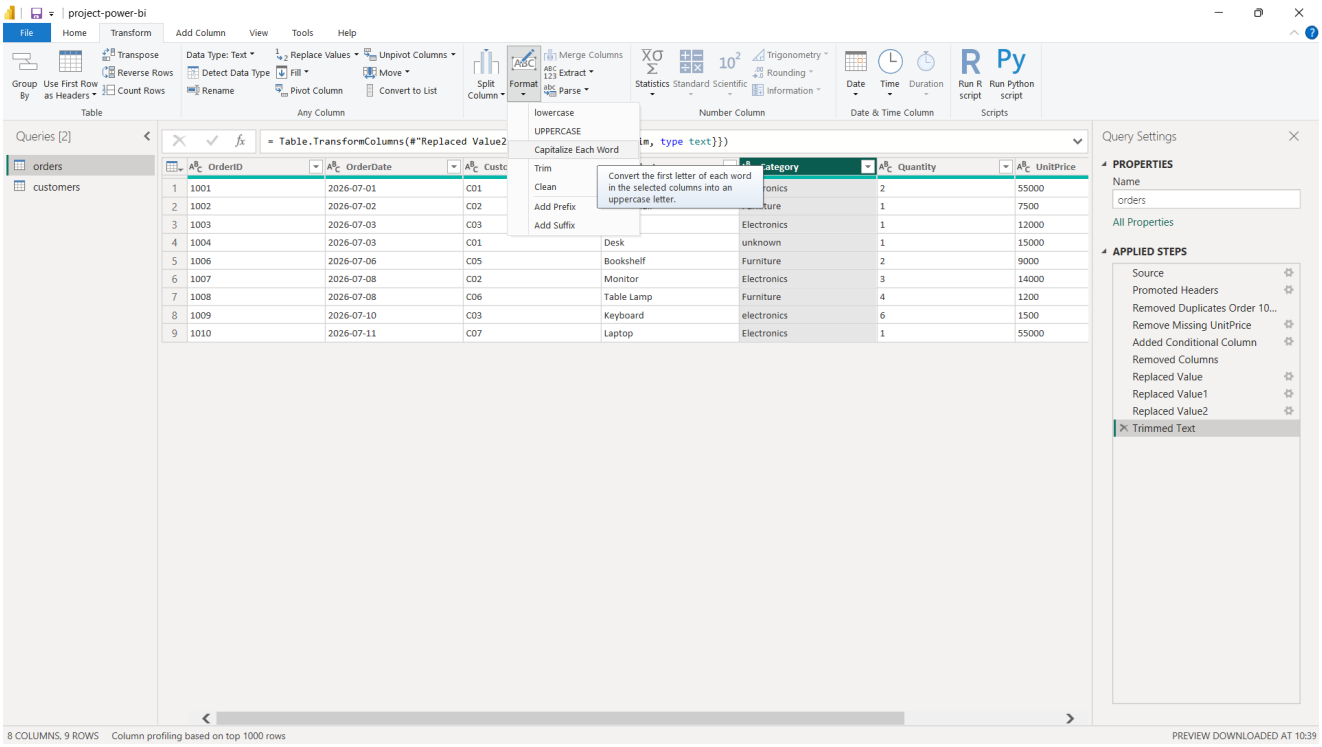
Step 9: Replaced the blank Region value with Unknown.



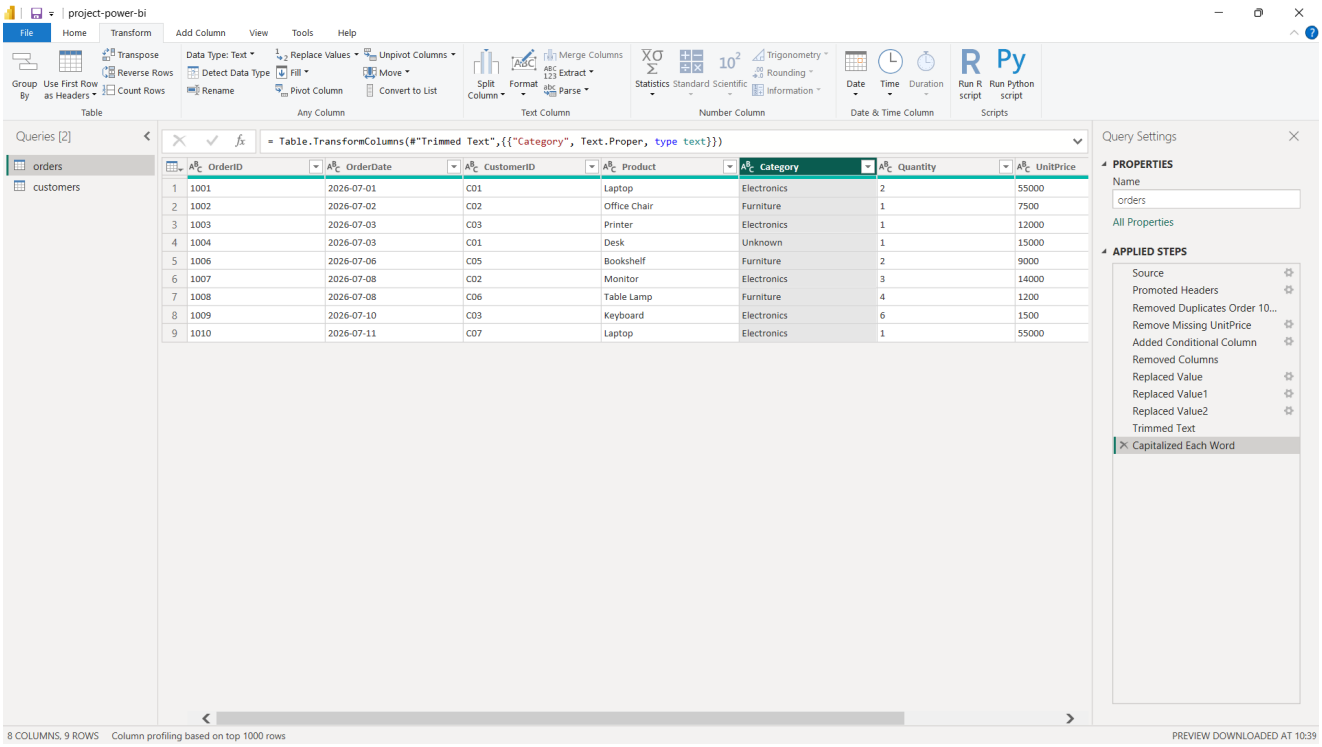
Step 10: Verified the Region replacement in the cleaned orders table.



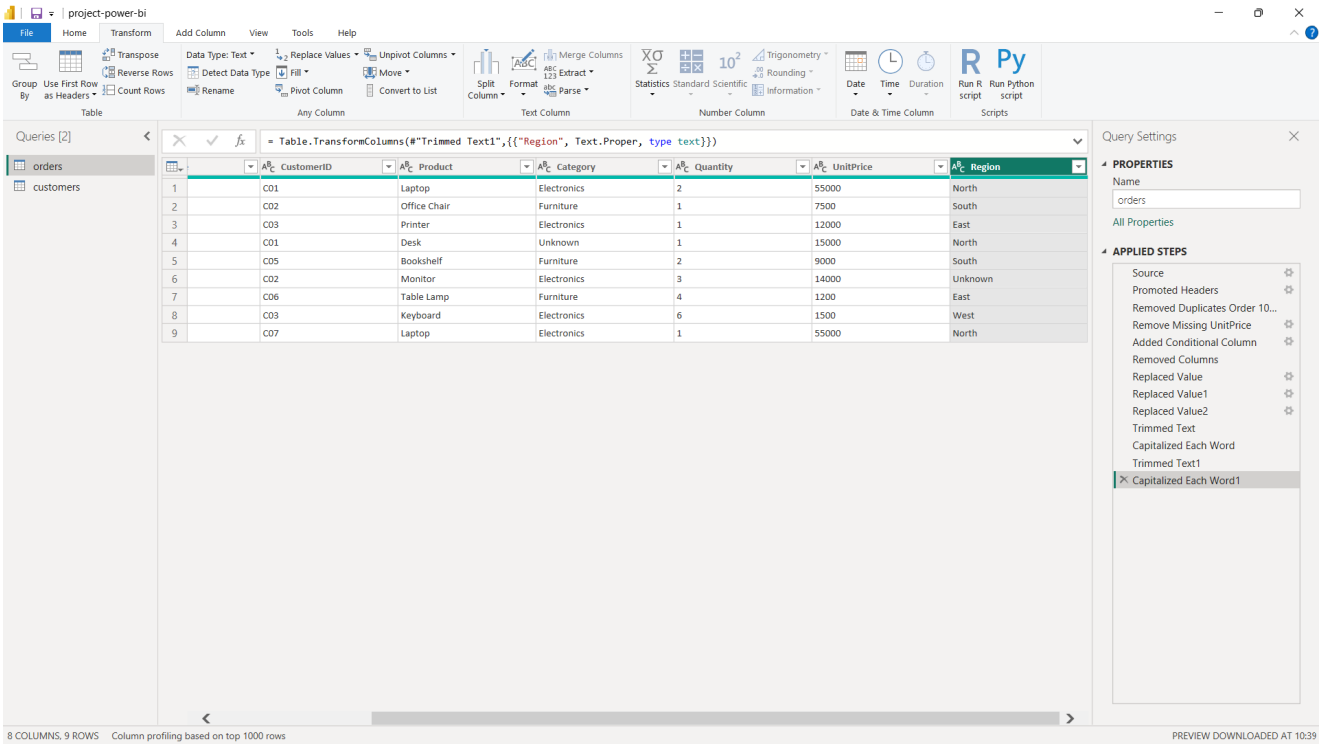
Step 11: Applied text trimming to the relevant order fields to remove unwanted whitespace.



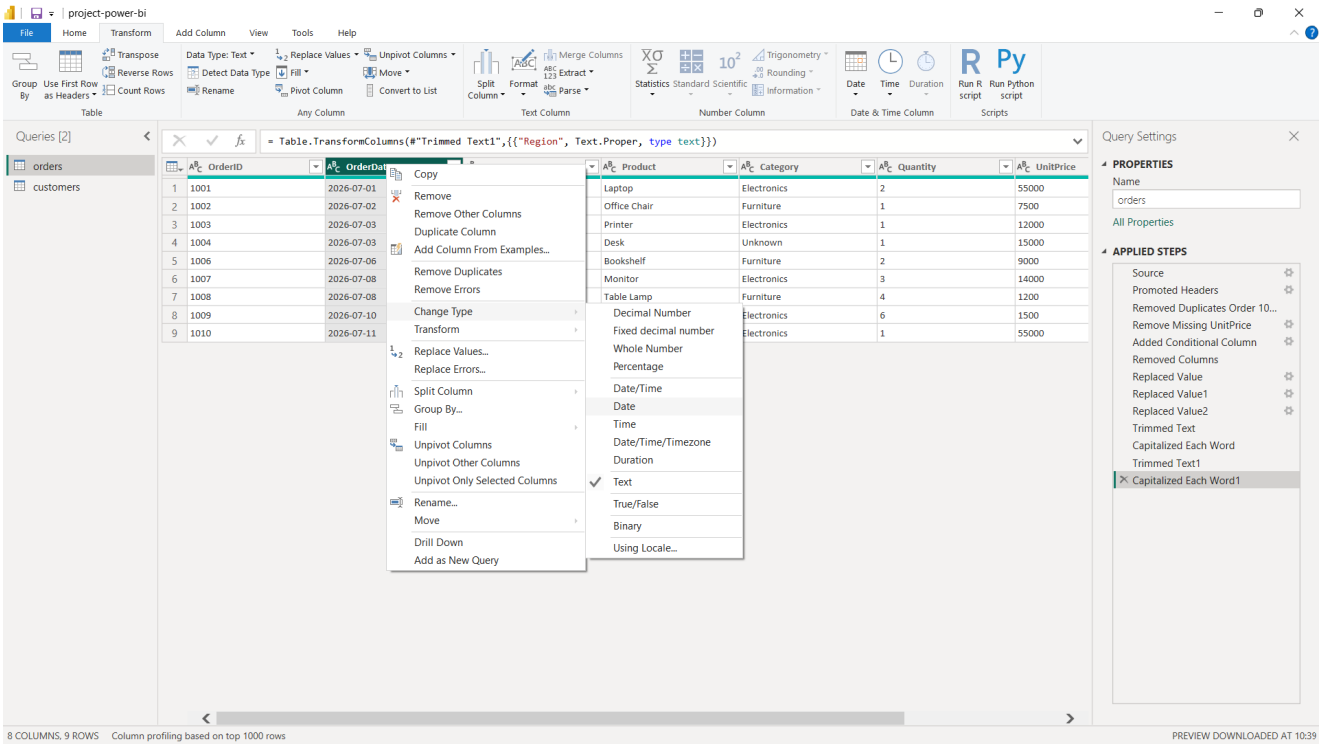
Step 12: Capitalized Category values consistently using proper-case formatting.



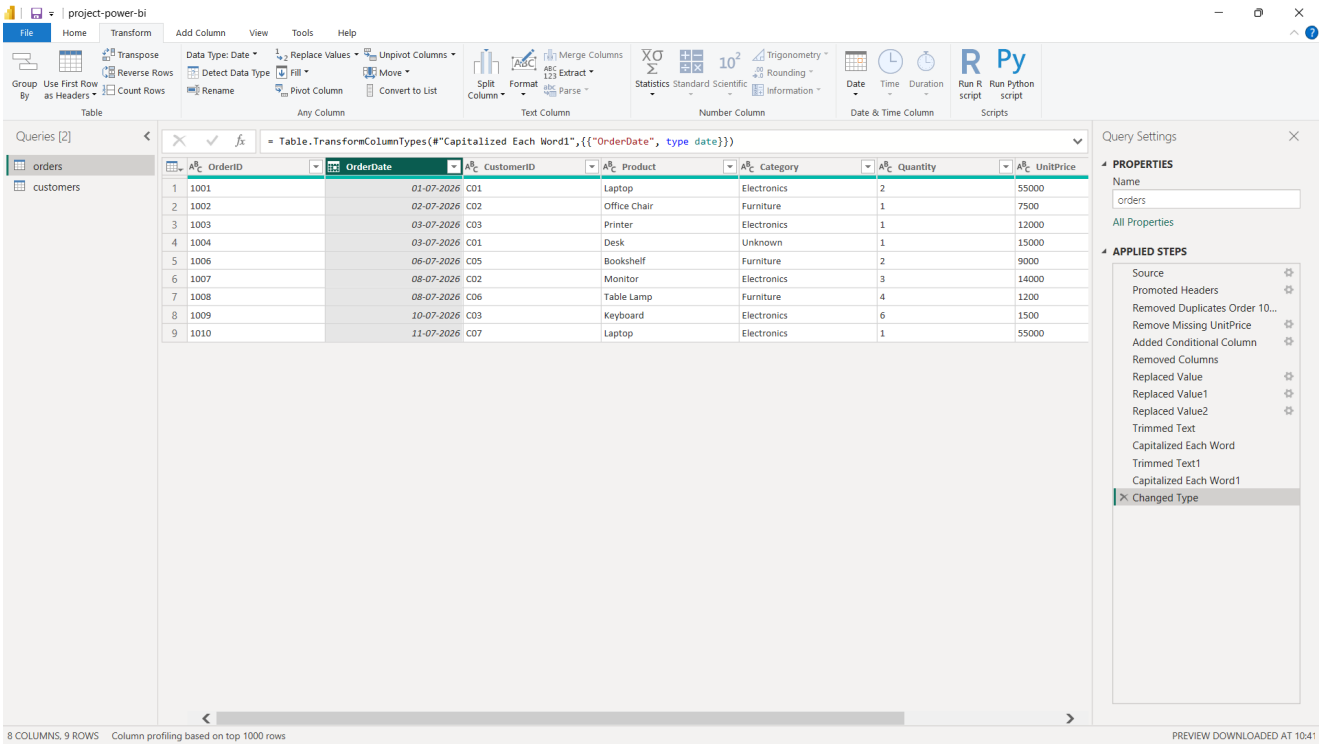
Step 13: Capitalized Region values consistently using proper-case formatting.



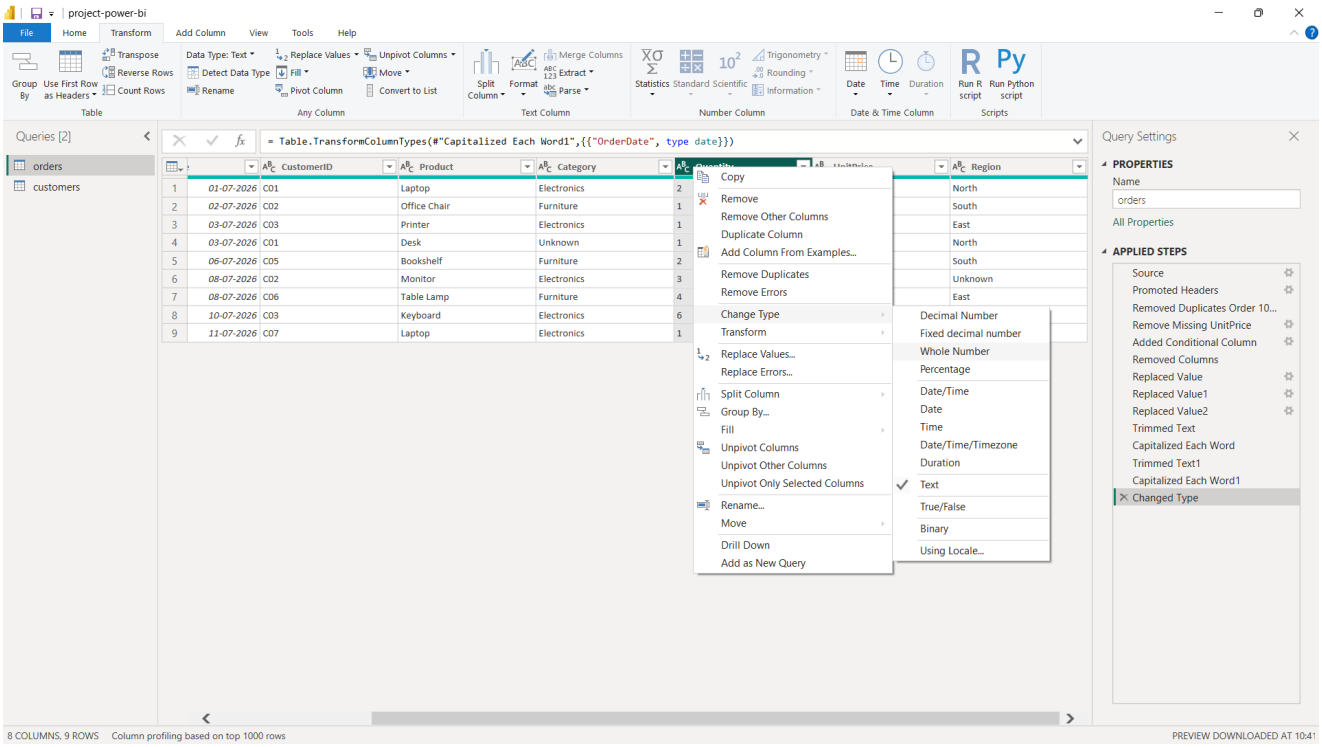
Step 14: Verified the cleaned orders table after the text-standardisation transformations.



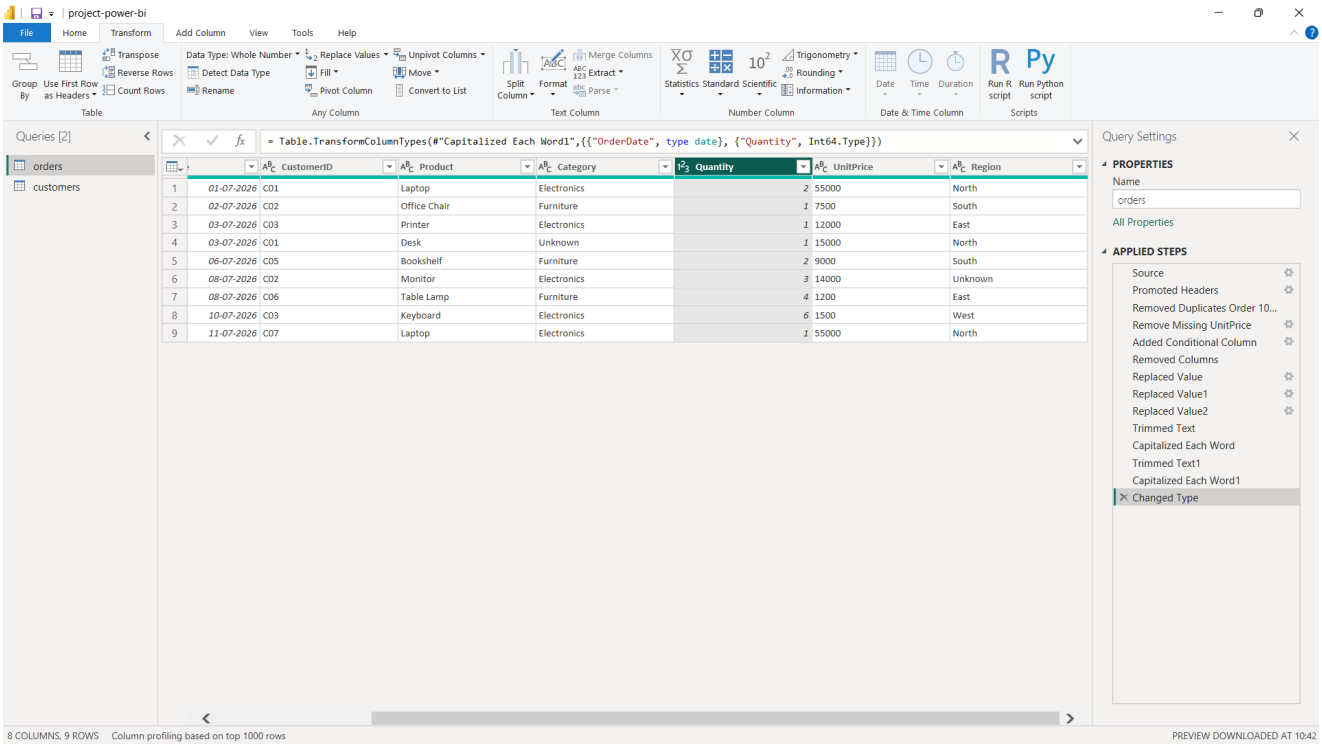
Step 15: Changed OrderDate to the Date data type.



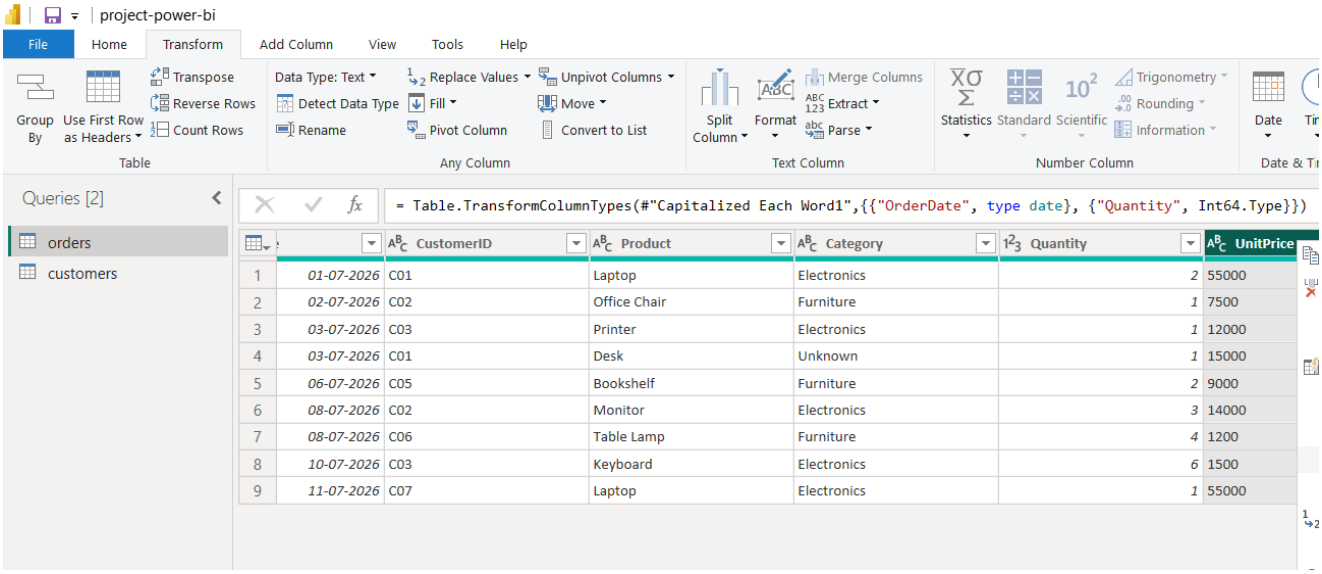
Step 16: Verified the cleaned order fields while continuing the required data-type transformations.



Step 17: Changed Quantity to Whole Number.

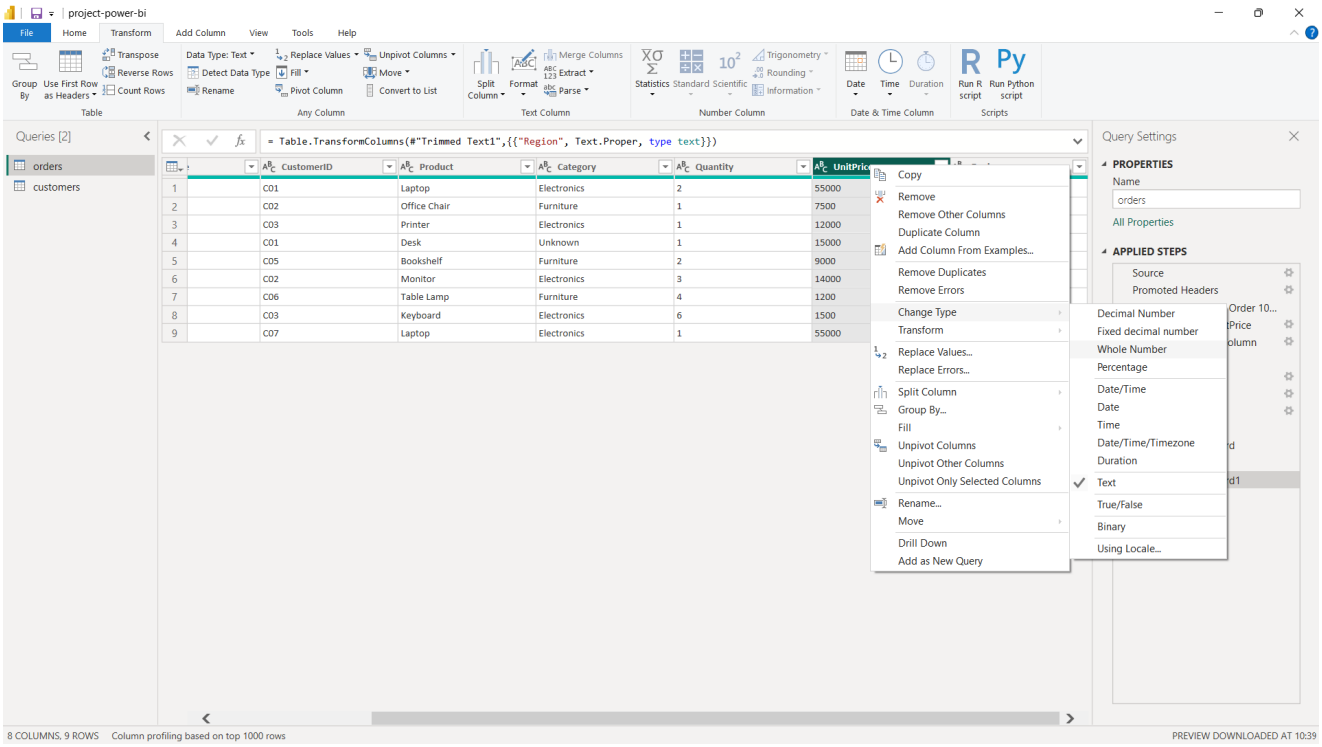


Step 18: Verified the cleaned orders table after the Quantity data-type change.

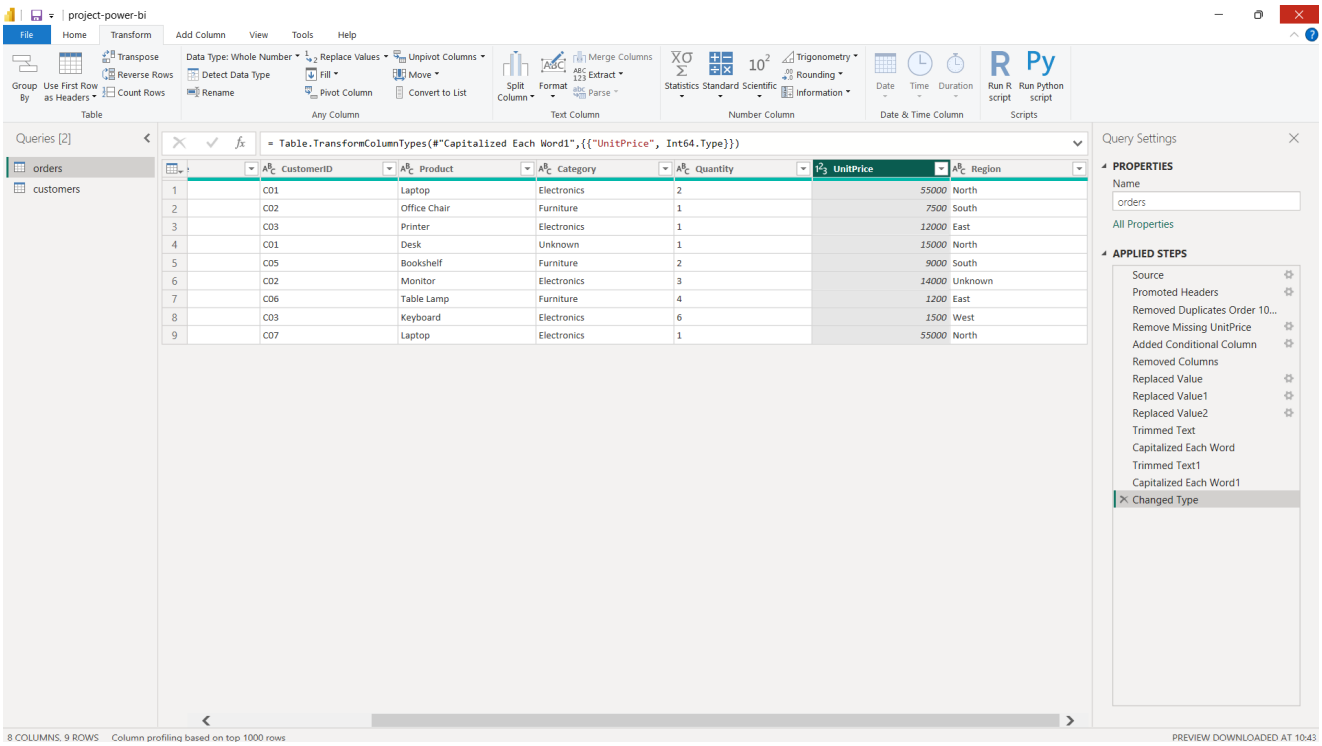


		CustomerID	Product	Category	Quantity	UnitPrice
1	01-07-2026	C01	Laptop	Electronics	2	55000
2	02-07-2026	C02	Office Chair	Furniture	1	7500
3	03-07-2026	C03	Printer	Electronics	1	12000
4	03-07-2026	C01	Desk	Unknown	1	15000
5	06-07-2026	C05	Bookshelf	Furniture	2	9000
6	08-07-2026	C02	Monitor	Electronics	3	14000
7	08-07-2026	C06	Table Lamp	Furniture	4	1200
8	10-07-2026	C03	Keyboard	Electronics	6	1500
9	11-07-2026	C07	Laptop	Electronics	1	55000

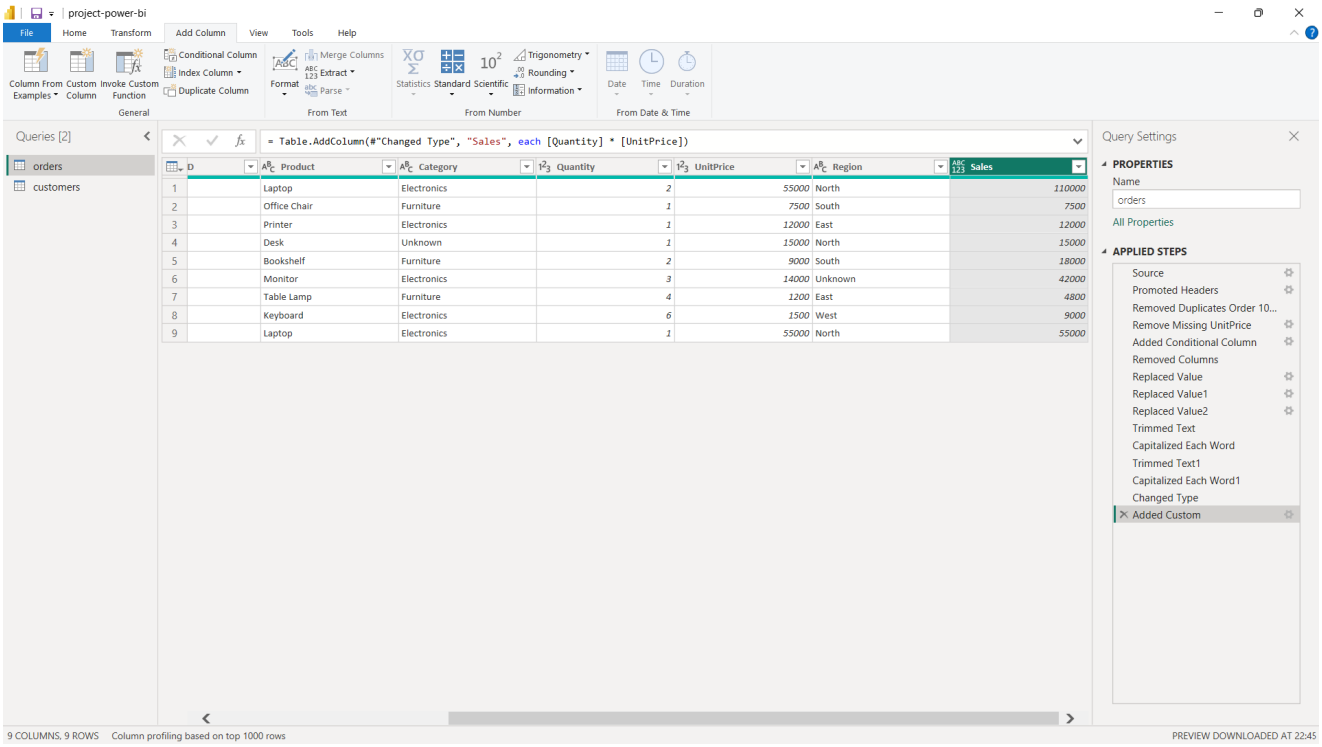
Step 19: Applied the remaining text/data-type transformation to the cleaned orders table.



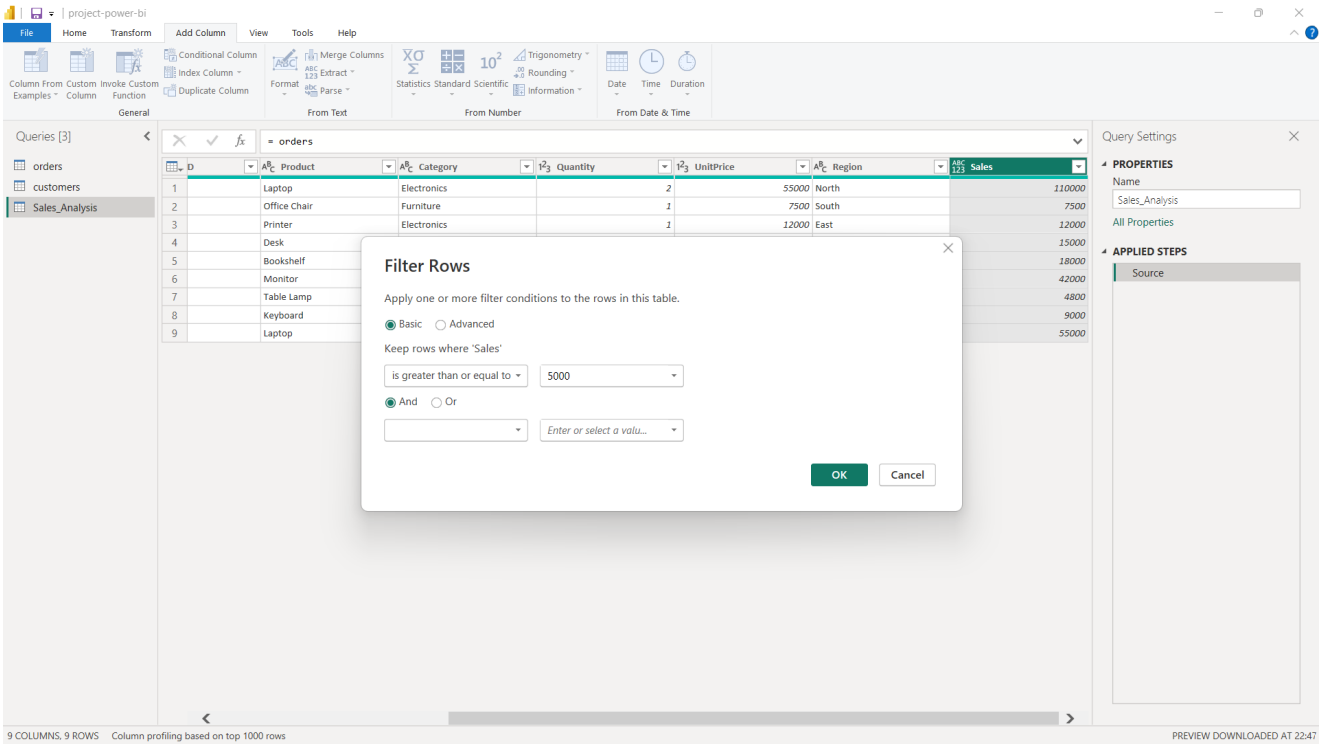
Step 20: Changed UnitPrice to the required numeric data type.



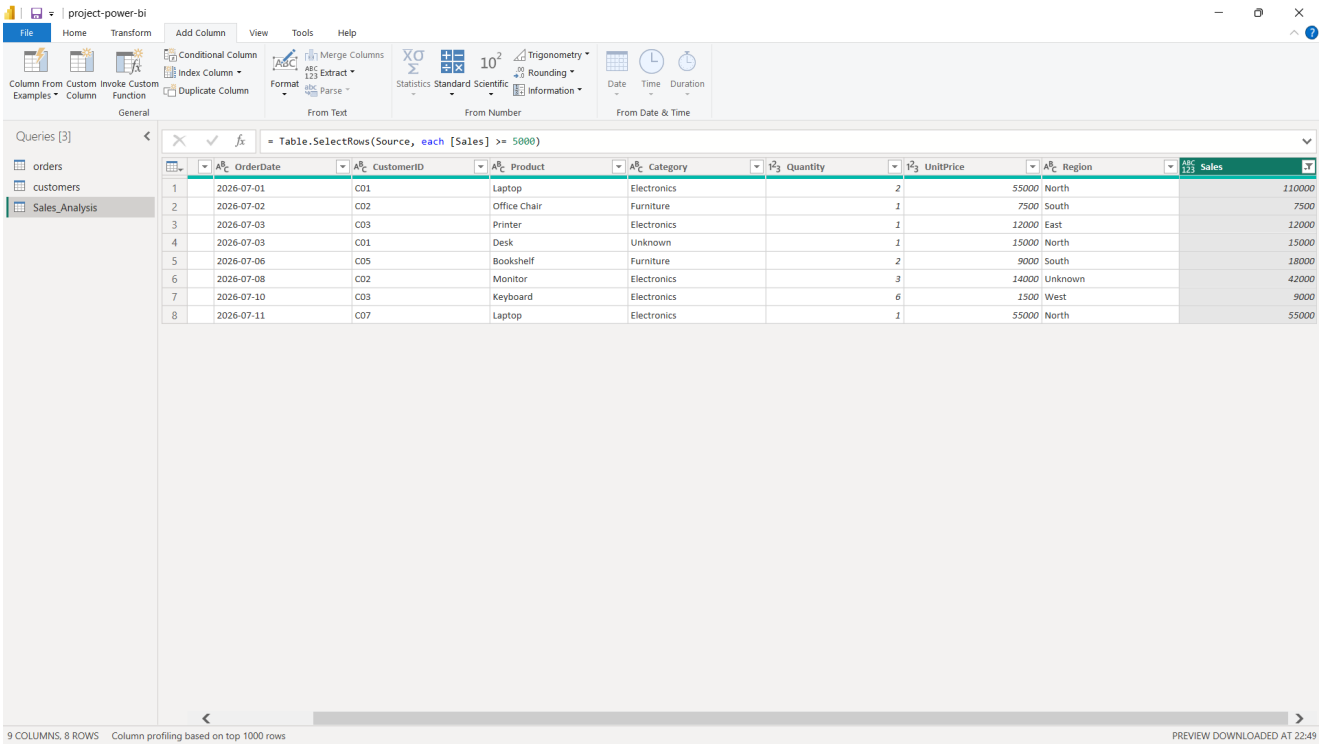
Step 21: Added the Sales column using Quantity × UnitPrice.



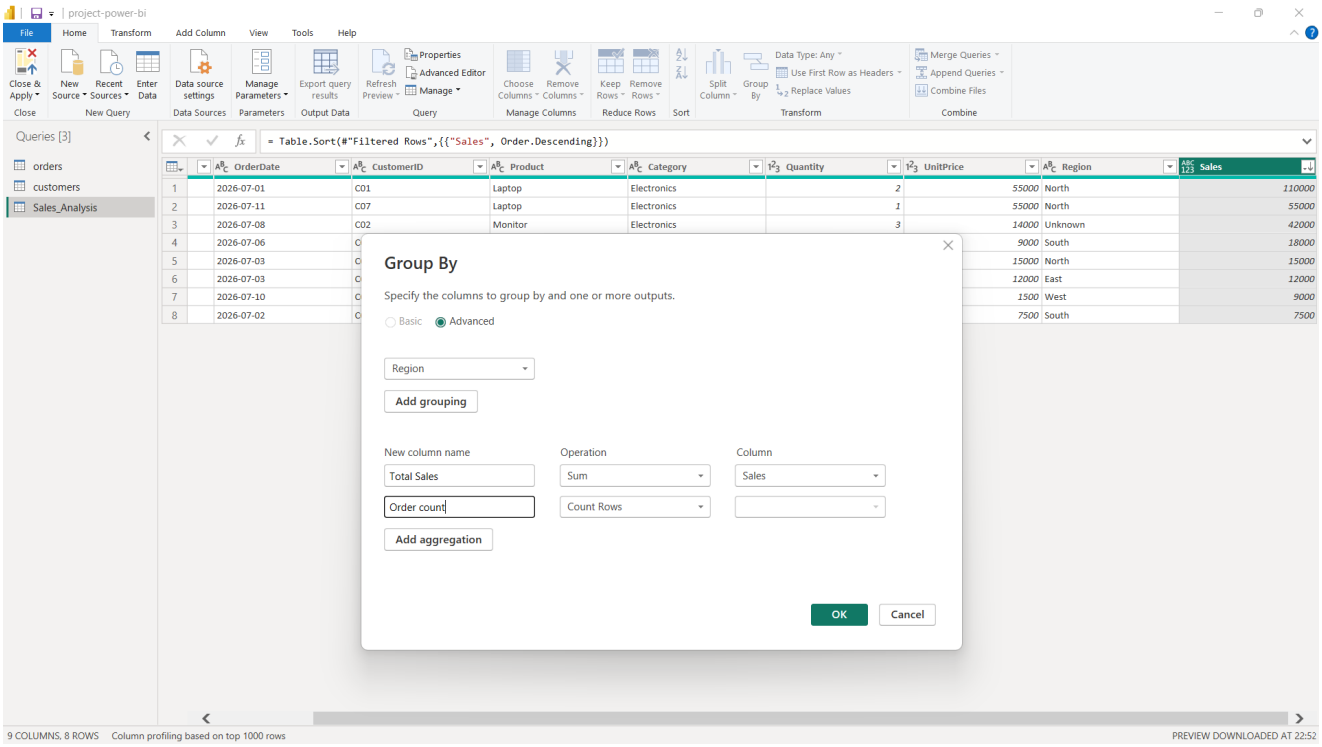
Step 22: Verified the calculated Sales values in the orders table.



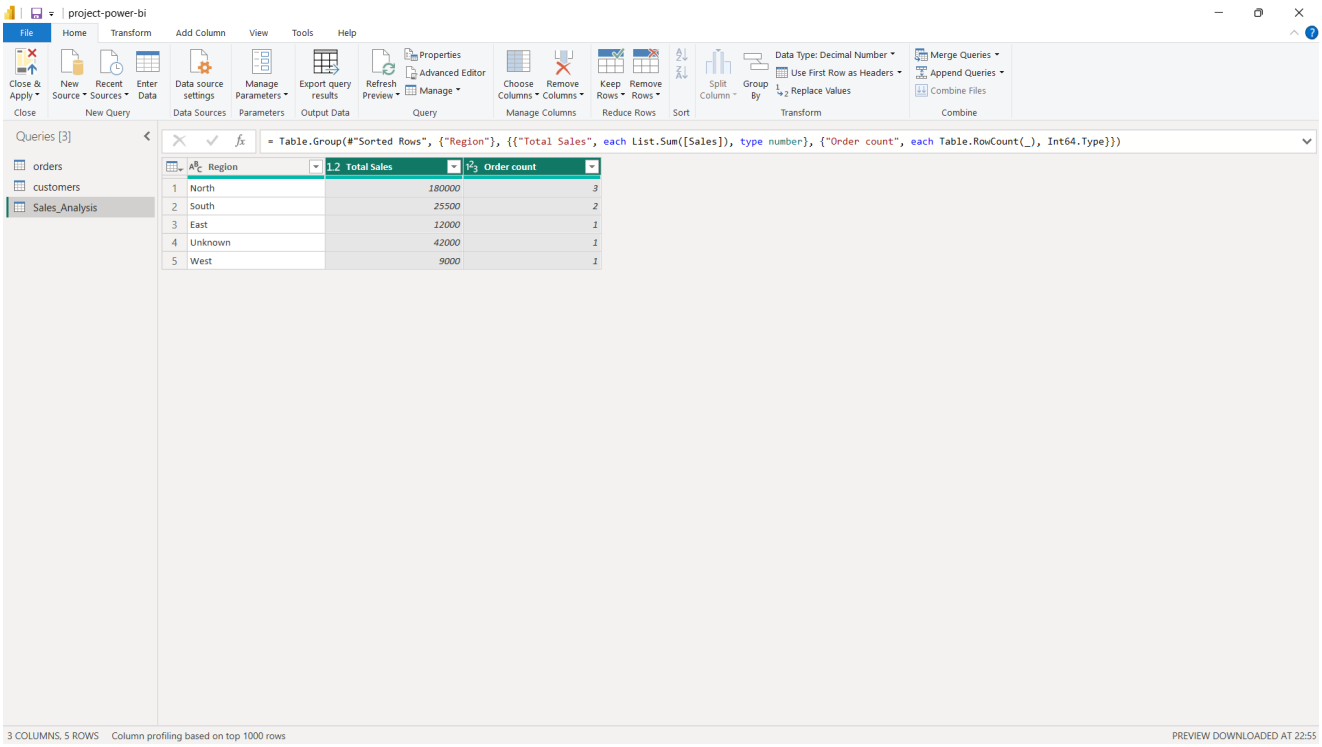
Step 23: Created the filtered copy containing orders with Sales greater than or equal to 5,000.



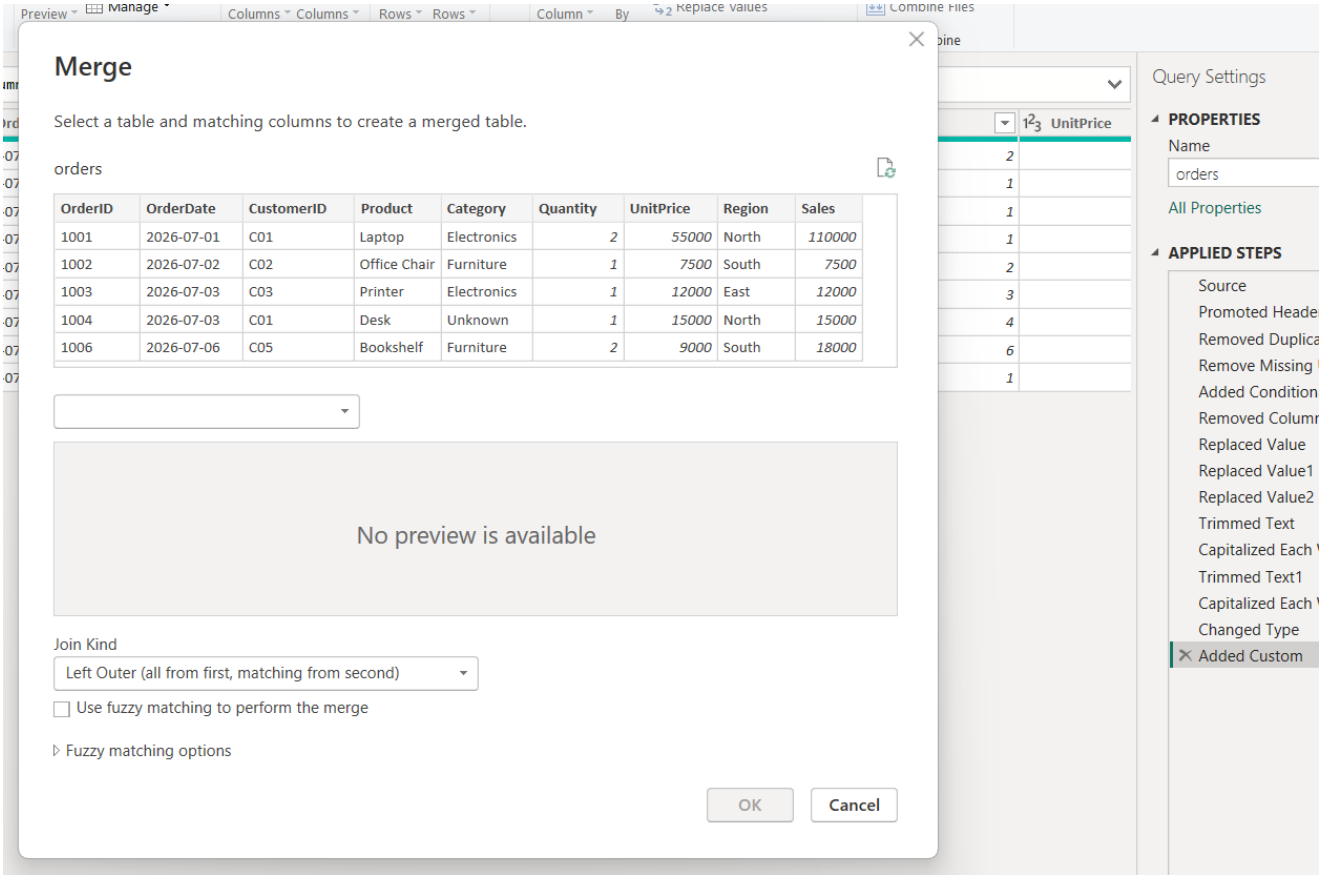
Step 24: Sorted the copied table by Sales in descending order.



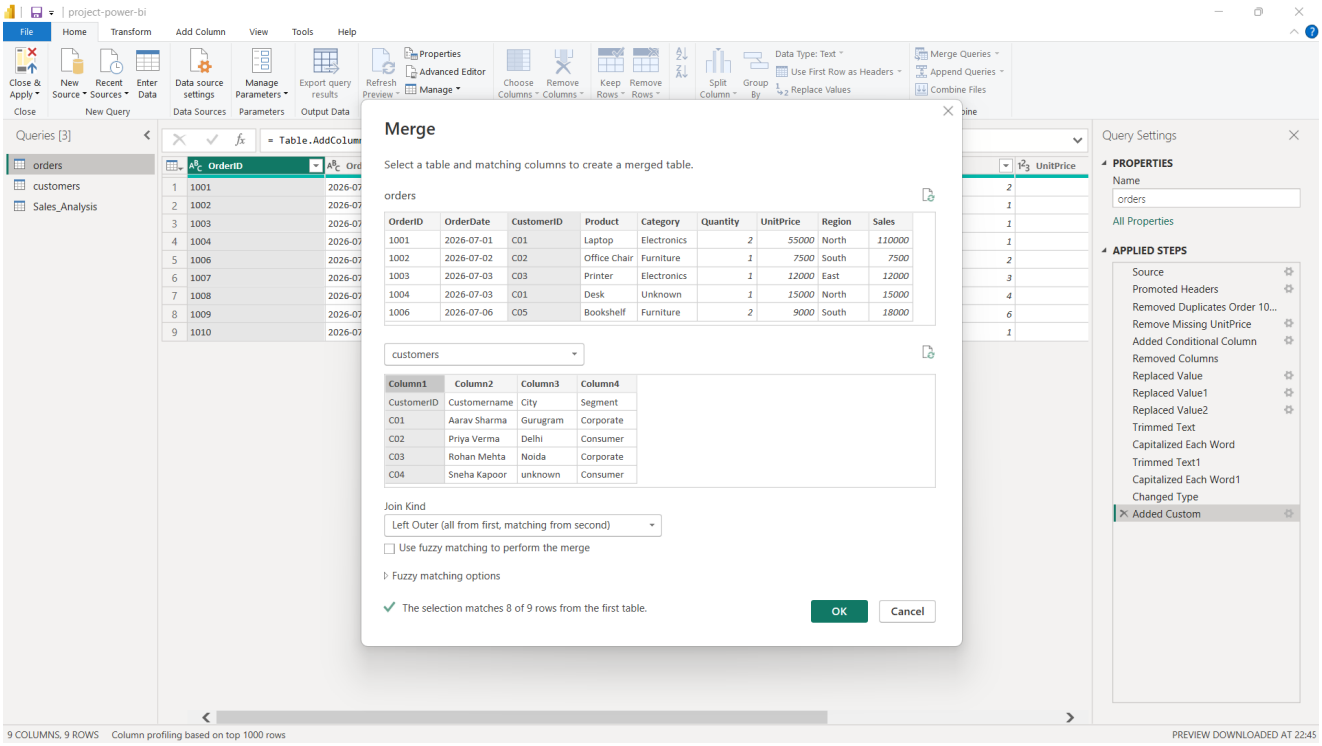
Step 25: Grouped the copied data by Region to prepare the regional summary.



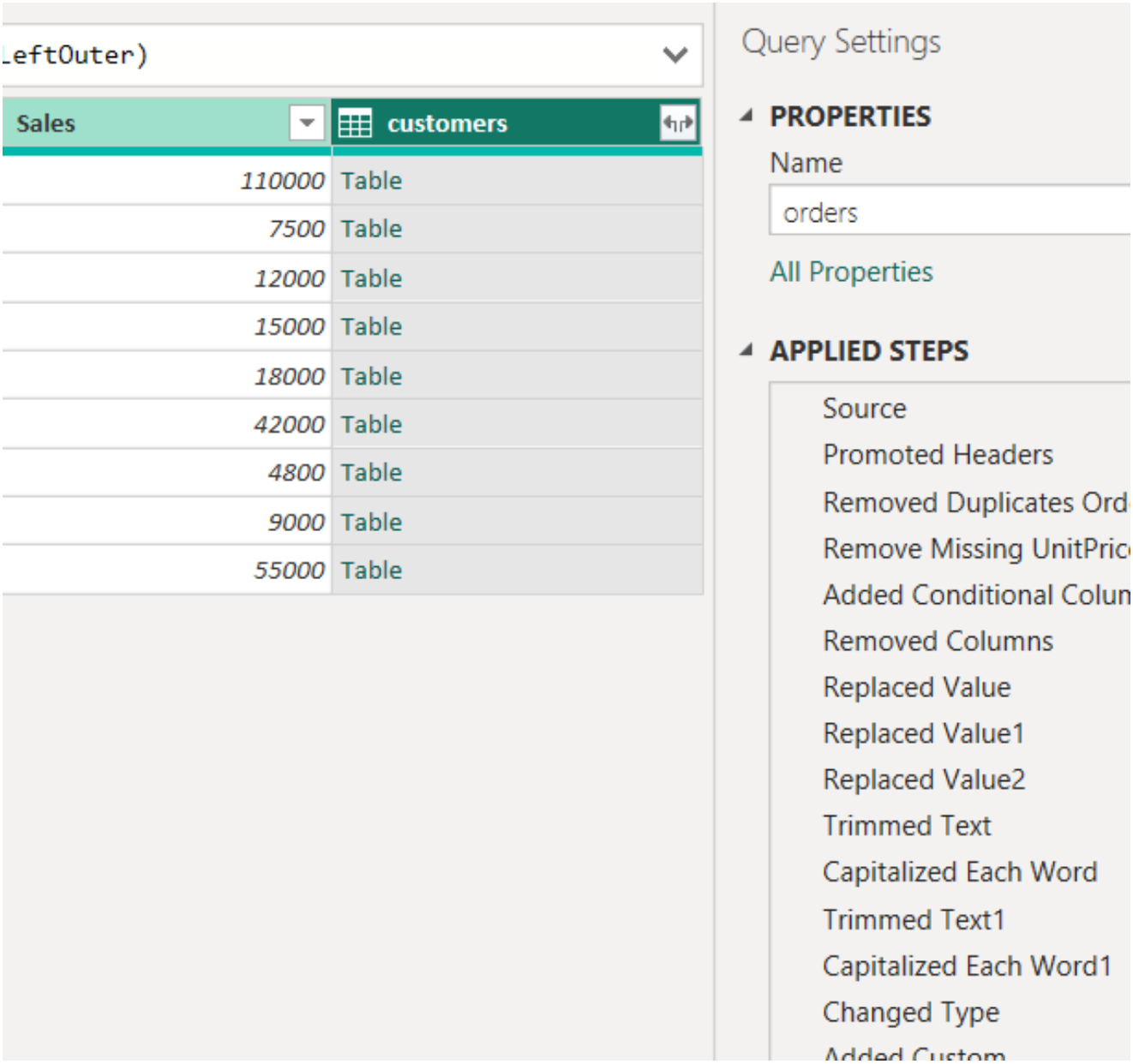
Step 26: Opened the Merge Queries operation to combine orders with customers using CustomerID.



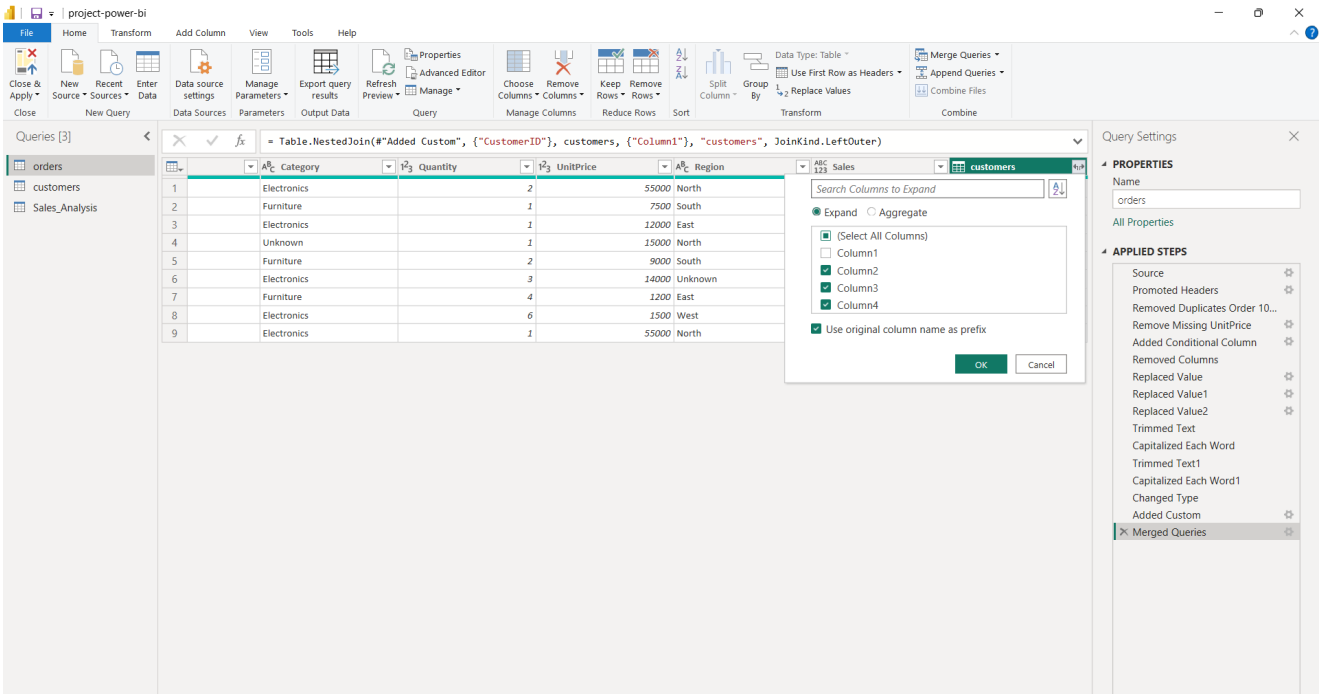
Step 27: Configured the successful Left Outer merge and matched CustomerID between the orders and customers tables.



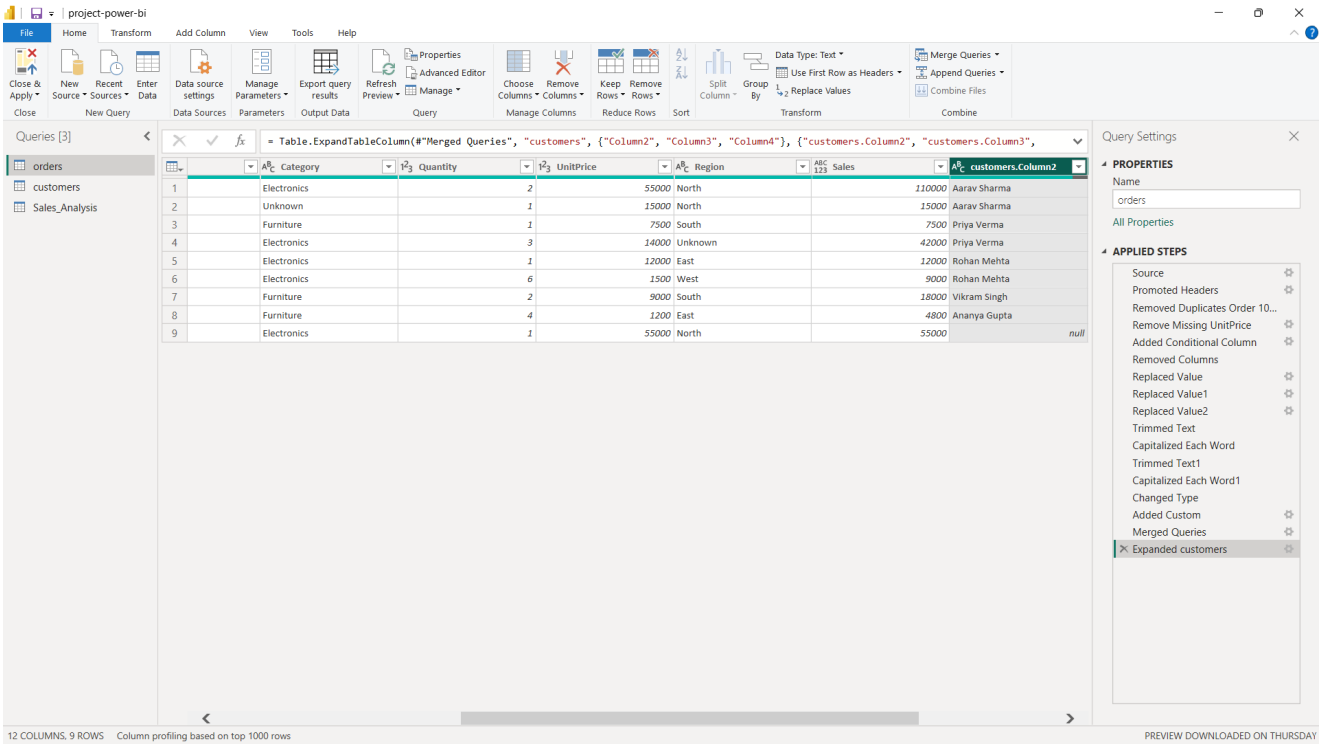
Step 28: Verified the merged table containing the nested customers table column.



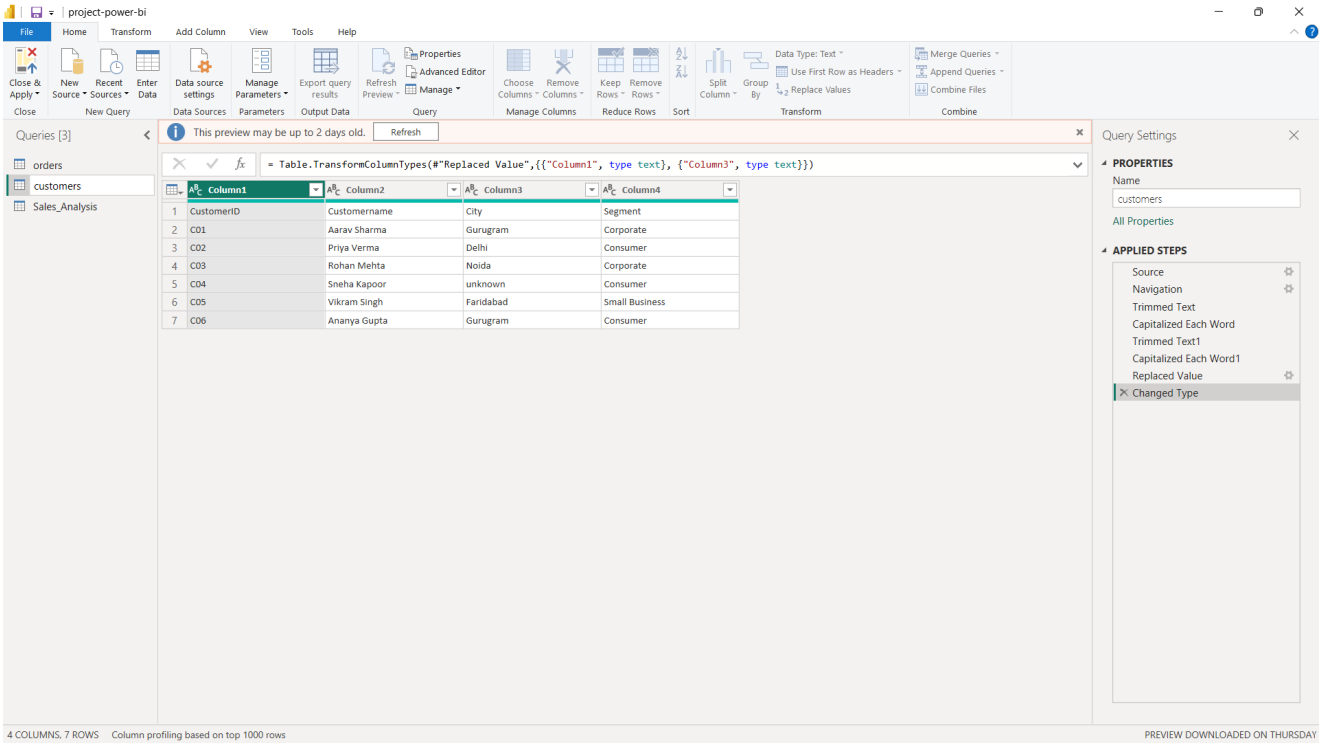
Step 29: Completed the merge step and generated the expanded customer data structure.



Step 30: Expanded the customer fields CustomerName, City and Segment into the orders table.

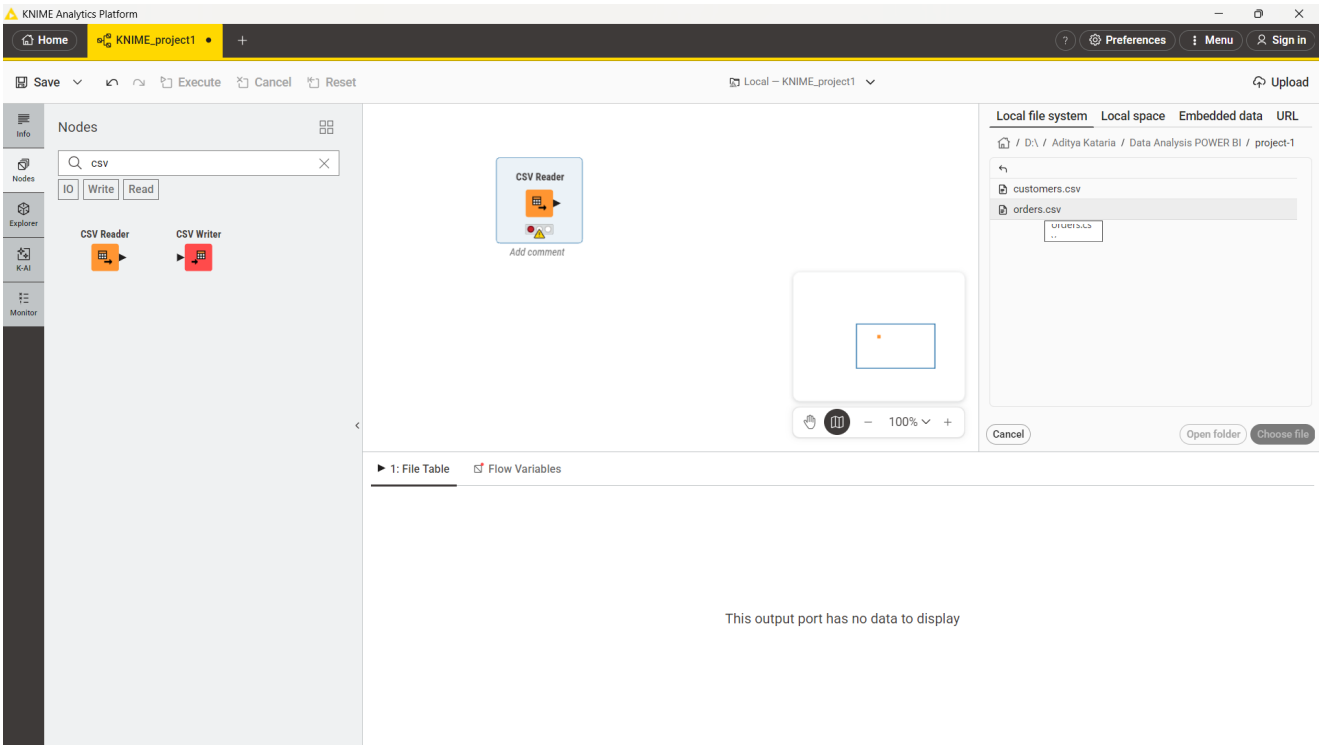


Step 31: Verified the cleaned customers table after trimming, capitalization and missing-City handling.

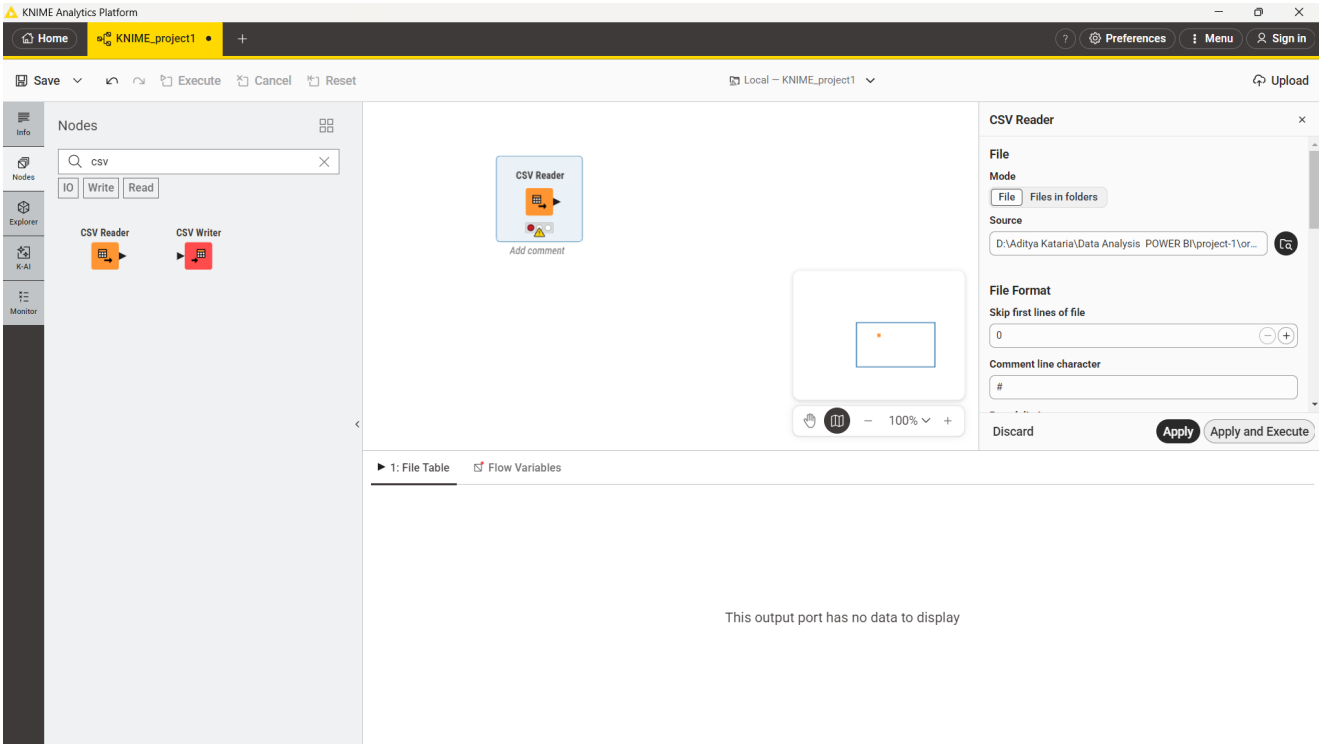


Data Cleaning and Preparation using KNIME

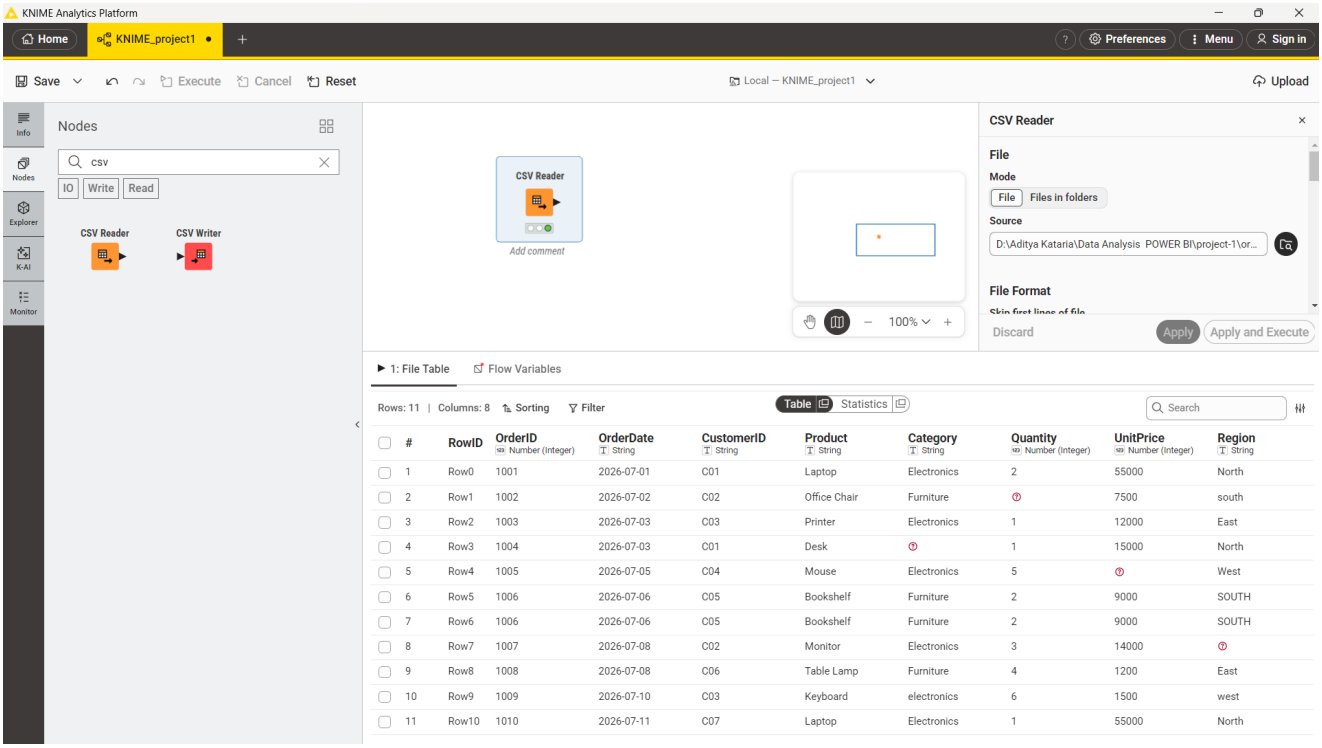
Step 1: Opened the KNIME workflow and prepared the nodes for reading the orders CSV and producing the cleaned output.



Step 2: Configured the CSV Reader and selected the orders.csv file as the input dataset.

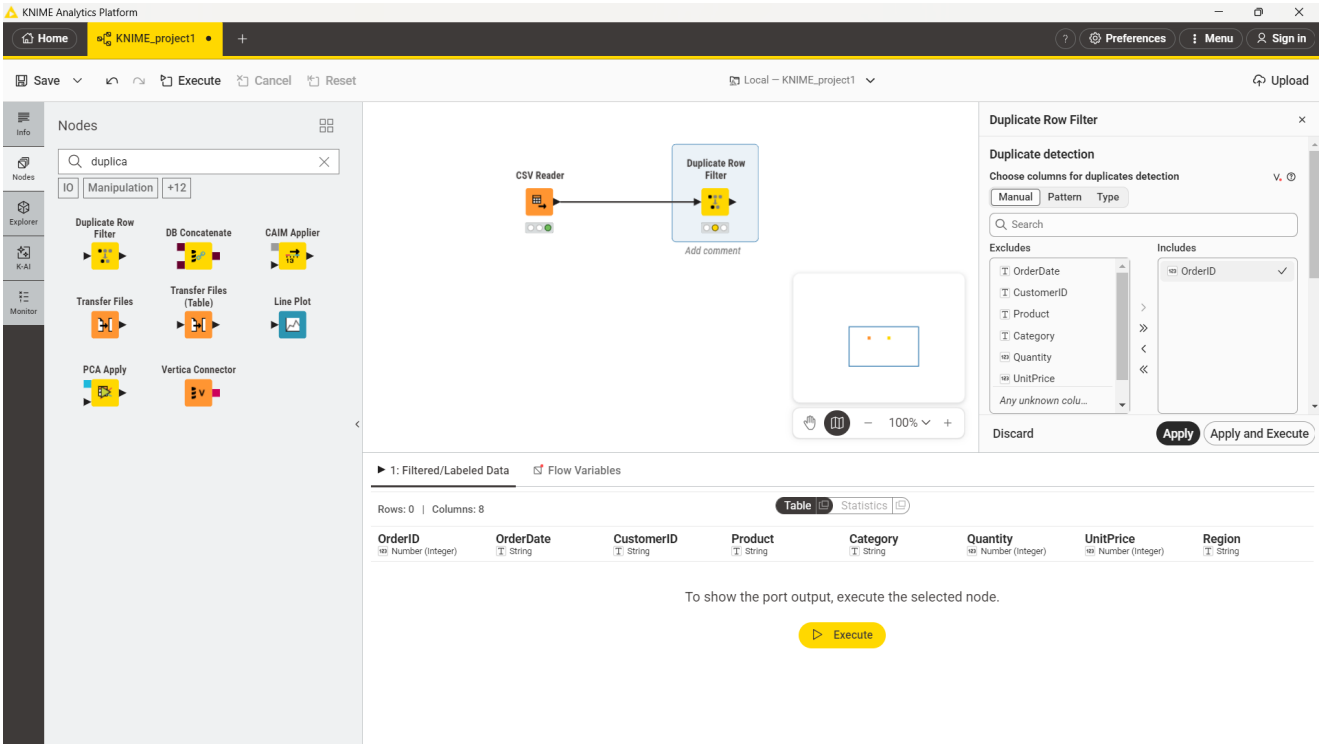


Step 3: Executed the CSV Reader and checked the imported orders table and its columns.

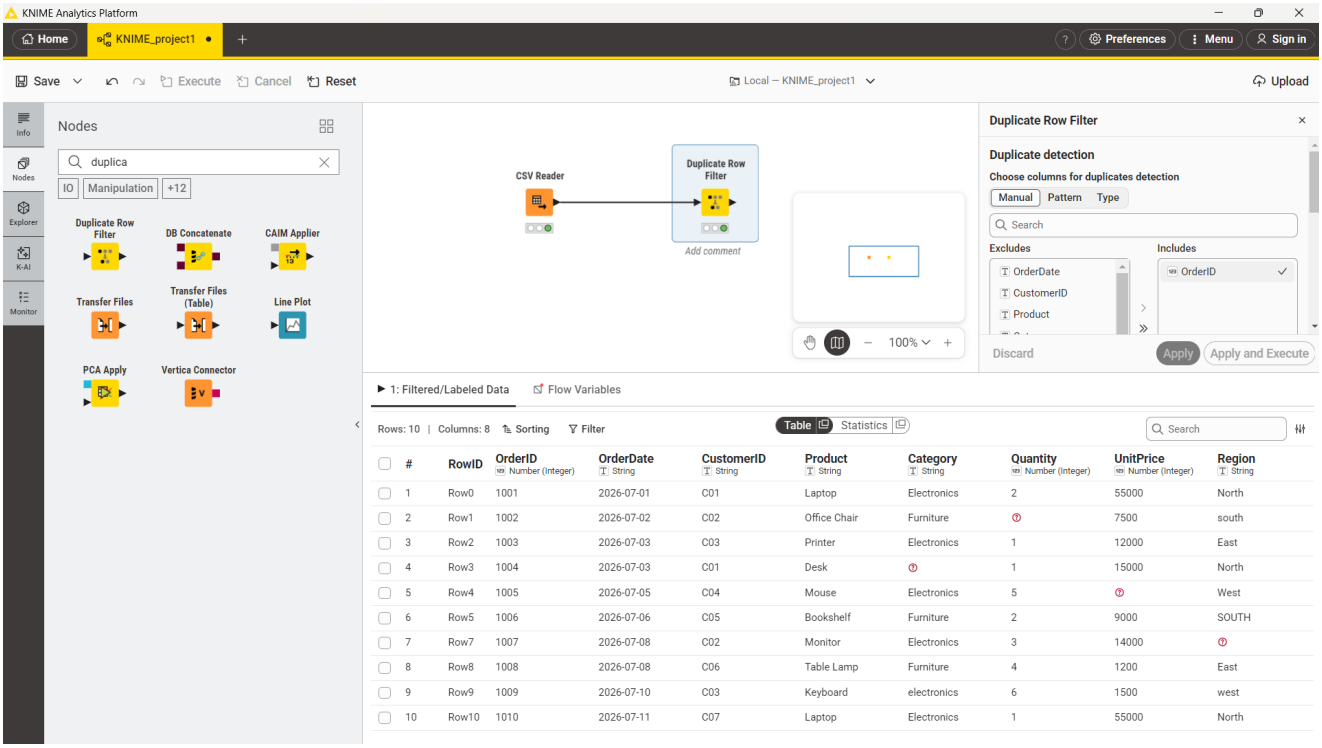


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region
		Number (Integer)	String	String	String	String	Number (Integer)	Number (Integer)	String
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	?	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row3	1004	2026-07-03	C01	Desk	?	1	15000	North
5	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
6	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
7	Row6	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
8	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
9	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
10	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
11	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

Step 4: Prepared the workflow for duplicate handling by adding the duplicate-row filtering step.

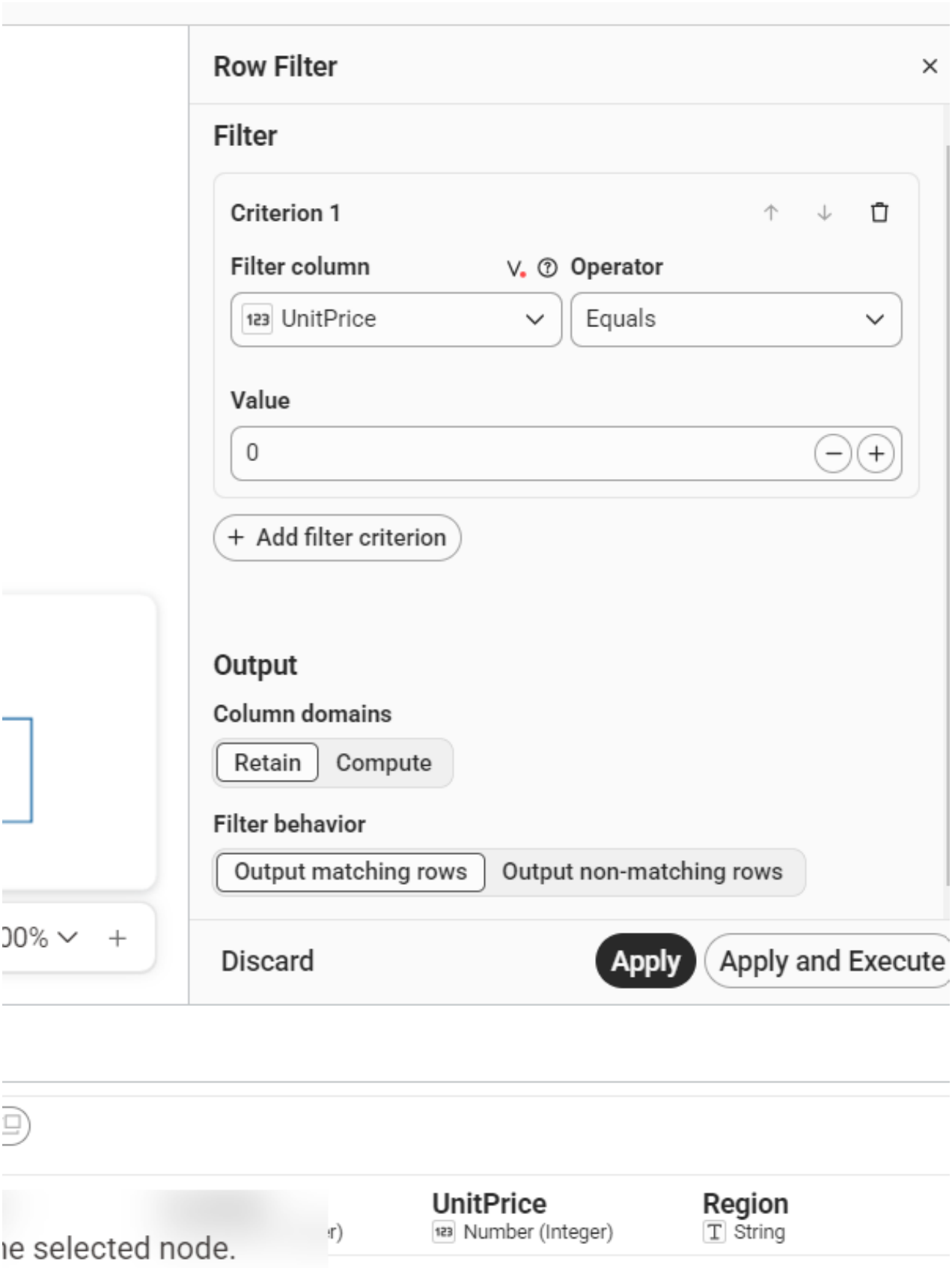


Step 5: Configured the duplicate-row operation to identify repeated order records.

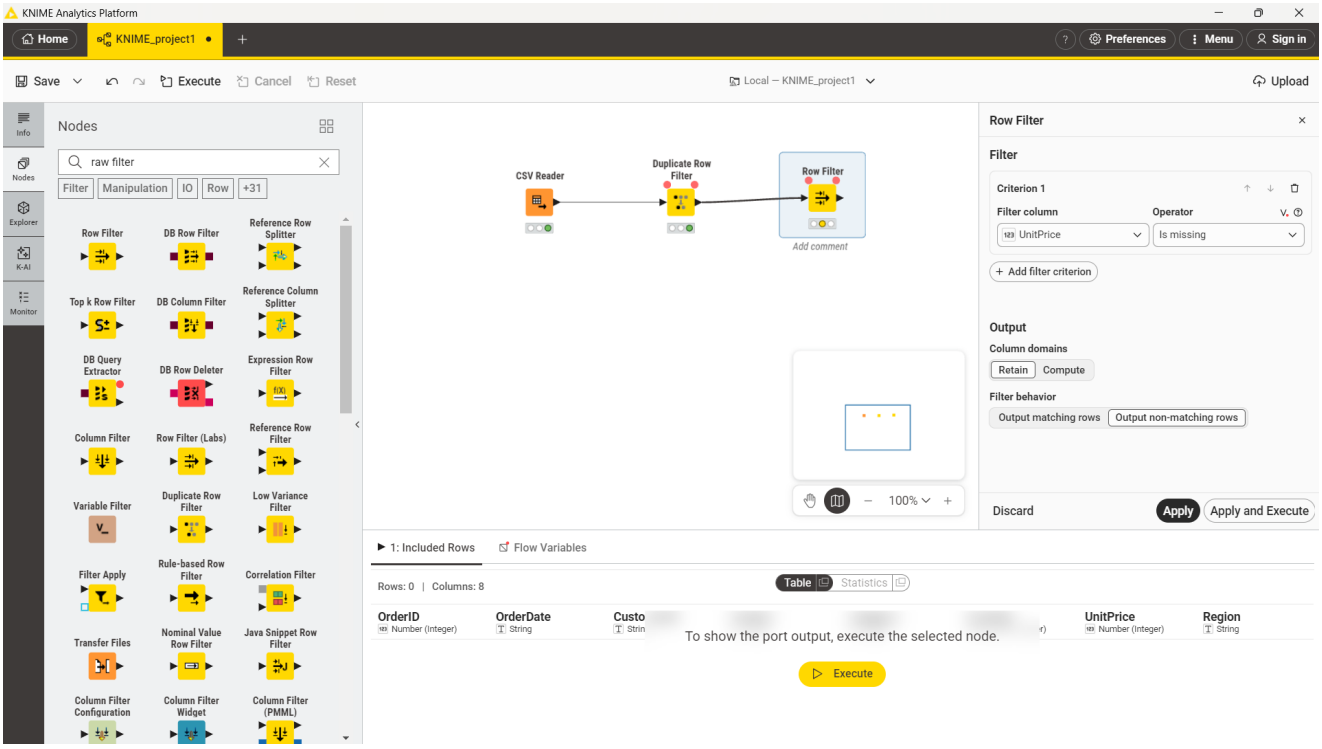


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region
		Number (Integer)	String	String	String	String	Number (Integer)	Number (Integer)	String
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	?	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row3	1004	2026-07-03	C01	Desk	?	1	15000	North
5	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
6	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
7	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
8	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
9	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
10	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

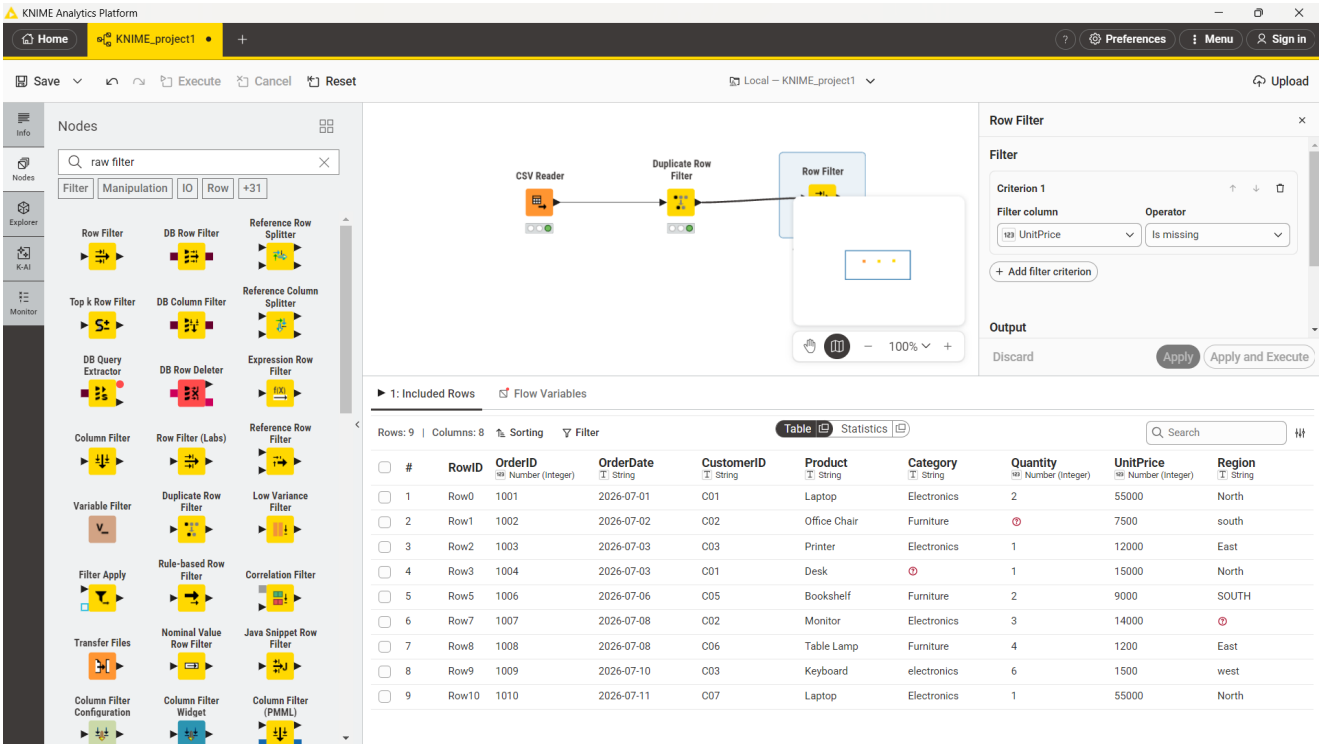
Step 6: Executed the duplicate-row step and inspected the resulting orders data.



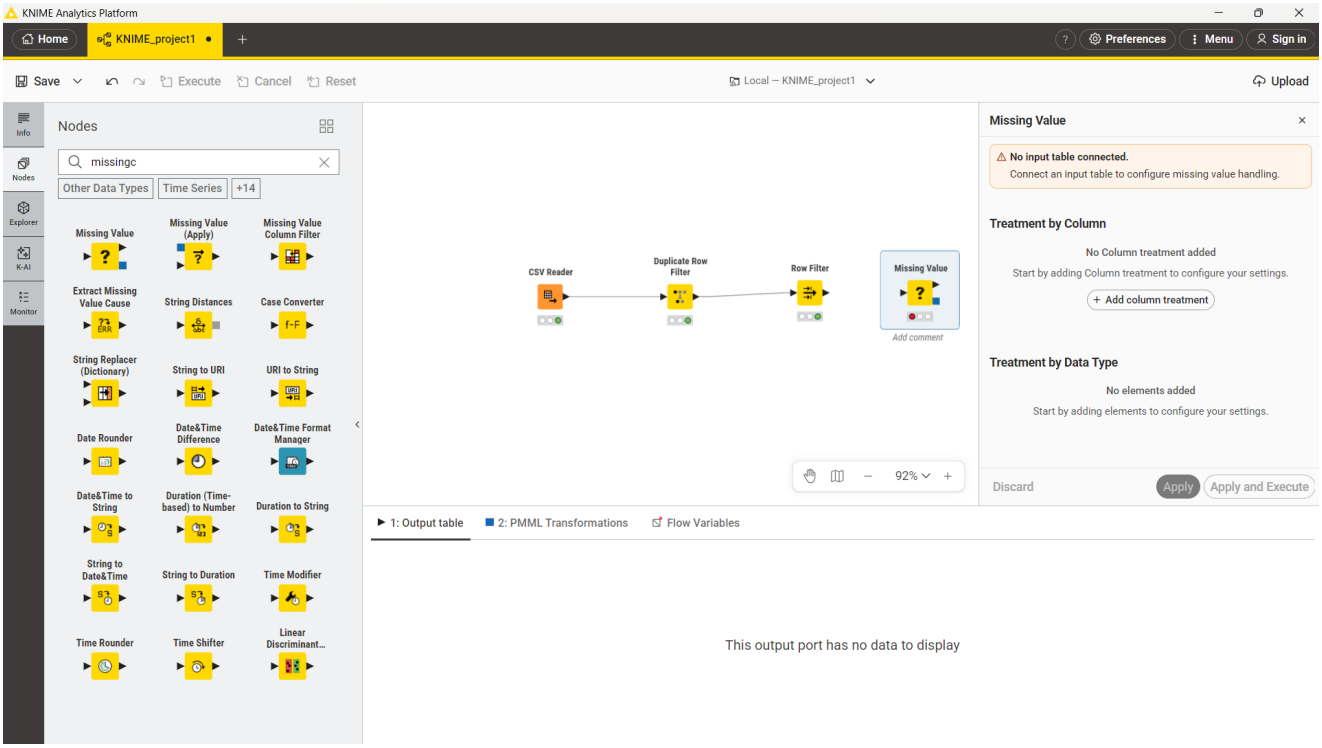
Step 7: Opened the Row Filter configuration for the required working-copy filter.



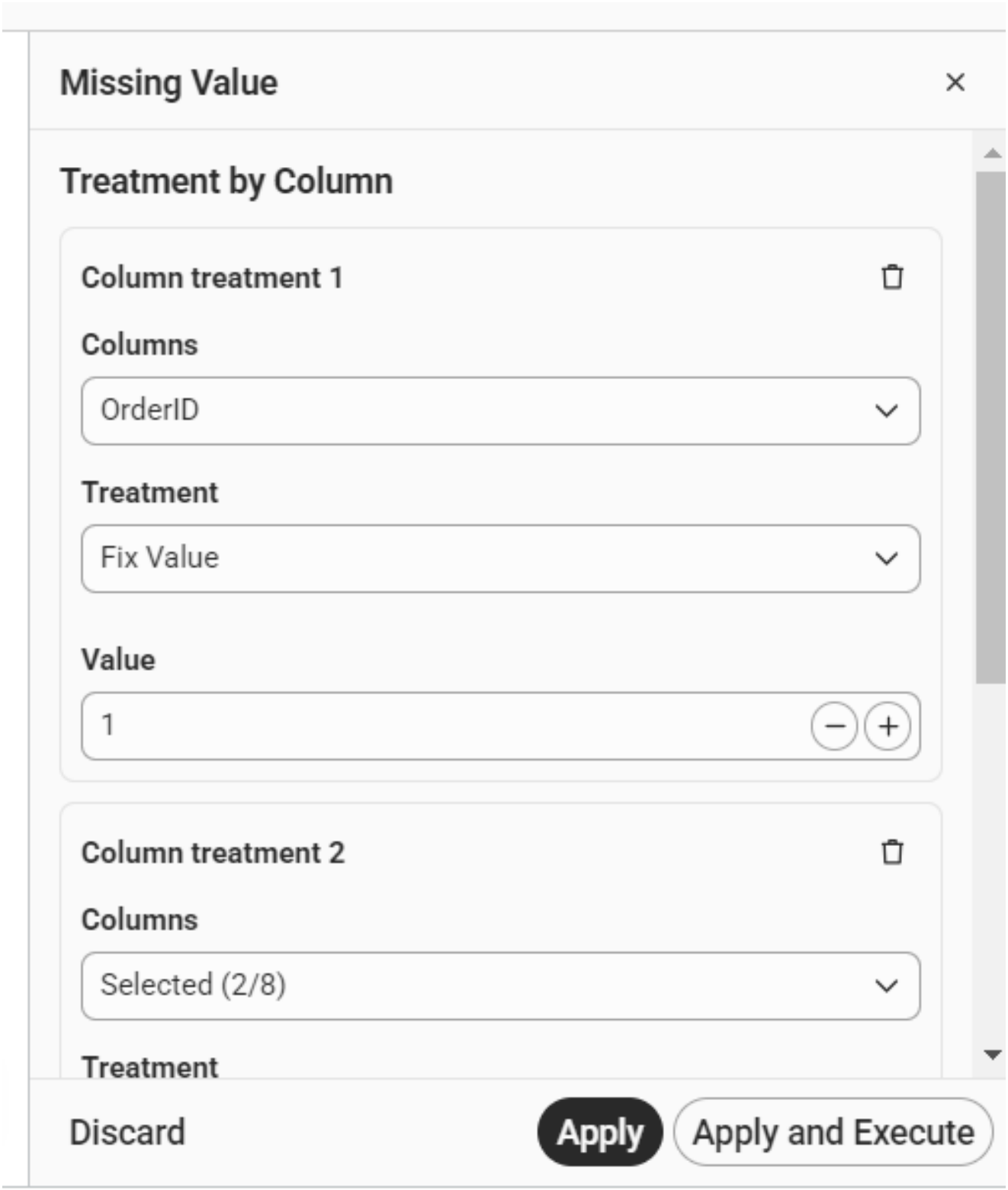
Step 8: Set the Row Filter condition for the requested subset of order records.



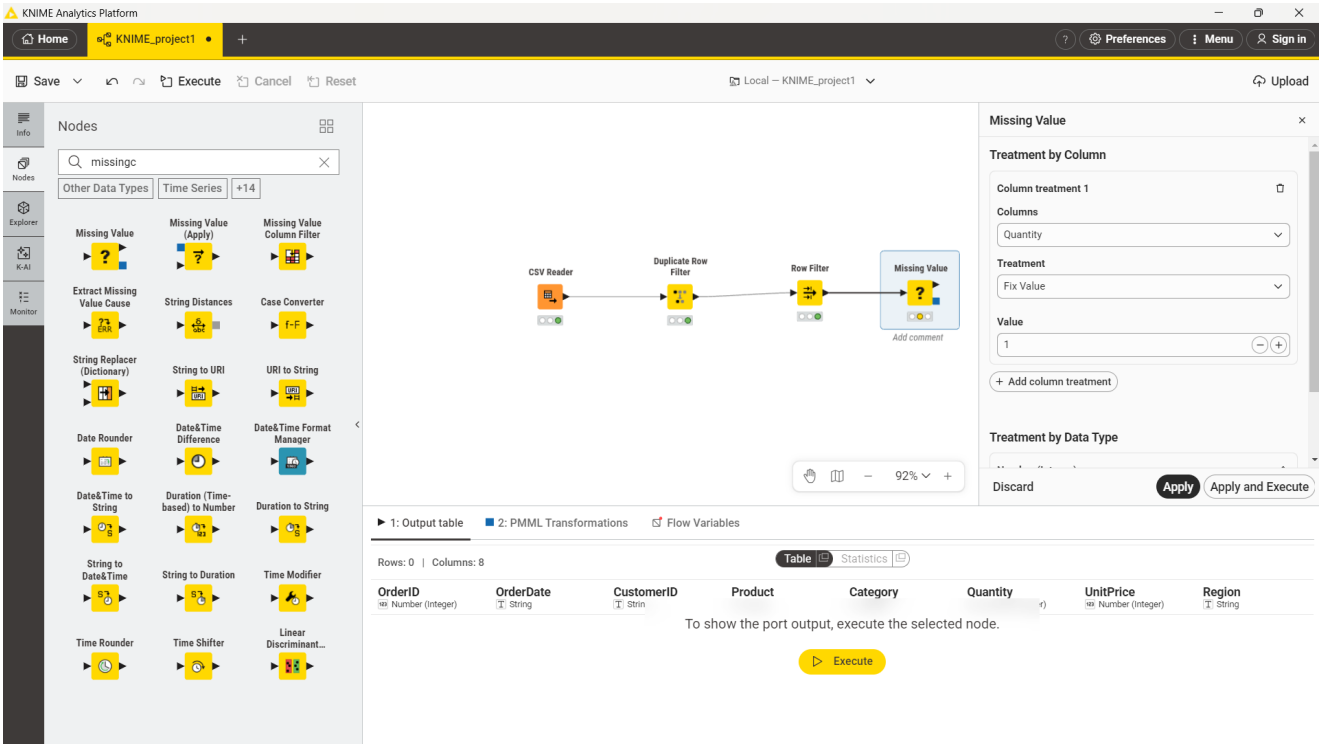
Step 9: Executed the filter and reviewed the filtered order table.



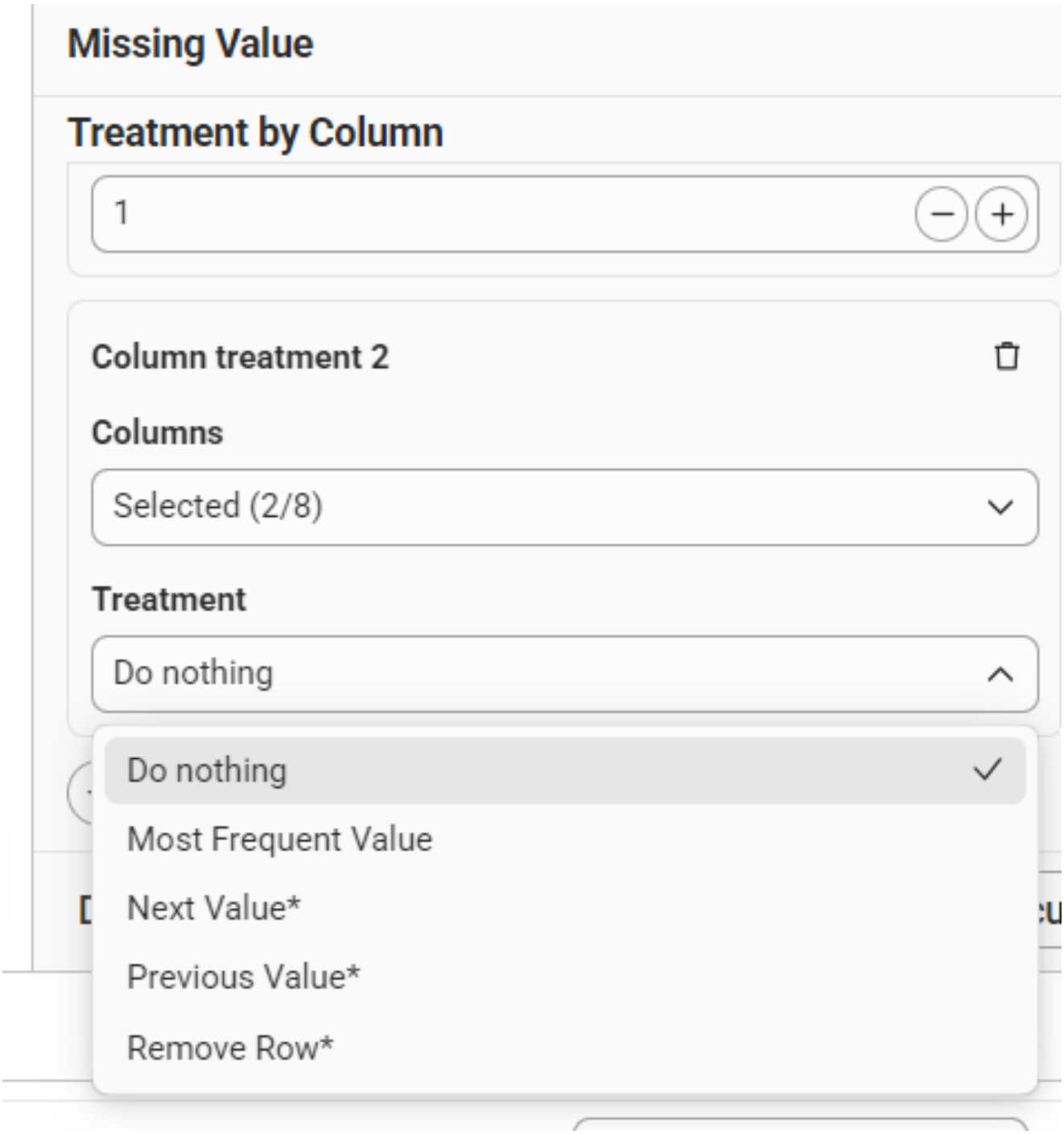
Step 10: Opened the Missing Value node to start cleaning blanks and missing entries.



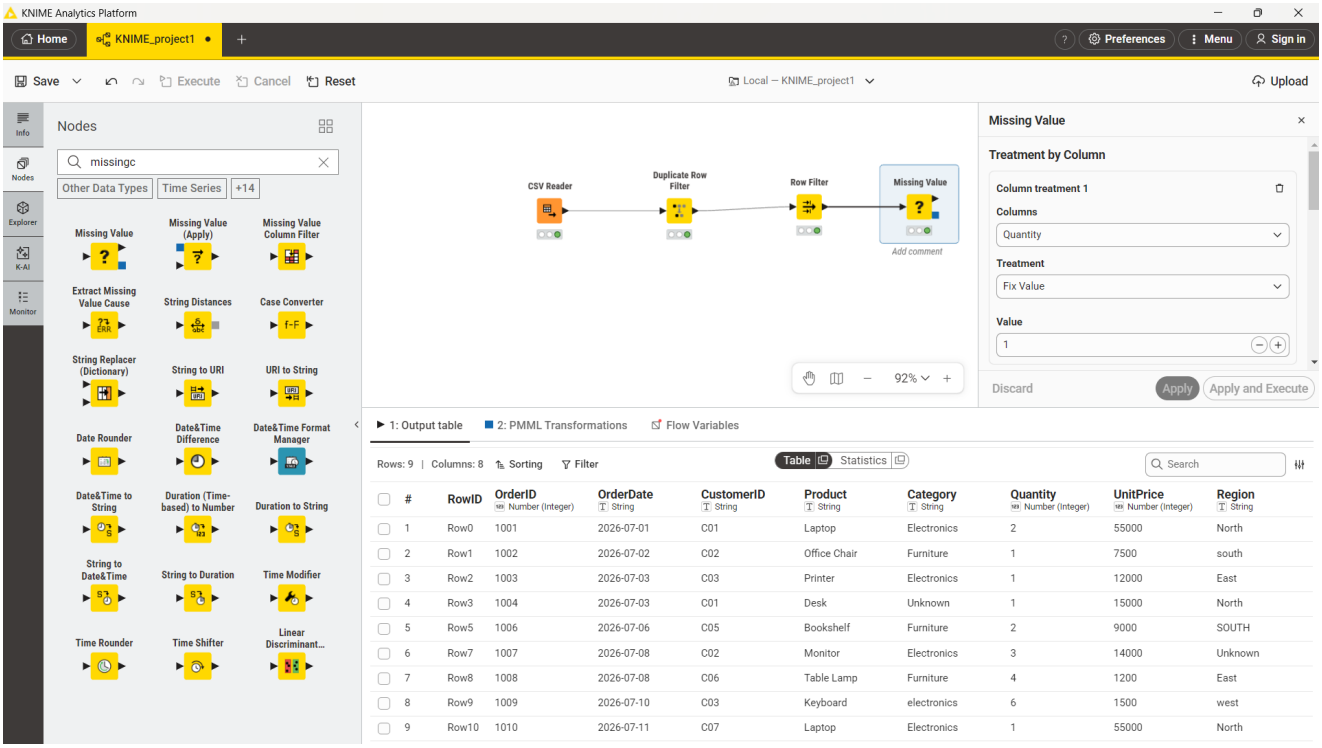
Step 11: Configured the Missing Value treatment for the selected columns.



Step 12: Adjusted the missing-value treatment options and prepared the node for execution.

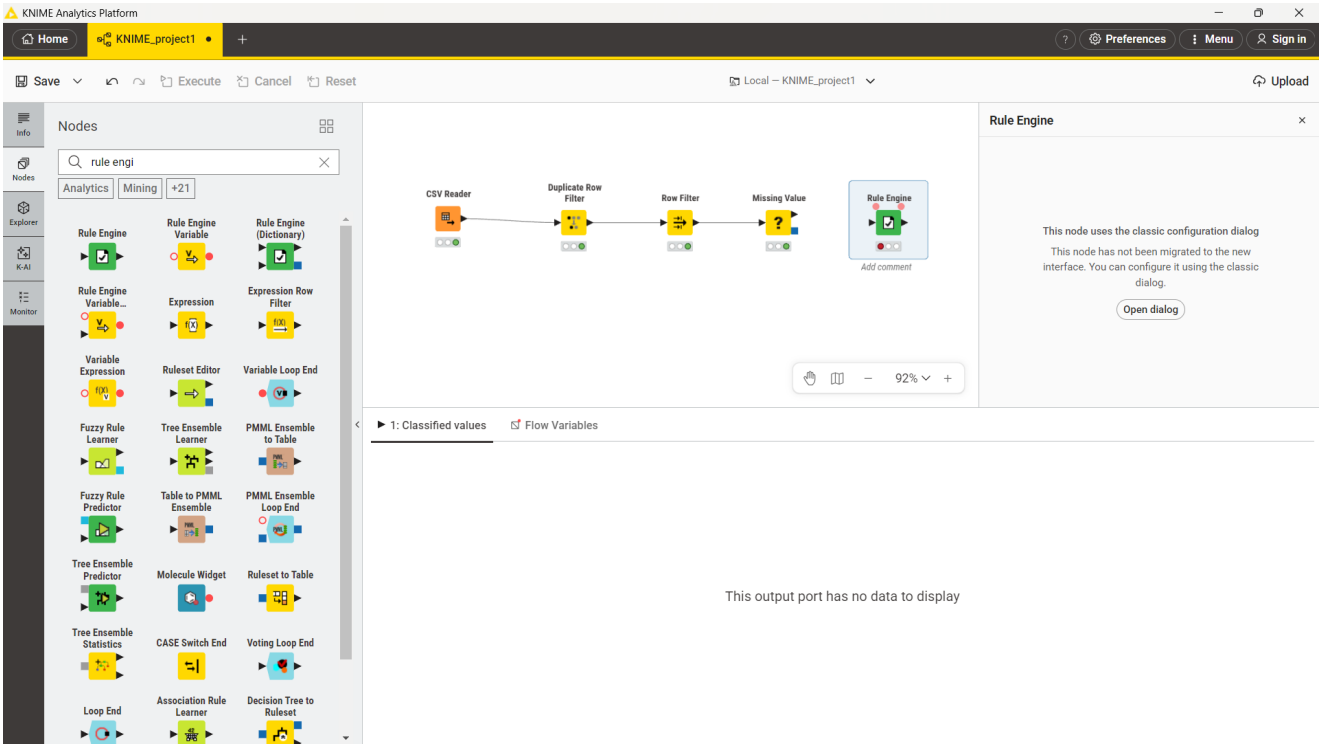


Step 13: Executed the missing-value cleaning and inspected the resulting table.

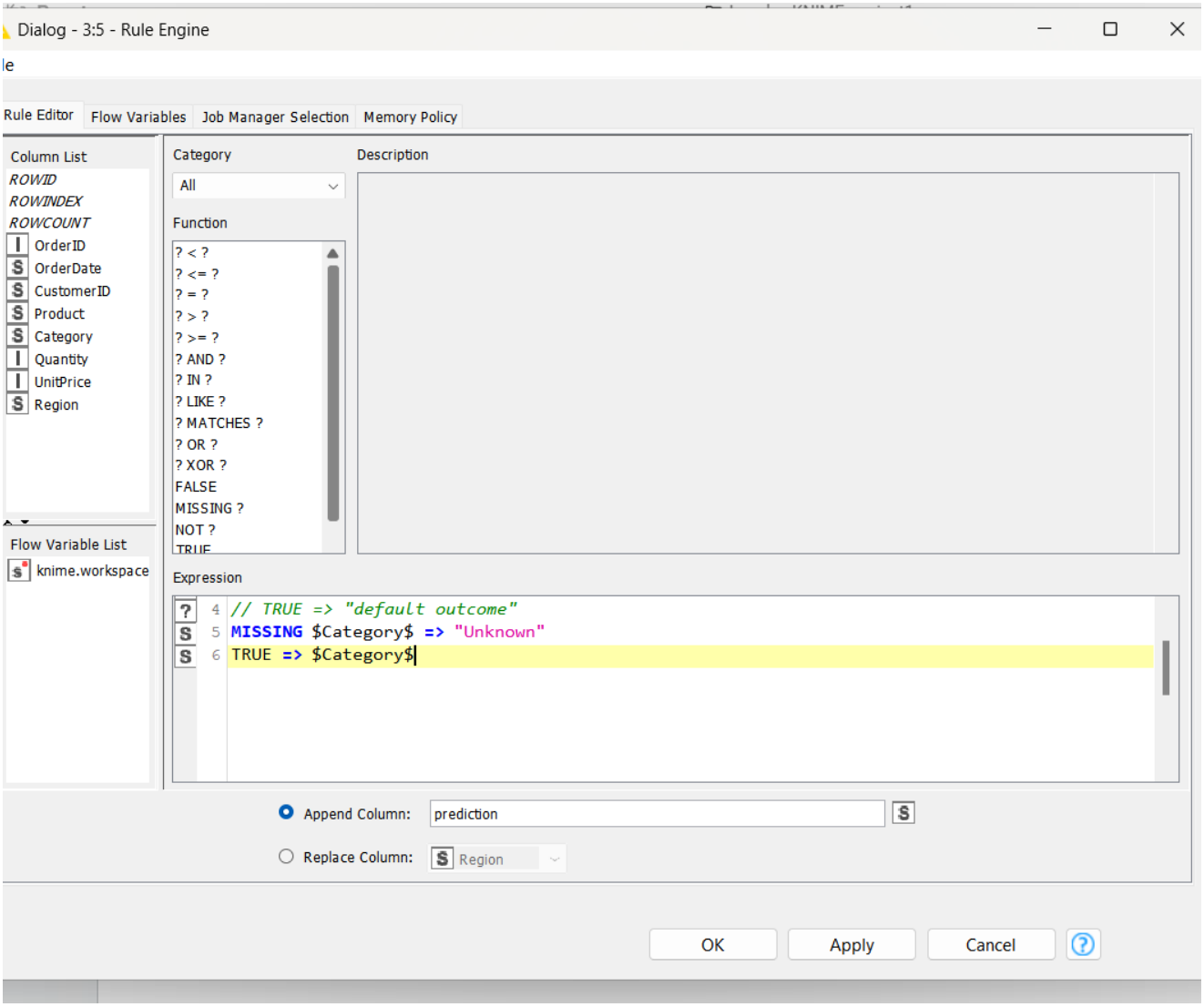


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row3	1004	2026-07-03	C01	Desk	Unknown	1	15000	North
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

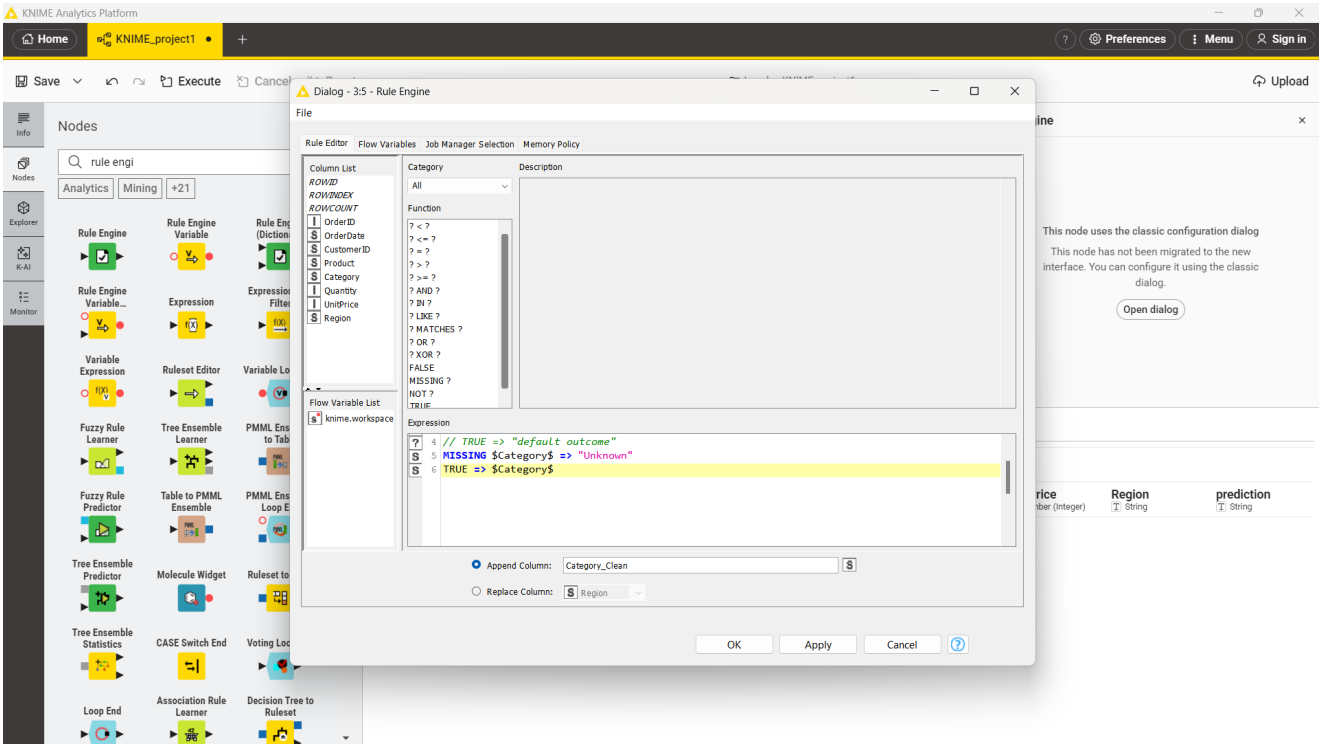
Step 14: Verified the cleaned table after missing-value treatment.



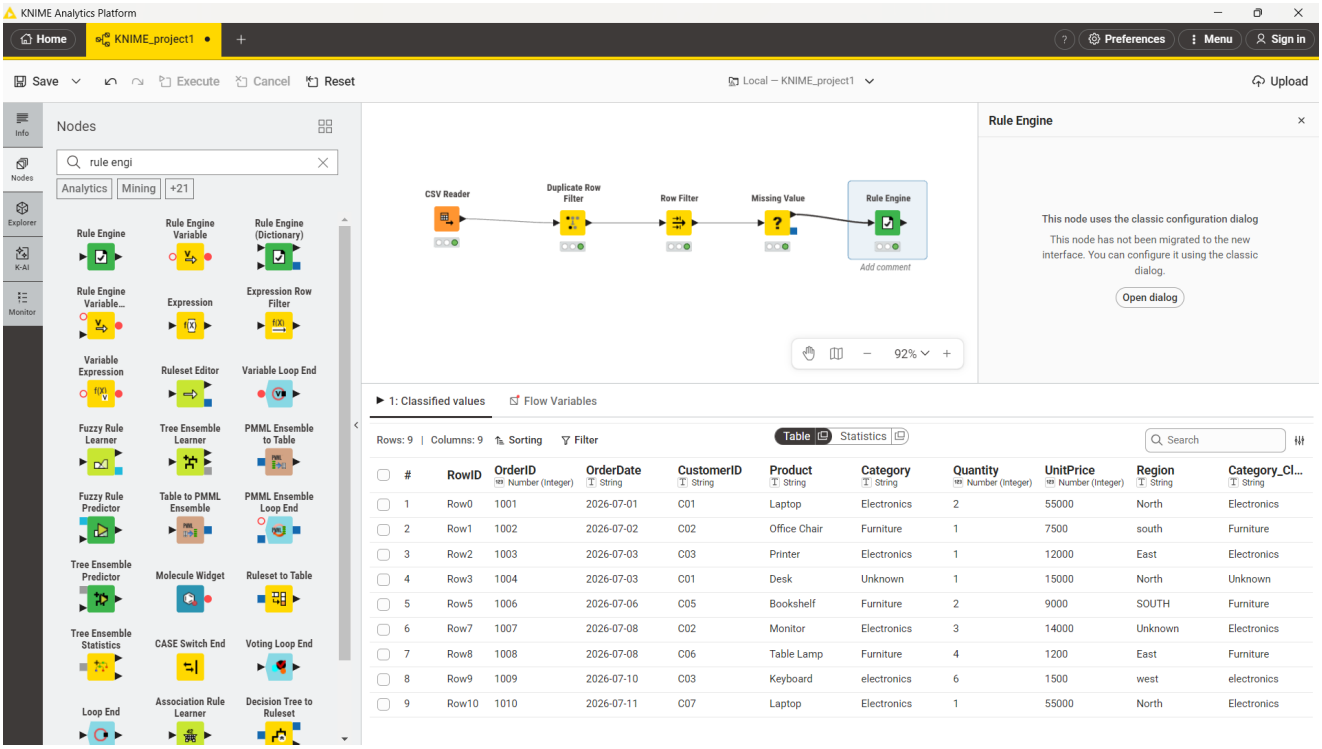
Step 15: Opened the Rule Engine configuration to standardize inconsistent text/category values.



Step 16: Entered the rules used to convert inconsistent values into the required standard form.

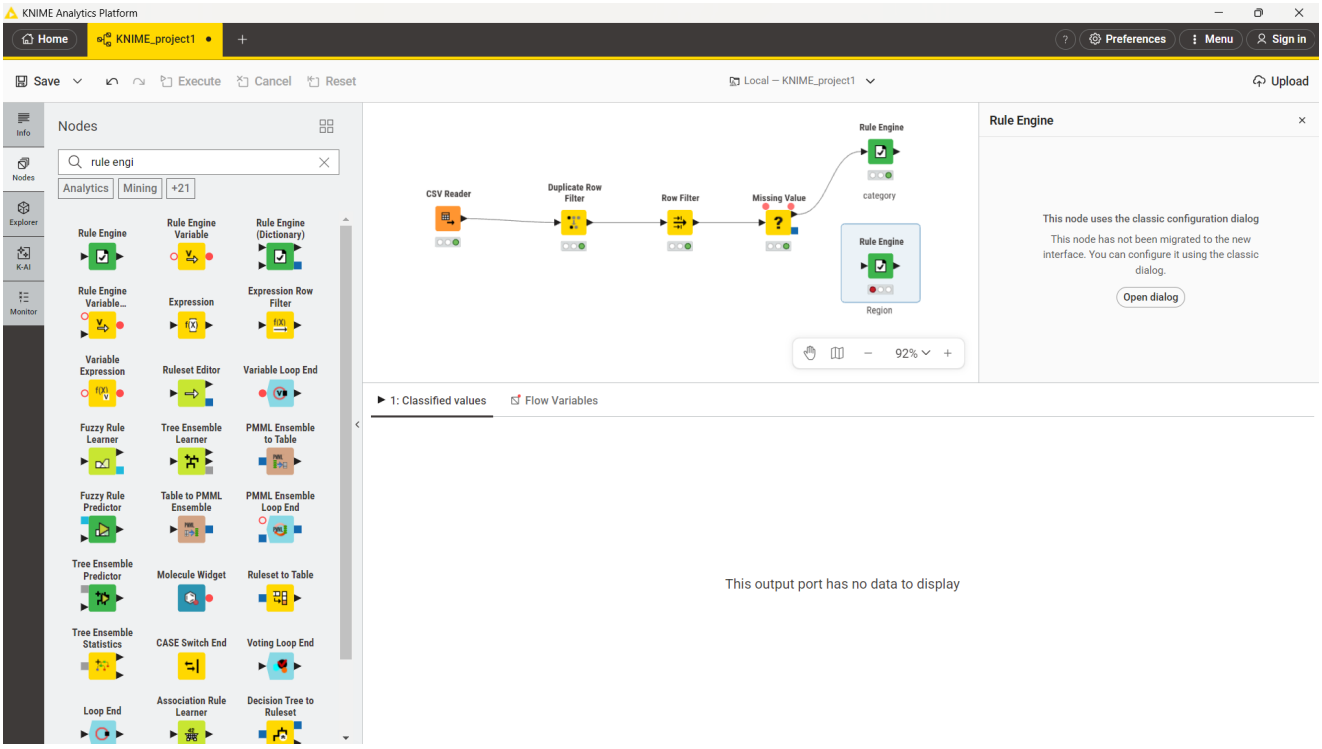


Step 17: Applied the Rule Engine and checked the standardized output.

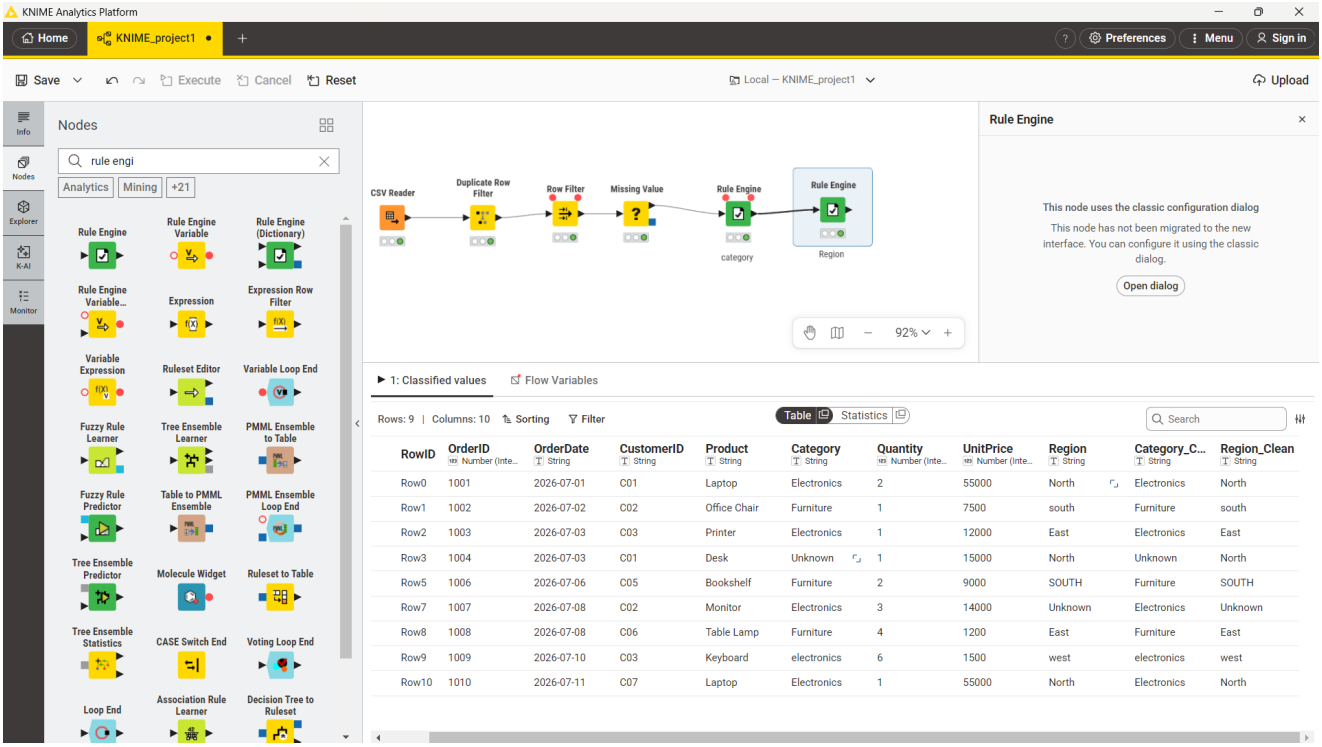


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)	Category_Cl... (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	Electronics
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	Furniture
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	Electronics
4	Row3	1004	2026-07-03	C01	Desk	Unknown	1	15000	North	Unknown
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	Furniture
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Electronics
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	Furniture
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	electronics
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North	Electronics

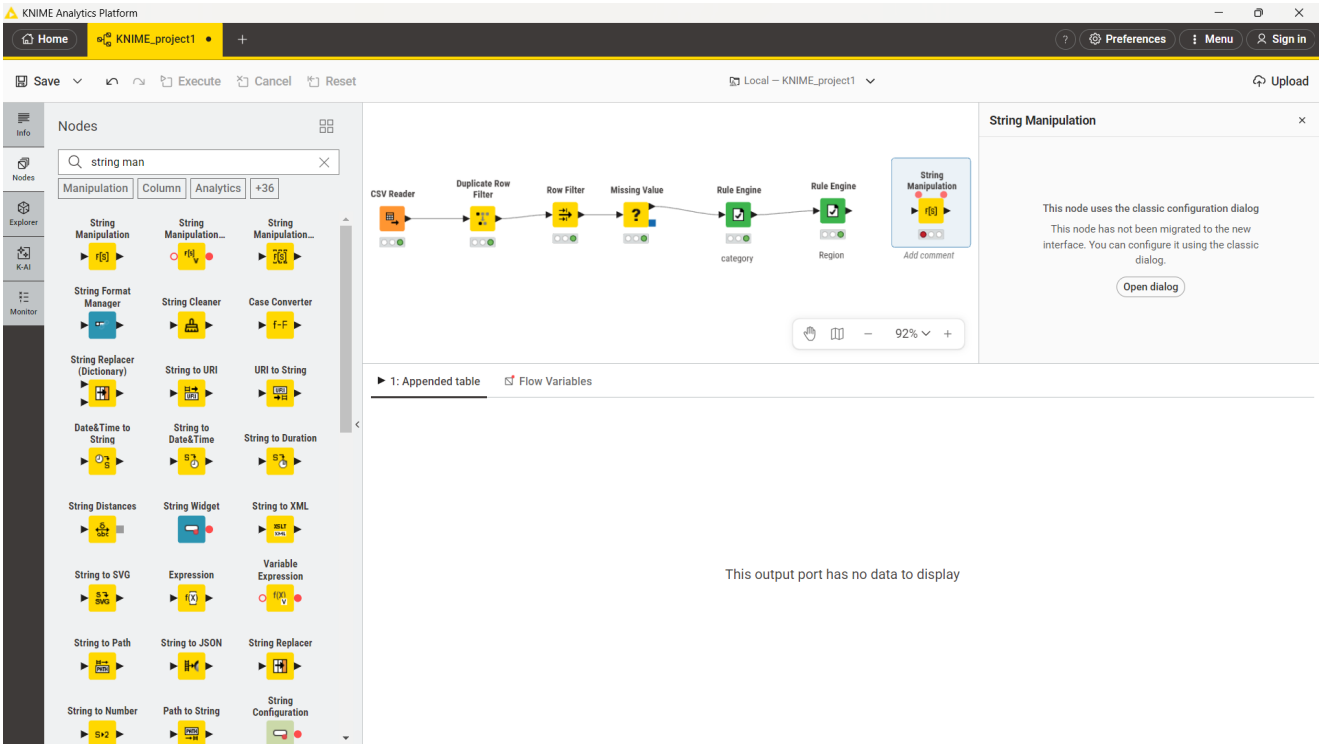
Step 18: Reviewed the cleaned orders table after text/category standardization.



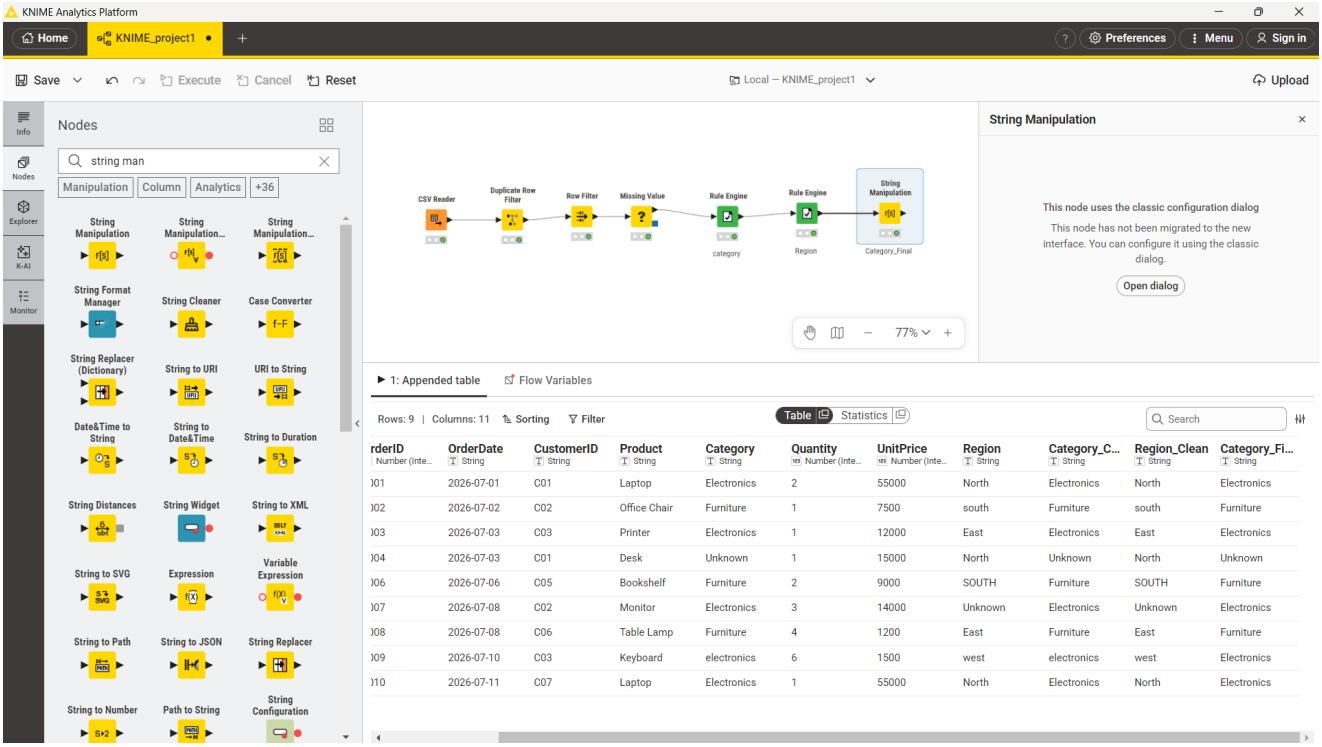
Step 19: Checked the workflow and confirmed the cleaned orders data is progressing through the required transformations.



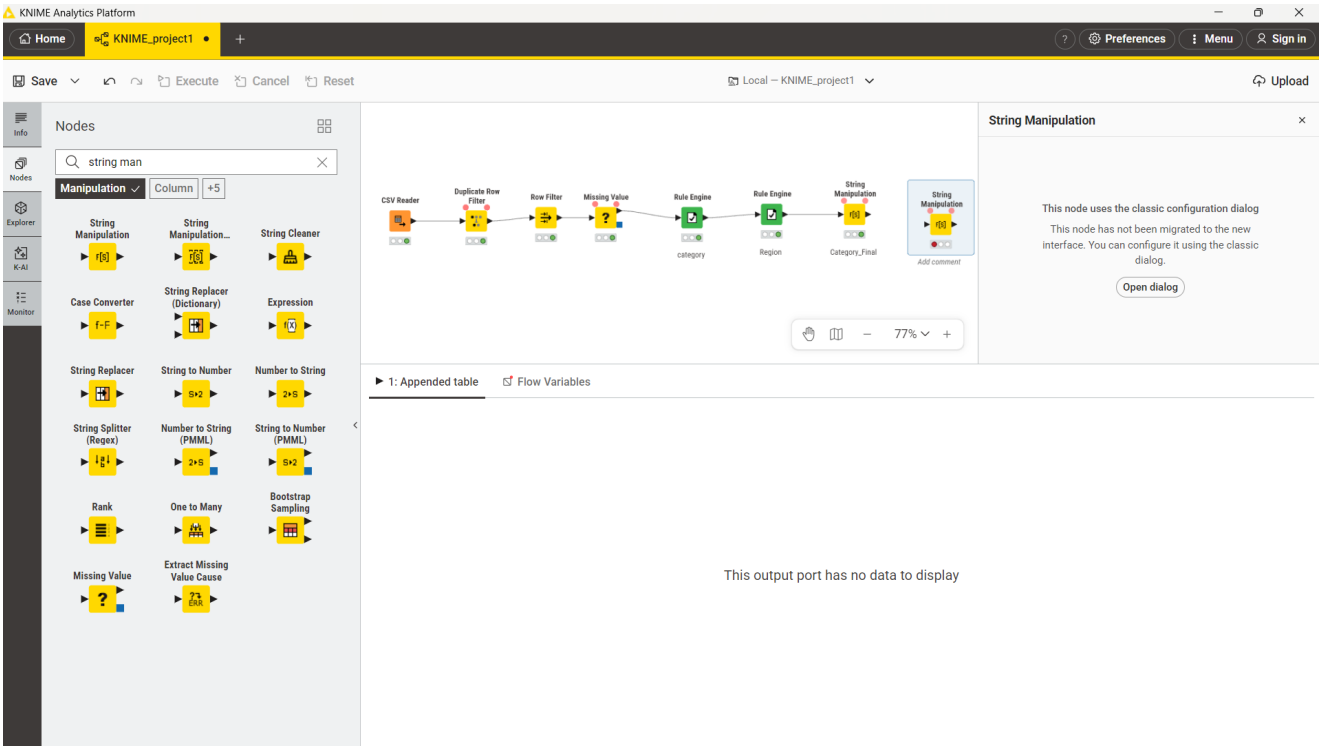
Step 20: Verified the cleaned dataset and preserved the main cleaned table for later merging.



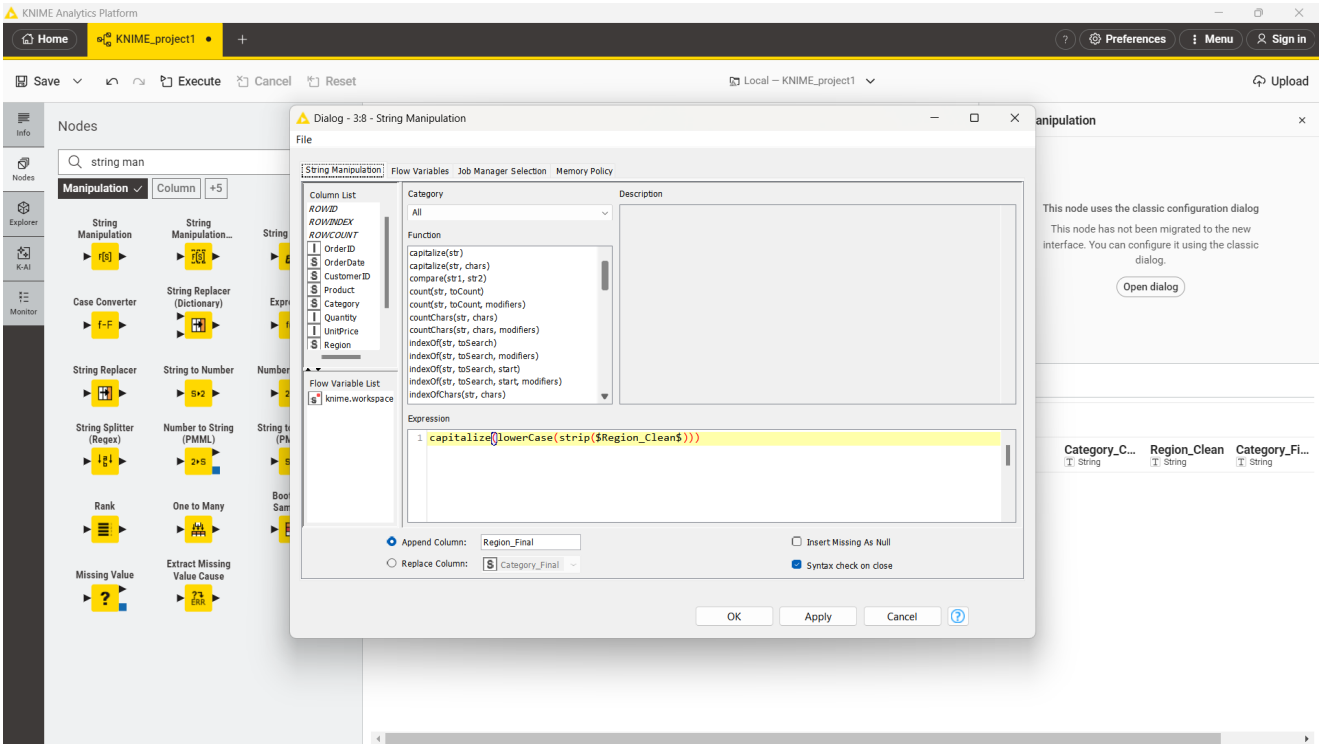
Step 21: Opened the working-copy branch so later filter, sort and group operations do not change the main cleaned 9-row table.



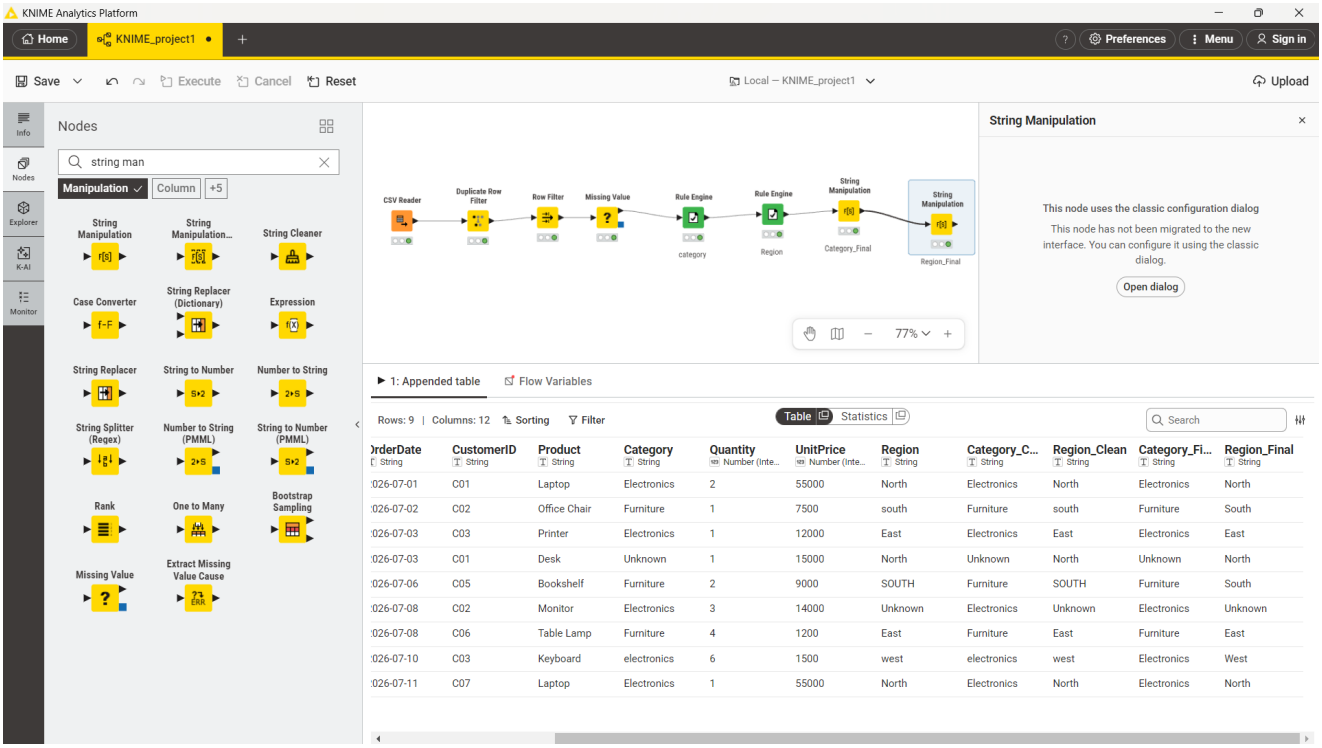
Step 22: Configured the Column Filter for the working copy and selected the fields needed for the next analysis operation.



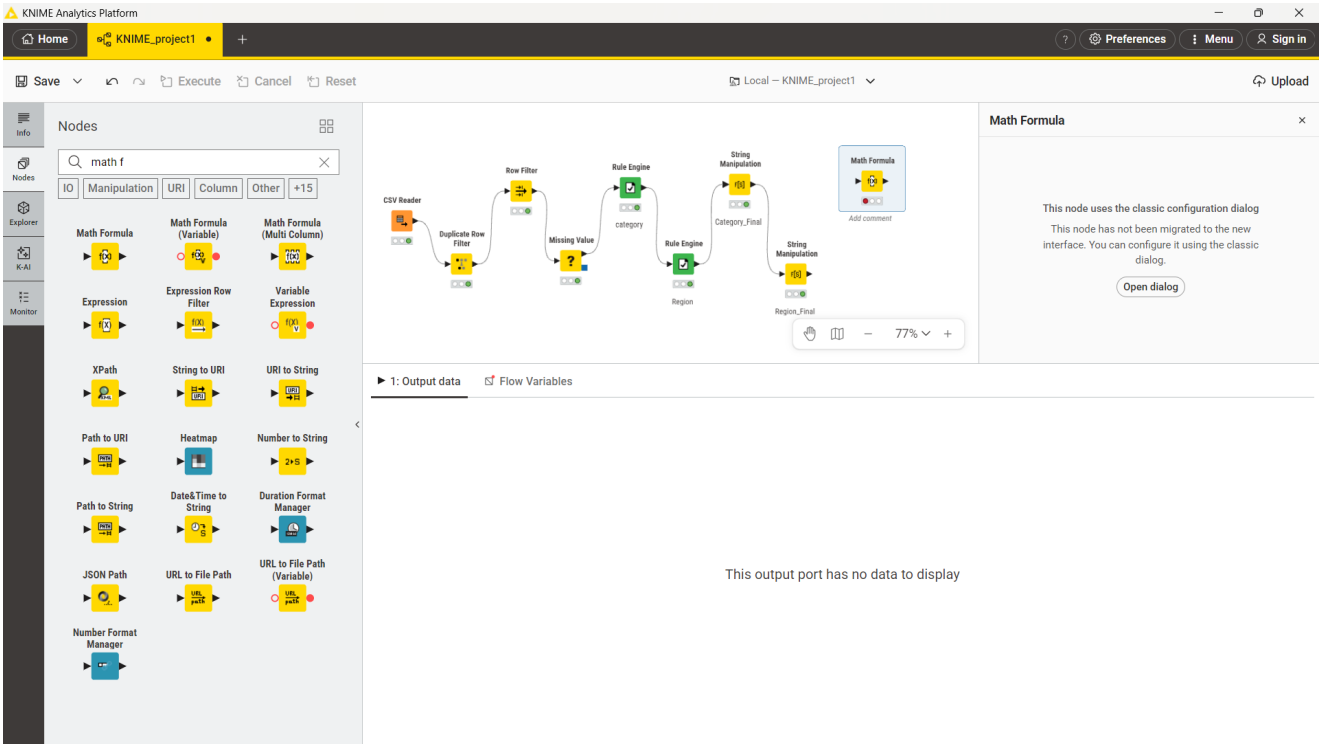
Step 23: Executed the Column Filter and inspected the reduced working-copy table.



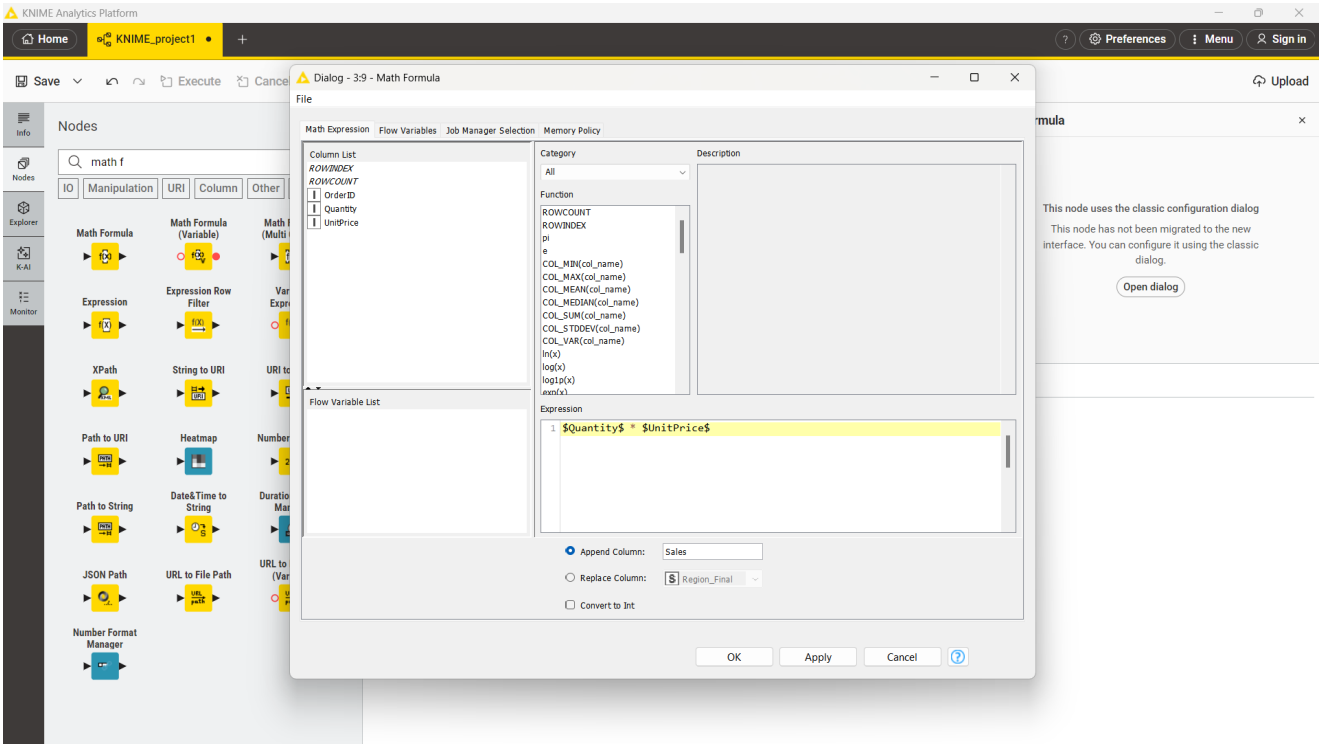
Step 24: Configured the Row Filter on the working copy for the required filter condition.



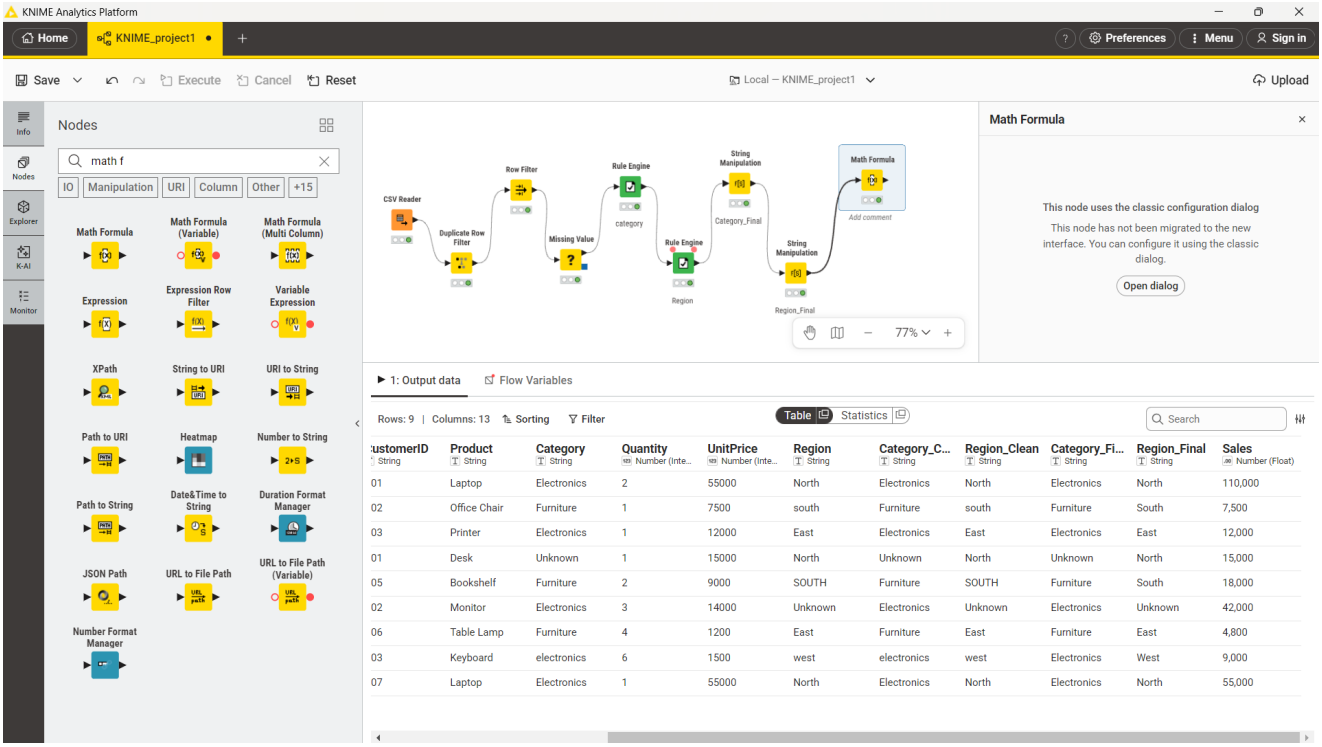
Step 25: Executed the working-copy Row Filter and reviewed its result.



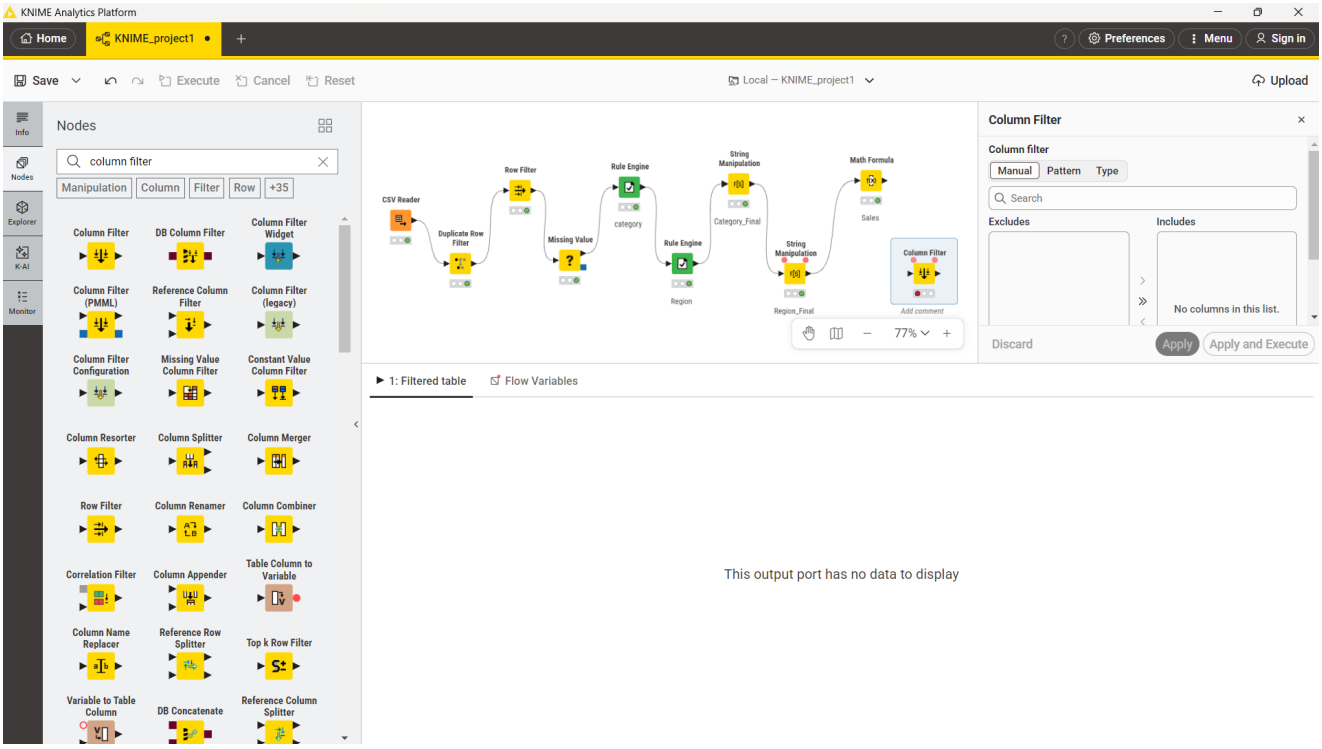
Step 26: Confirmed the filtered working-copy output before applying the next transformation.



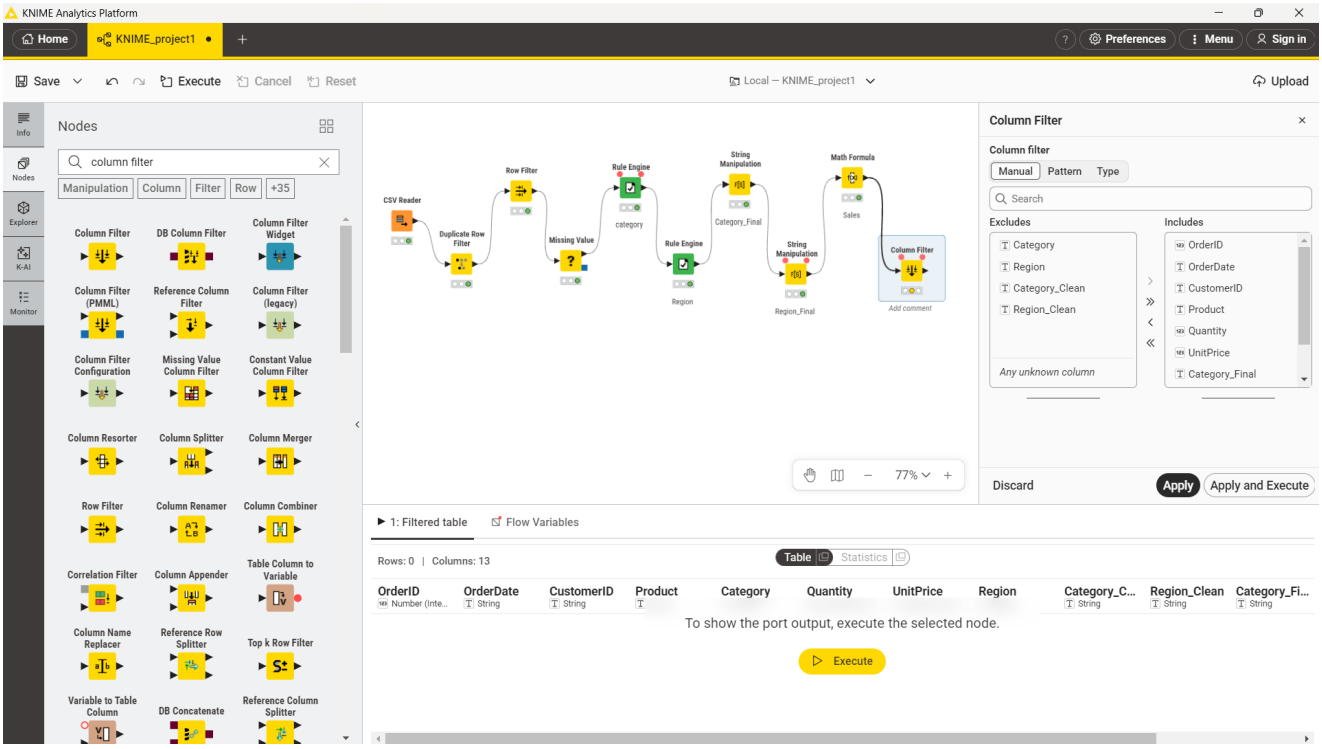
Step 27: Opened the Sorter node and selected the column and order required for sorting the working copy.



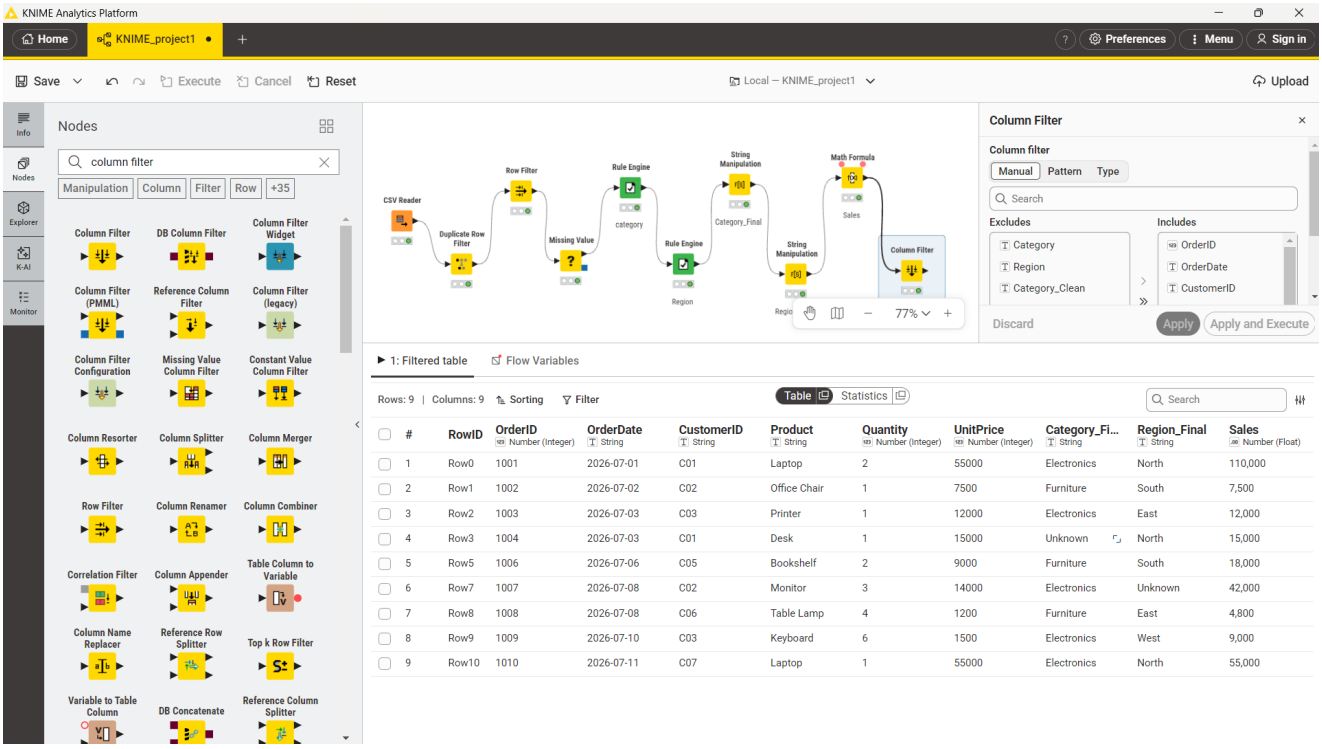
Step 28: Configured the Sorter settings and prepared the working-copy data for execution.



Step 29: Executed the Sorter and checked the rows in the requested order.

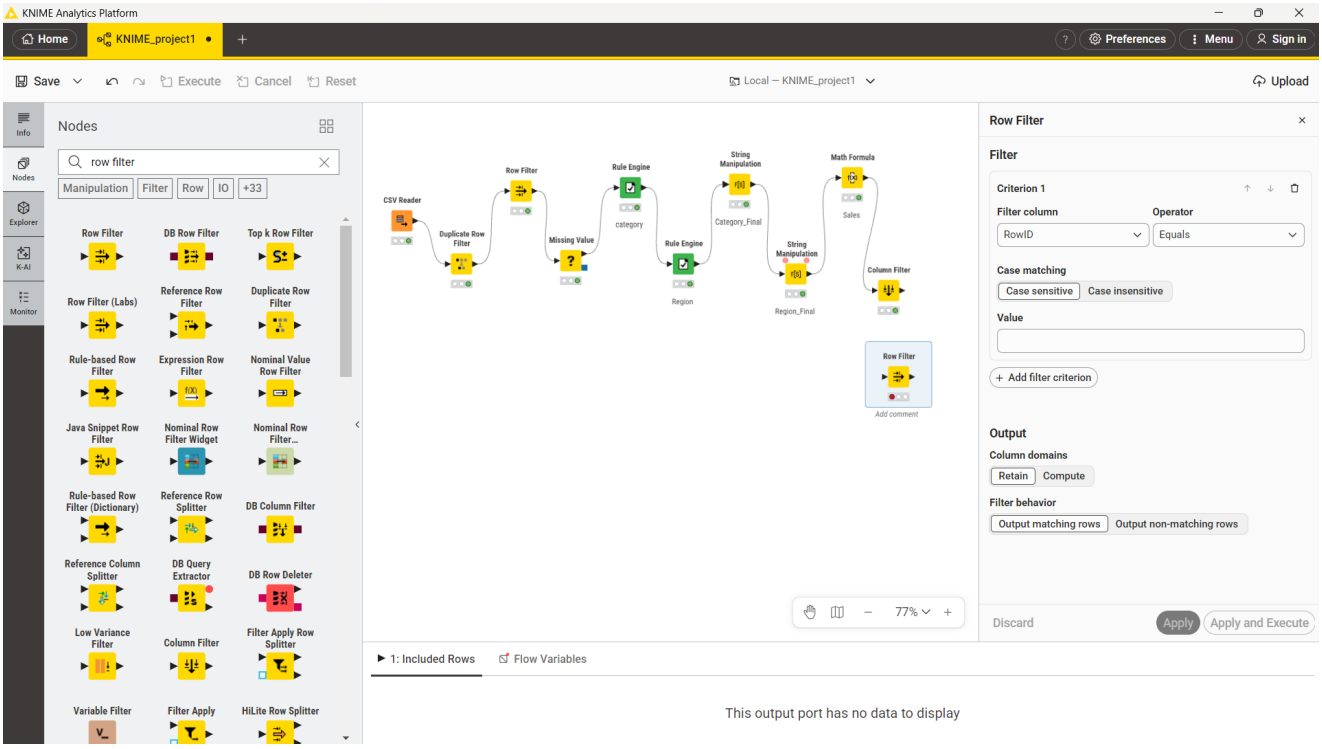


Step 30: Reviewed the sorted working-copy output before grouping.

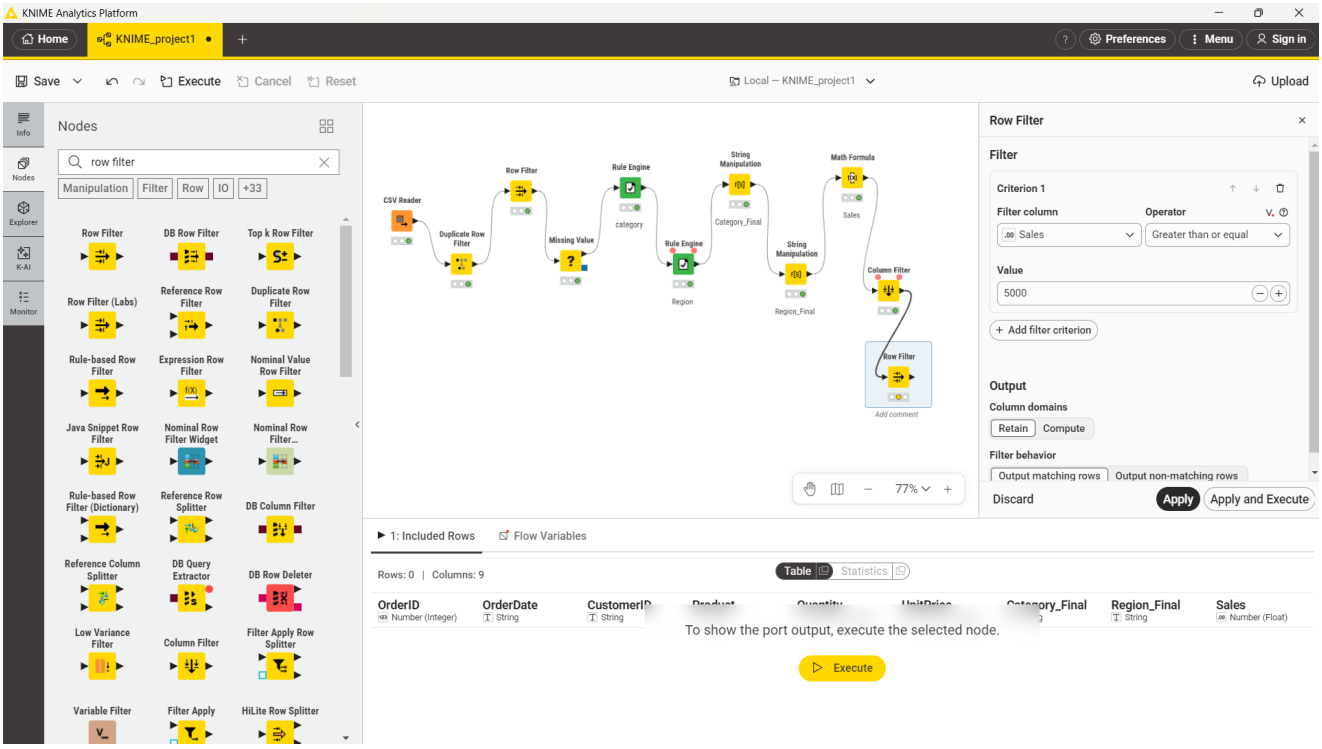


#	RowID	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	Category_Fi...	Region_Final	Sales
		Number (Integer)	String	String	String	Number (Integer)	Number (Integer)	String	String	Number (Float)
1	Row0	1001	2026-07-01	C01	Laptop	2	55000	Electronics	North	110,000
2	Row1	1002	2026-07-02	C02	Office Chair	1	7500	Furniture	South	7,500
3	Row2	1003	2026-07-03	C03	Printer	1	12000	Electronics	East	12,000
4	Row3	1004	2026-07-03	C01	Desk	1	15000	Unknown	North	15,000
5	Row5	1006	2026-07-06	C05	Bookshelf	2	9000	Furniture	South	18,000
6	Row7	1007	2026-07-08	C02	Monitor	3	14000	Electronics	Unknown	42,000
7	Row8	1008	2026-07-08	C06	Table Lamp	4	1200	Furniture	East	4,800
8	Row9	1009	2026-07-10	C03	Keyboard	6	1500	Electronics	West	9,000
9	Row10	1010	2026-07-11	C07	Laptop	1	55000	Electronics	North	55,000

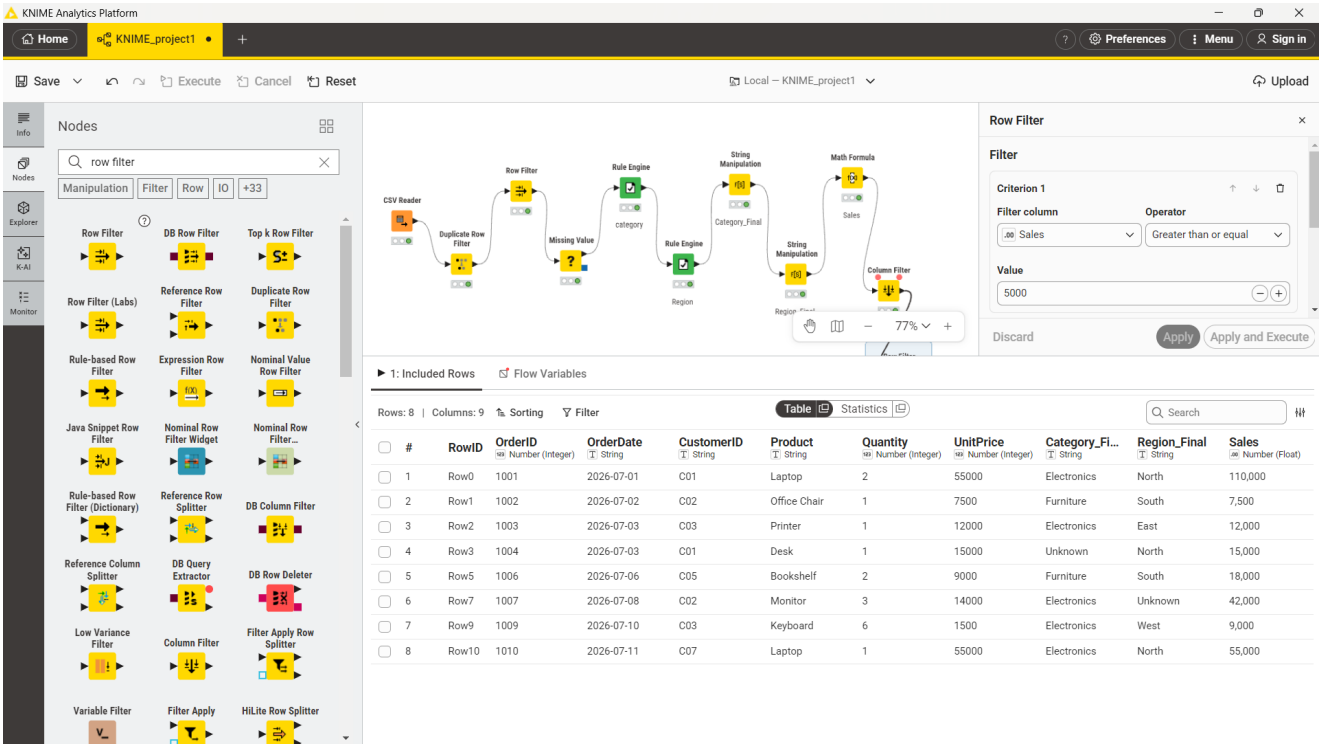
Step 31: Opened the GroupBy node to create the required grouped summary on the working copy.



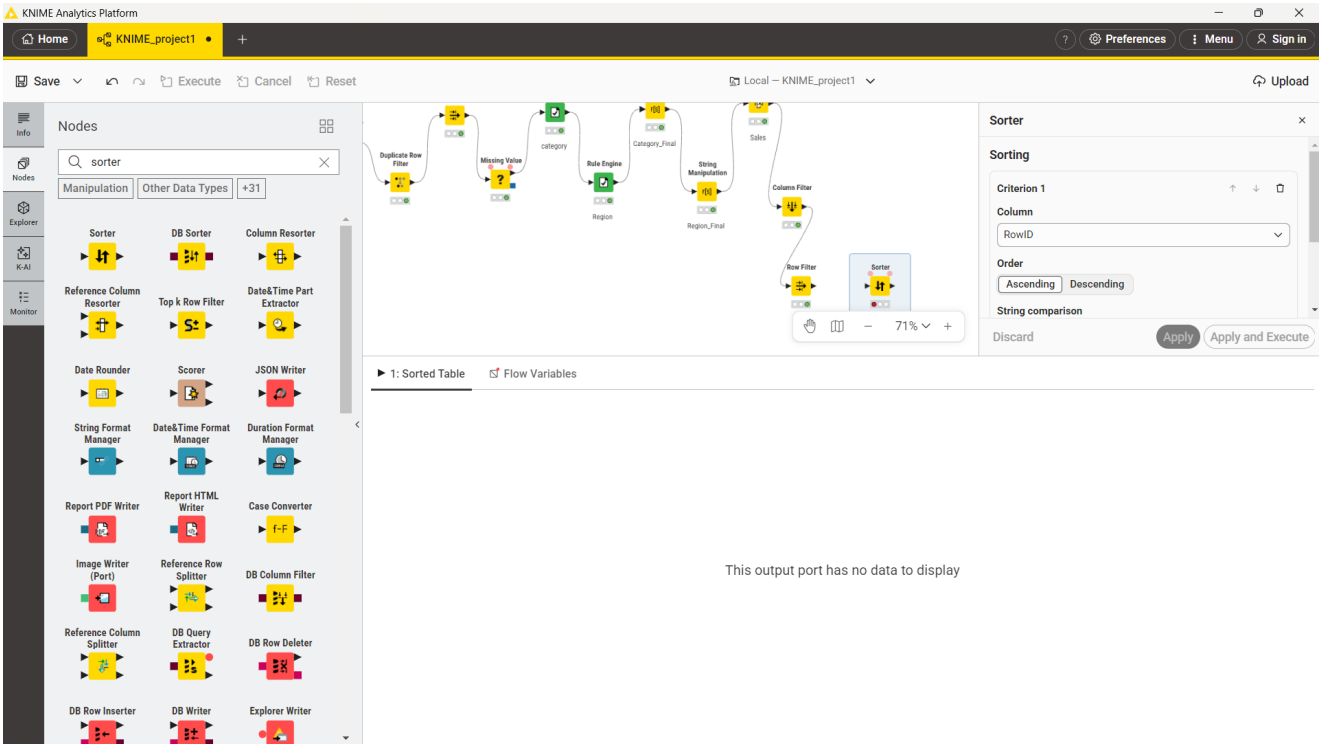
Step 32: Selected the grouping column(s) and configured the aggregation settings.



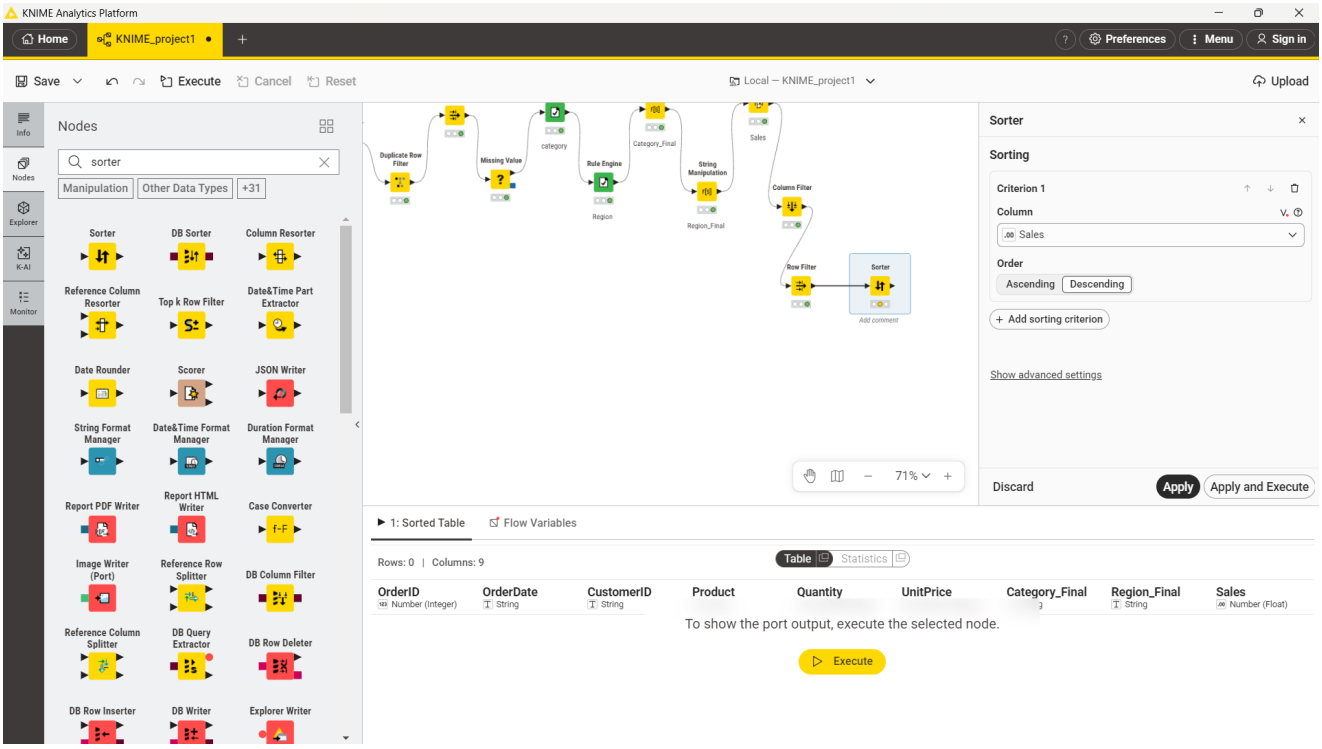
Step 33: Set the Sales aggregation to Sum and completed the GroupBy configuration.



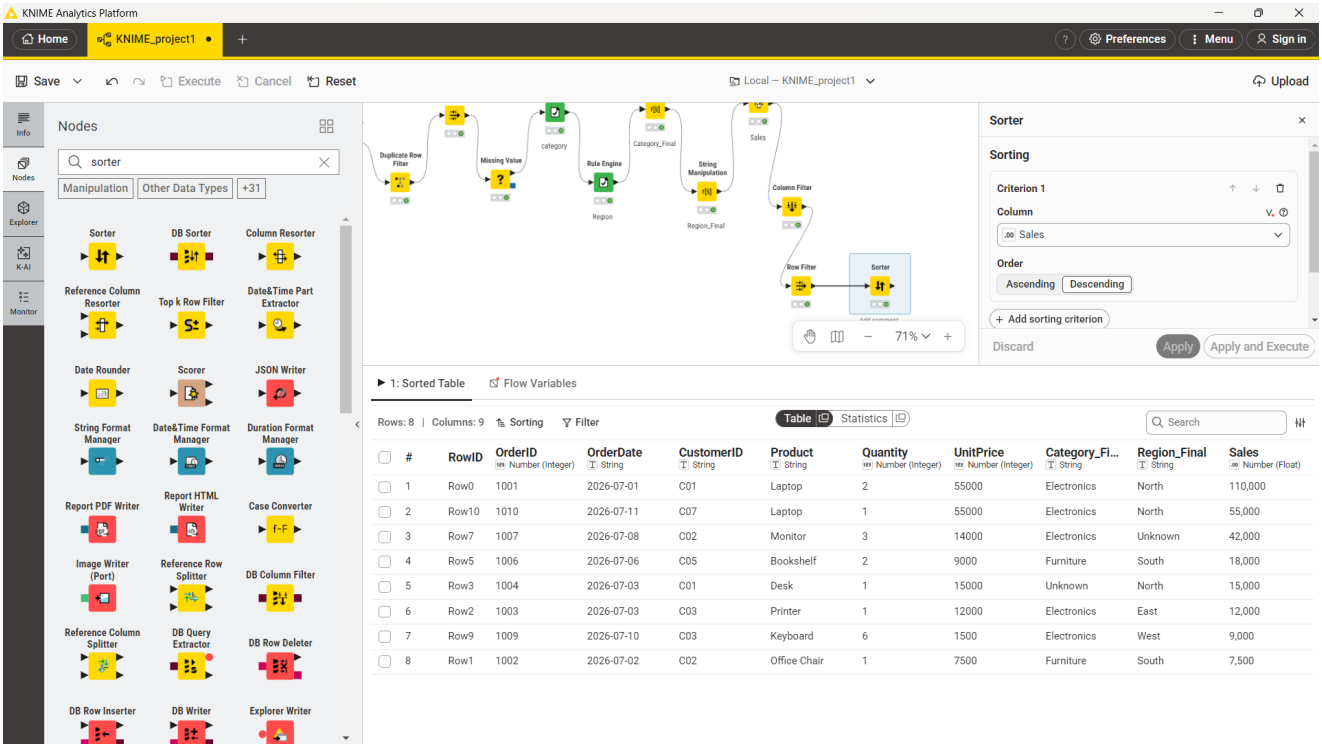
Step 34: Executed the GroupBy operation and inspected the grouped result.



Step 35: Started the customer-data branch by adding and configuring the Excel Reader node.

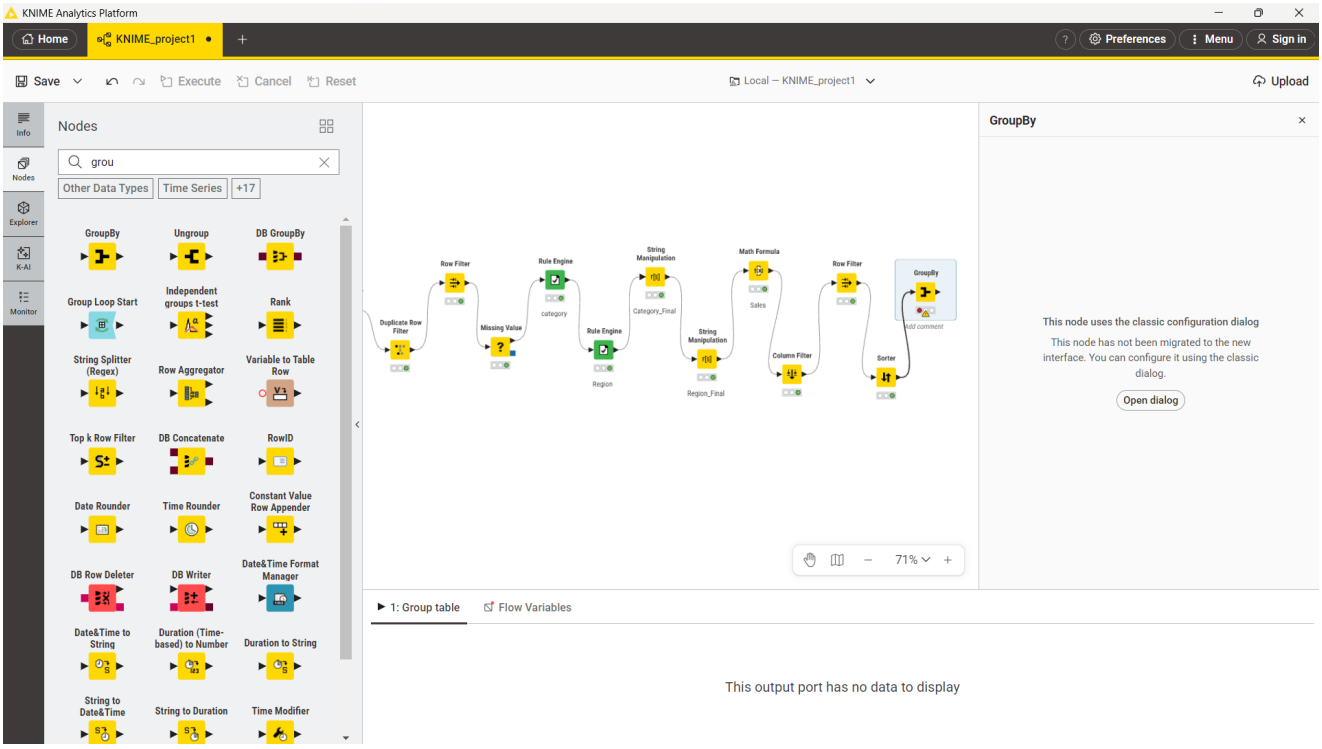


Step 36: Selected customers.xlsx as the Excel input file.

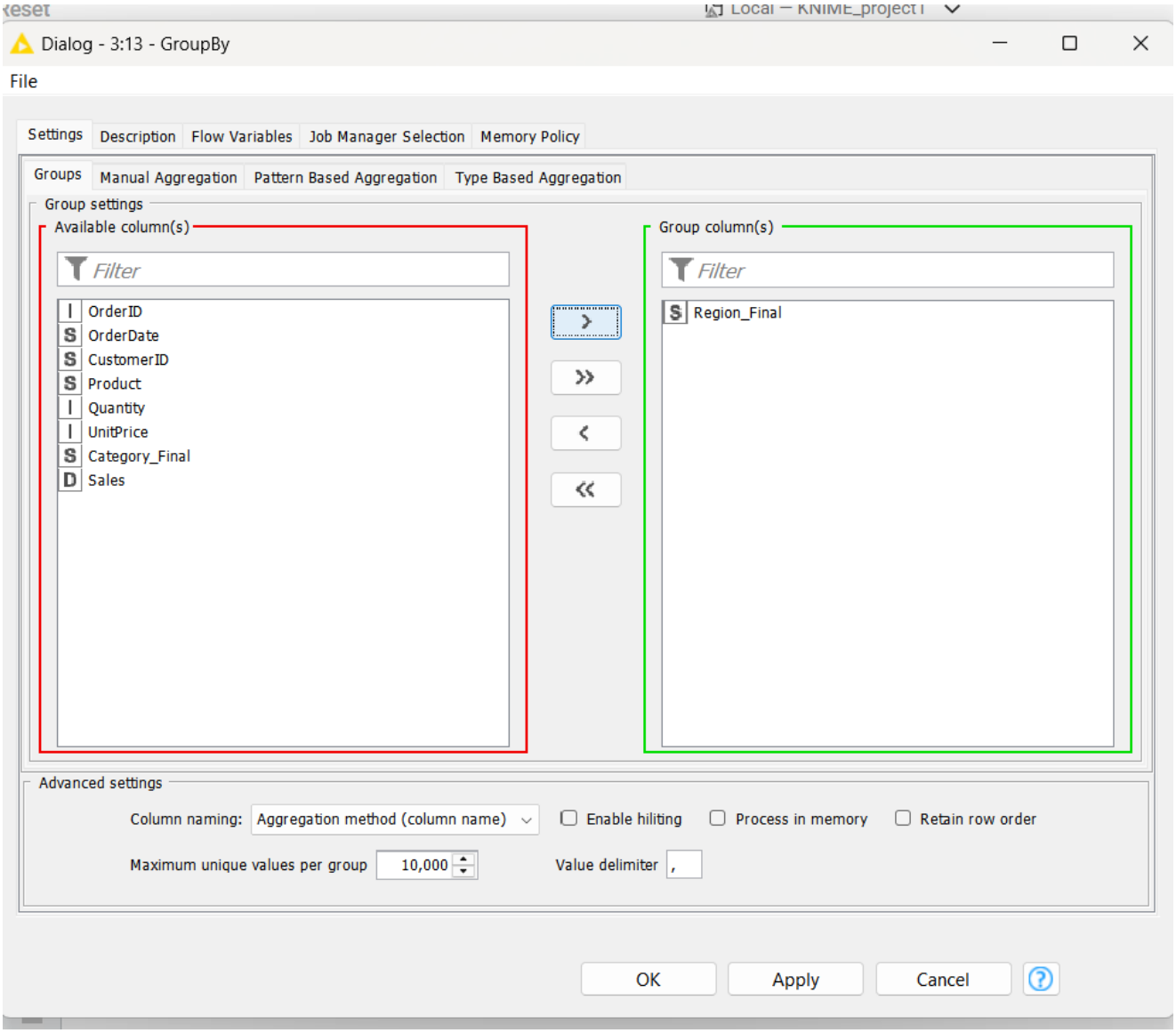


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Category_Fi... (String)	Region_Final (String)	Sales (Number (Float))
1	Row0	1001	2026-07-01	C01	Laptop	2	55000	Electronics	North	110,000
2	Row10	1010	2026-07-11	C07	Laptop	1	55000	Electronics	North	55,000
3	Row7	1007	2026-07-08	C02	Monitor	3	14000	Electronics	Unknown	42,000
4	Row5	1006	2026-07-06	C05	Bookshelf	2	9000	Furniture	South	18,000
5	Row3	1004	2026-07-03	C01	Desk	1	15000	Unknown	North	15,000
6	Row2	1003	2026-07-03	C03	Printer	1	12000	Electronics	East	12,000
7	Row9	1009	2026-07-10	C03	Keyboard	6	1500	Electronics	West	9,000
8	Row1	1002	2026-07-02	C02	Office Chair	1	7500	Furniture	South	7,500

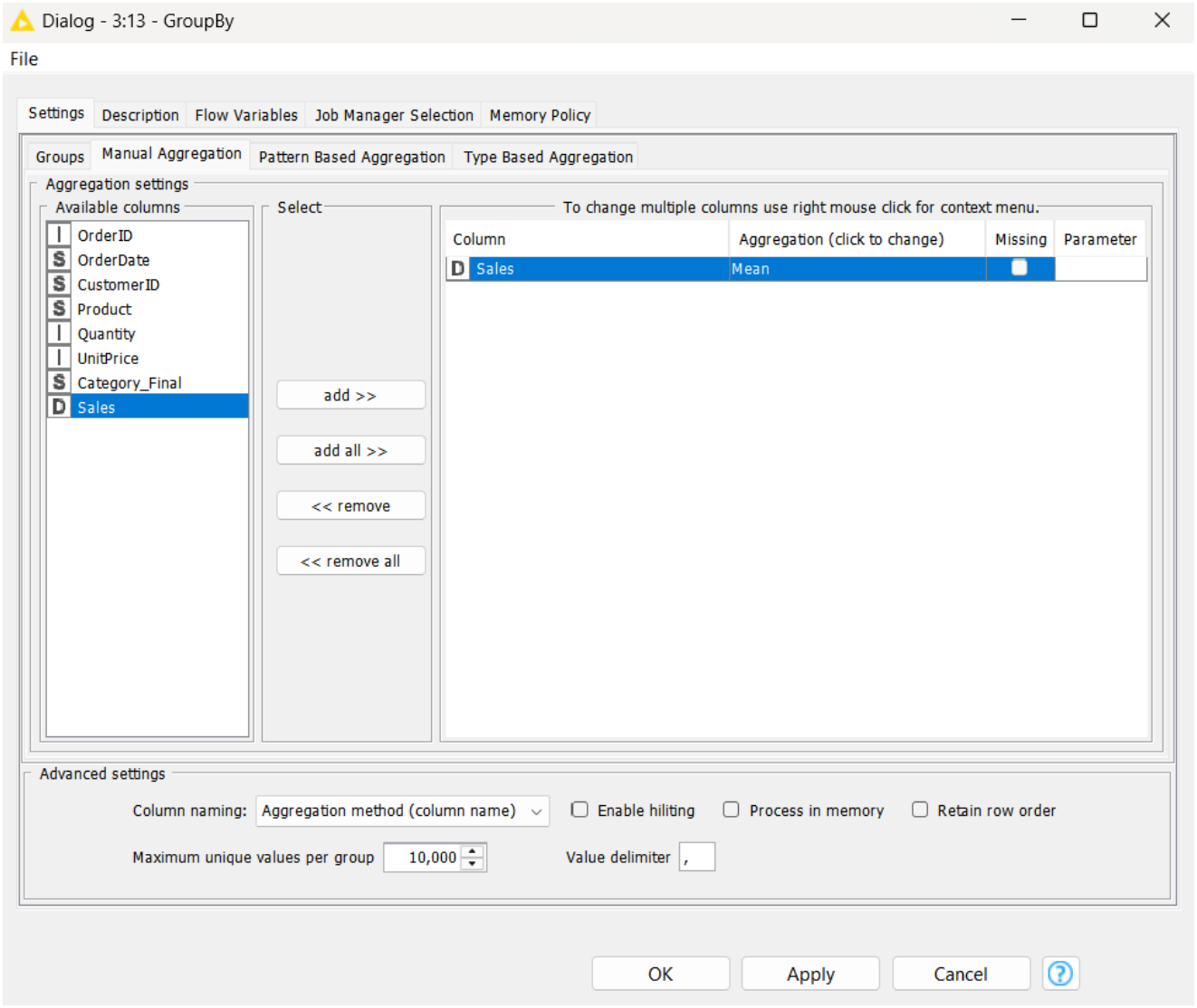
Step 37: Configured the Excel Reader sheet/range settings needed to import the customer list.



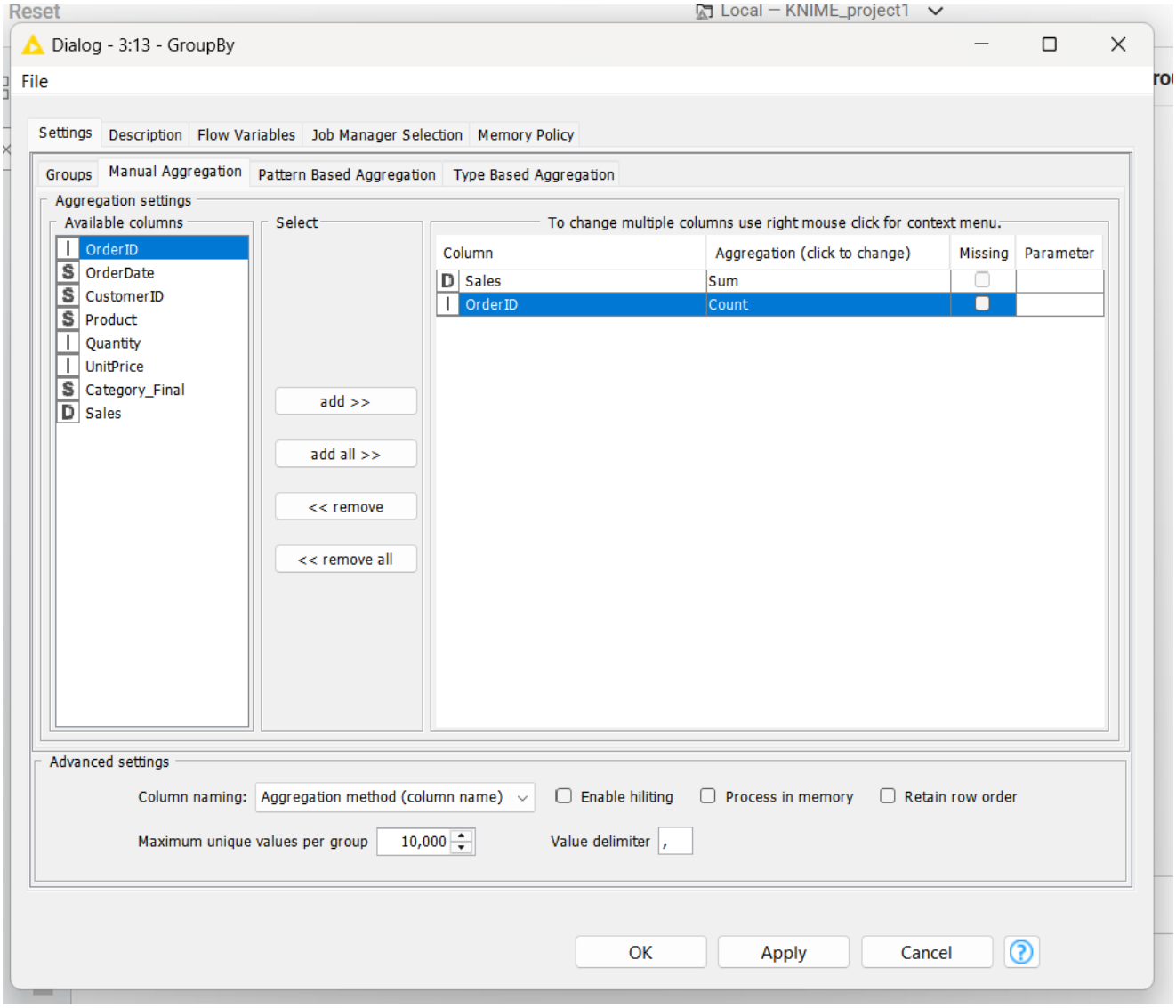
Step 38: Executed the Excel Reader and checked the imported customer table.



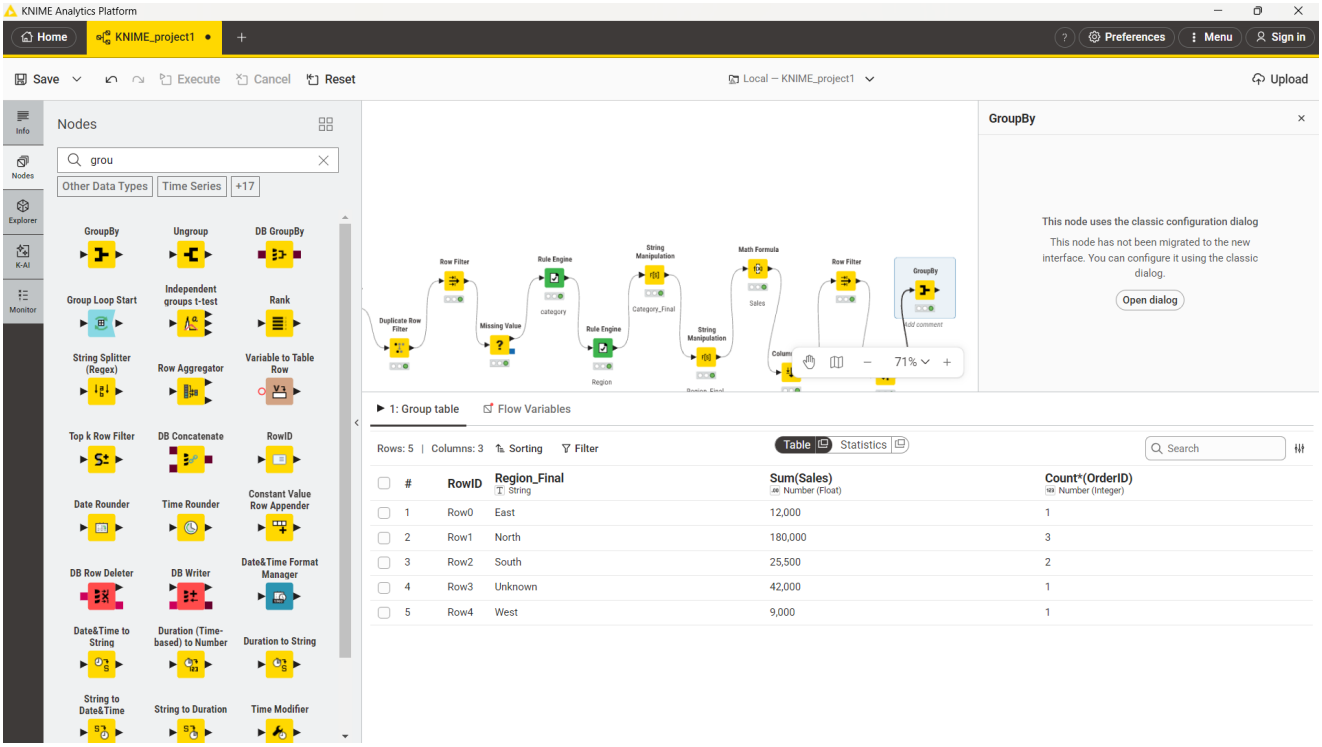
Step 39: Reviewed the customer data and identified fields requiring standardization.



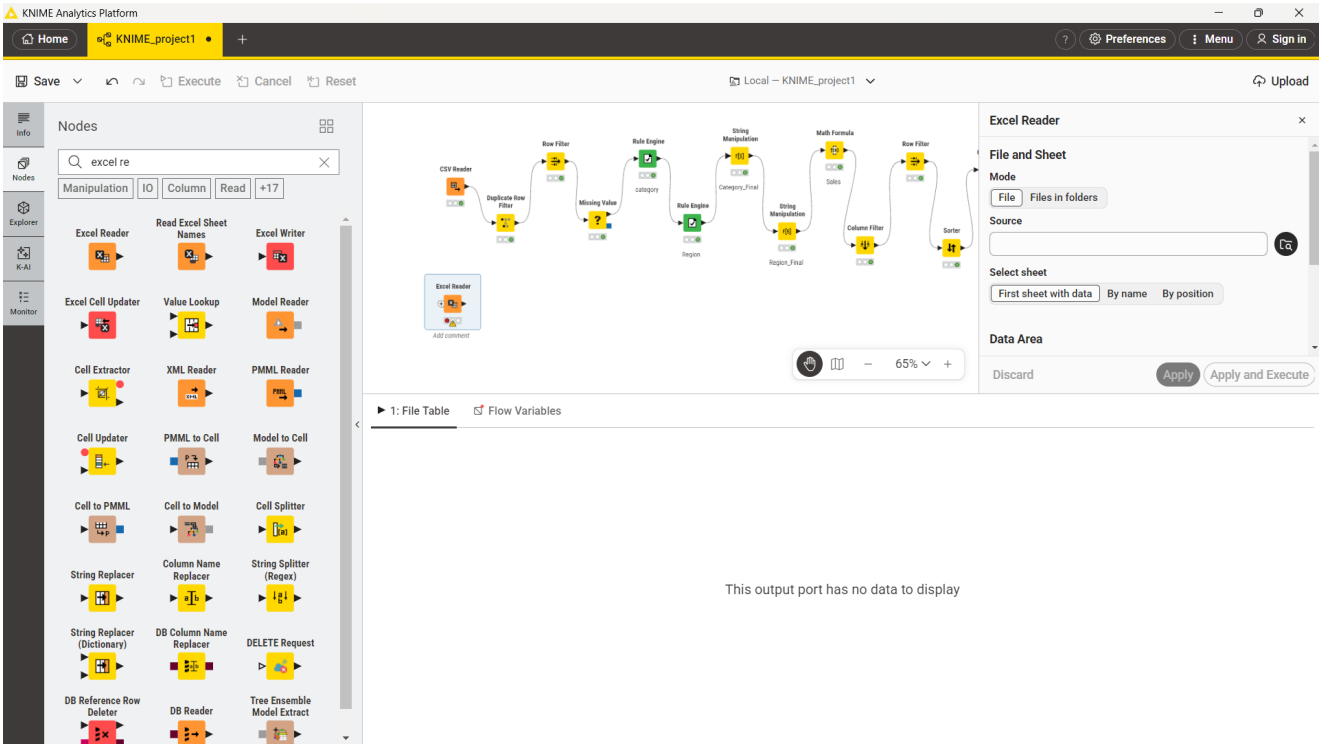
Step 40: Opened String Manipulation to clean and standardize customer-name text.



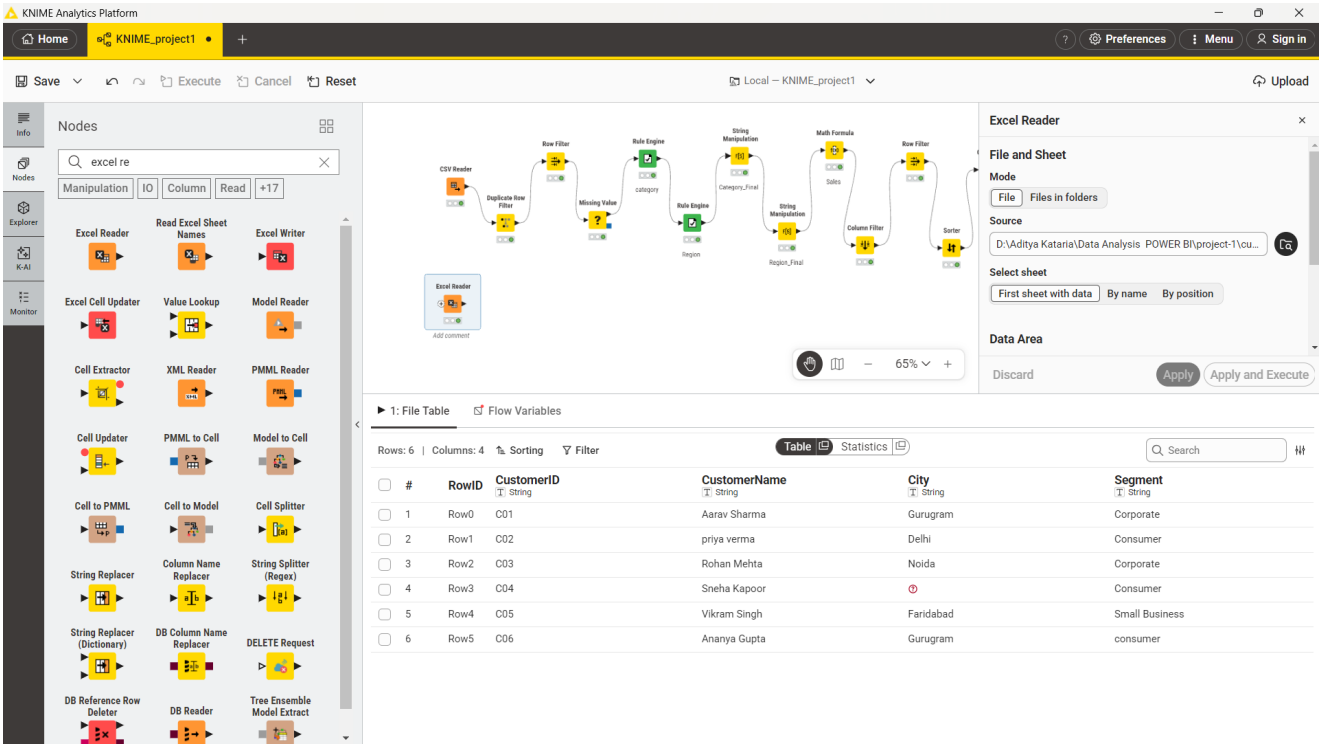
Step 41: Entered the String Manipulation expression used for customer-name cleaning.



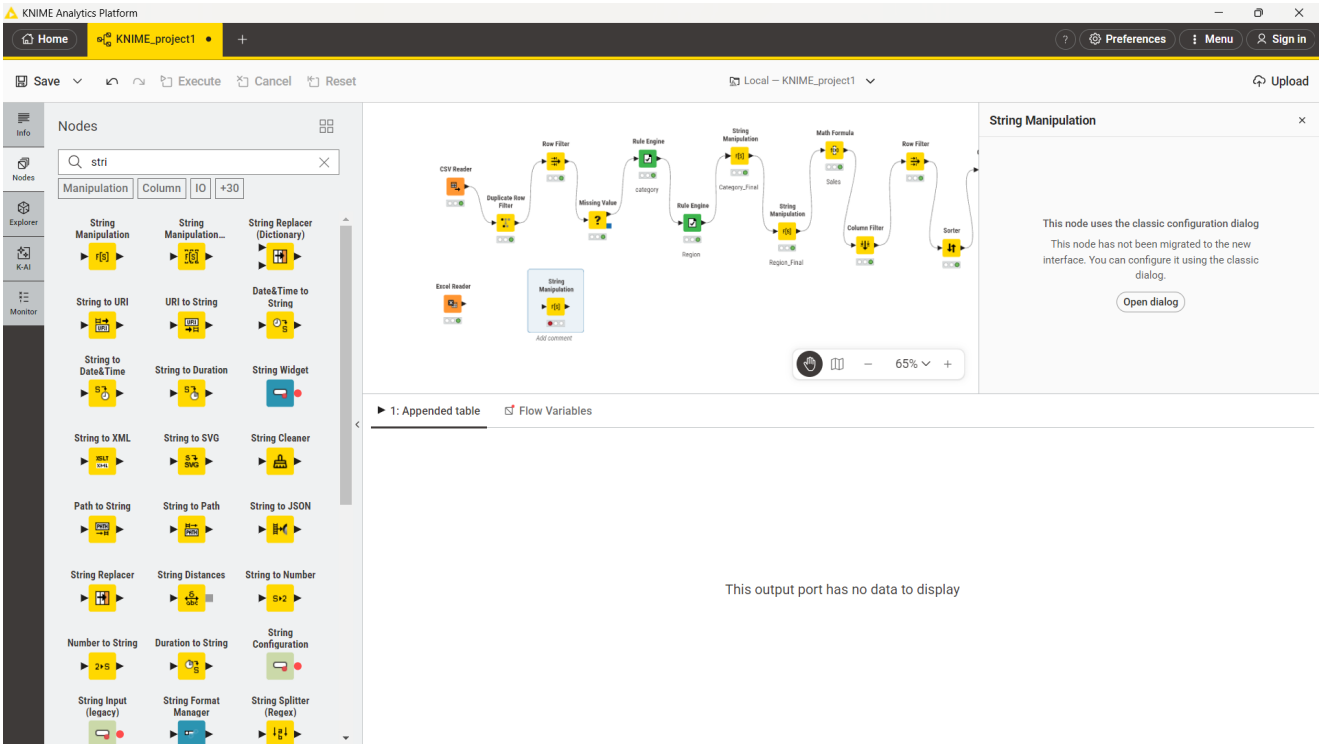
Step 42: Executed the customer-name cleaning step and reviewed the updated values.



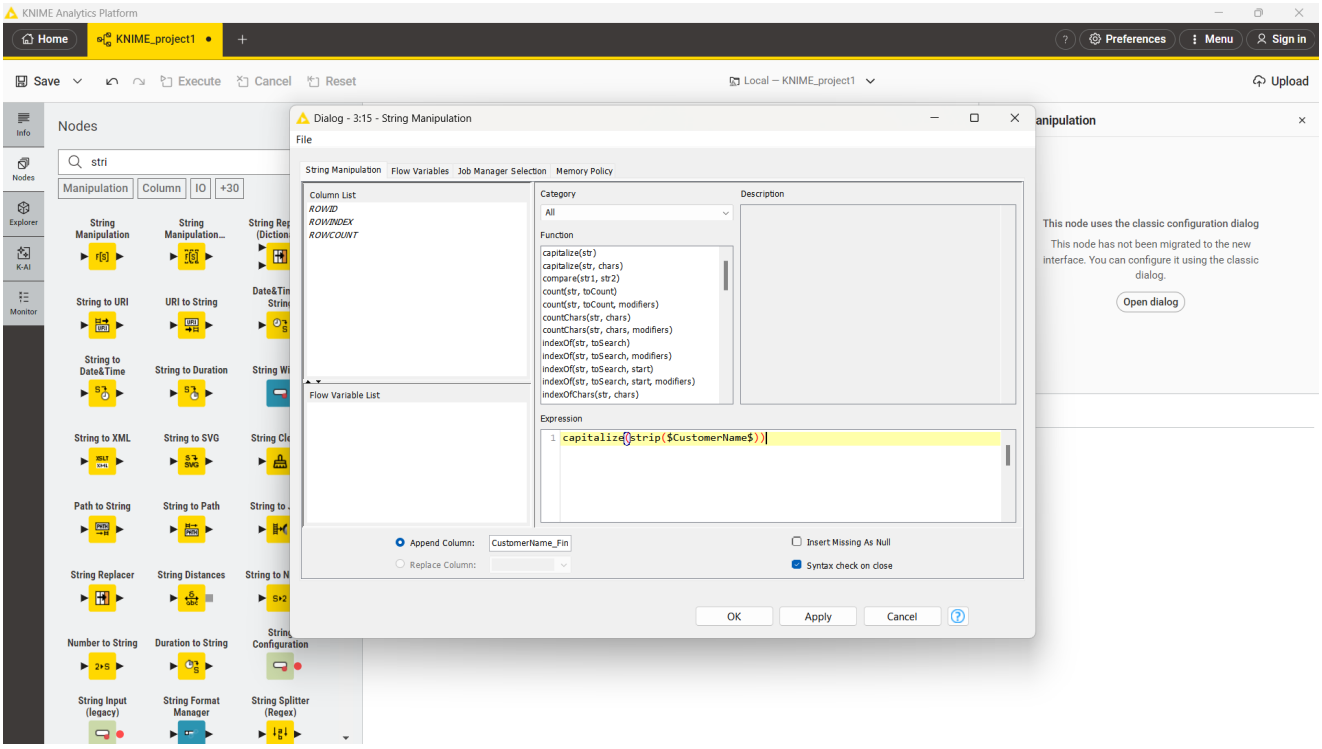
Step 43: Opened the customer-data Rule Engine to standardize remaining inconsistent values.



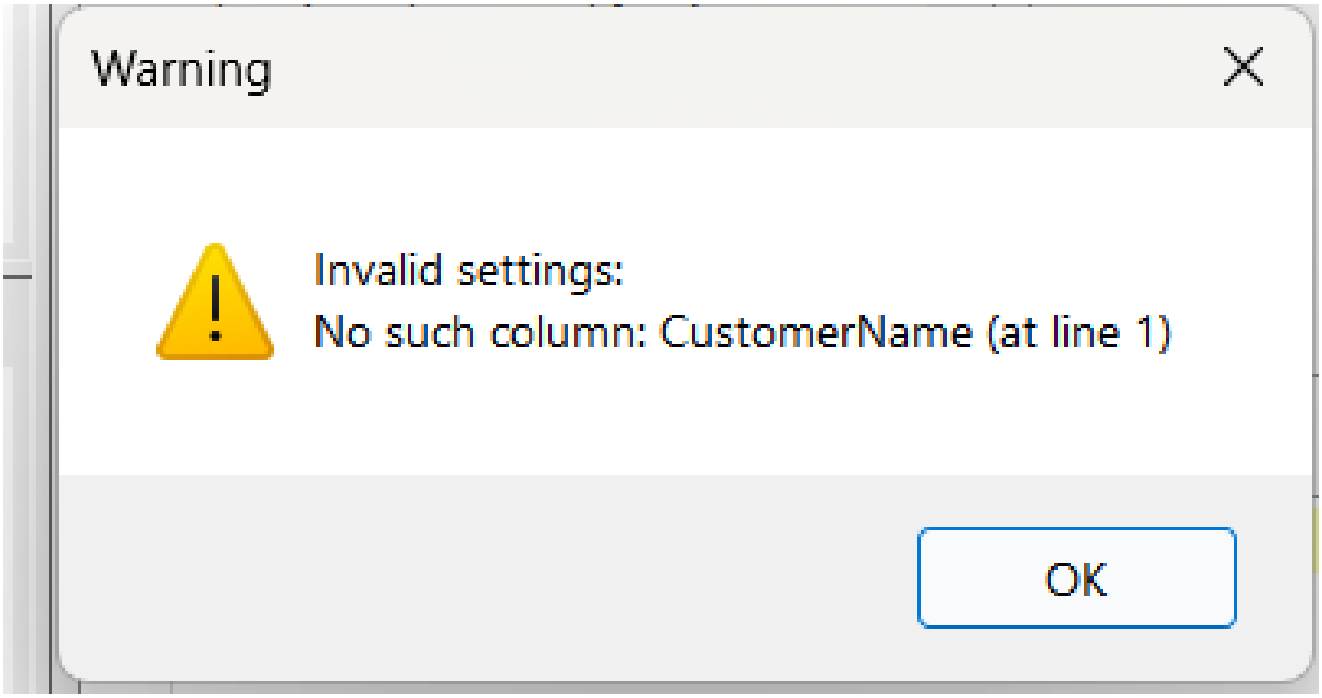
Step 44: Entered the customer-data rules and prepared the node for execution.



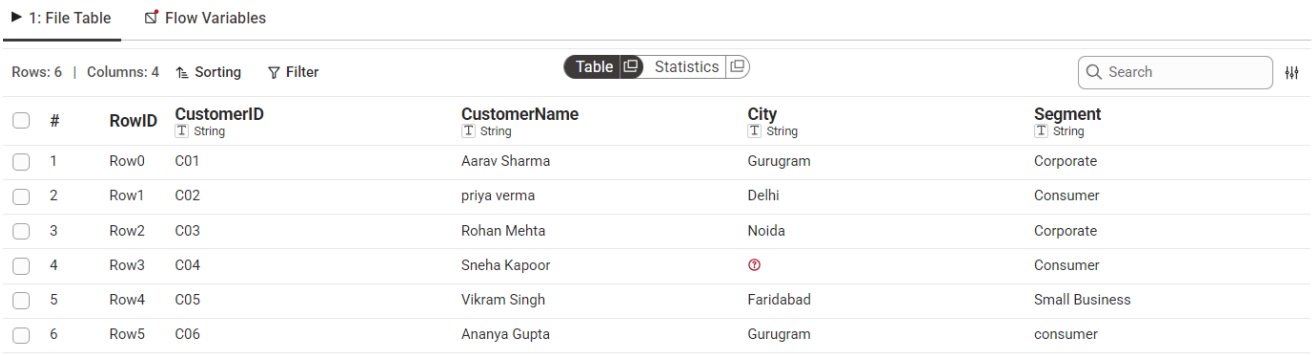
Step 45: Applied the customer-data rules and checked the cleaned customer table.



Step 46: Verified the customer-cleaning branch before combining it with the orders data.



Step 47: Configured the Joiner node to merge the cleaned orders and customer datasets.

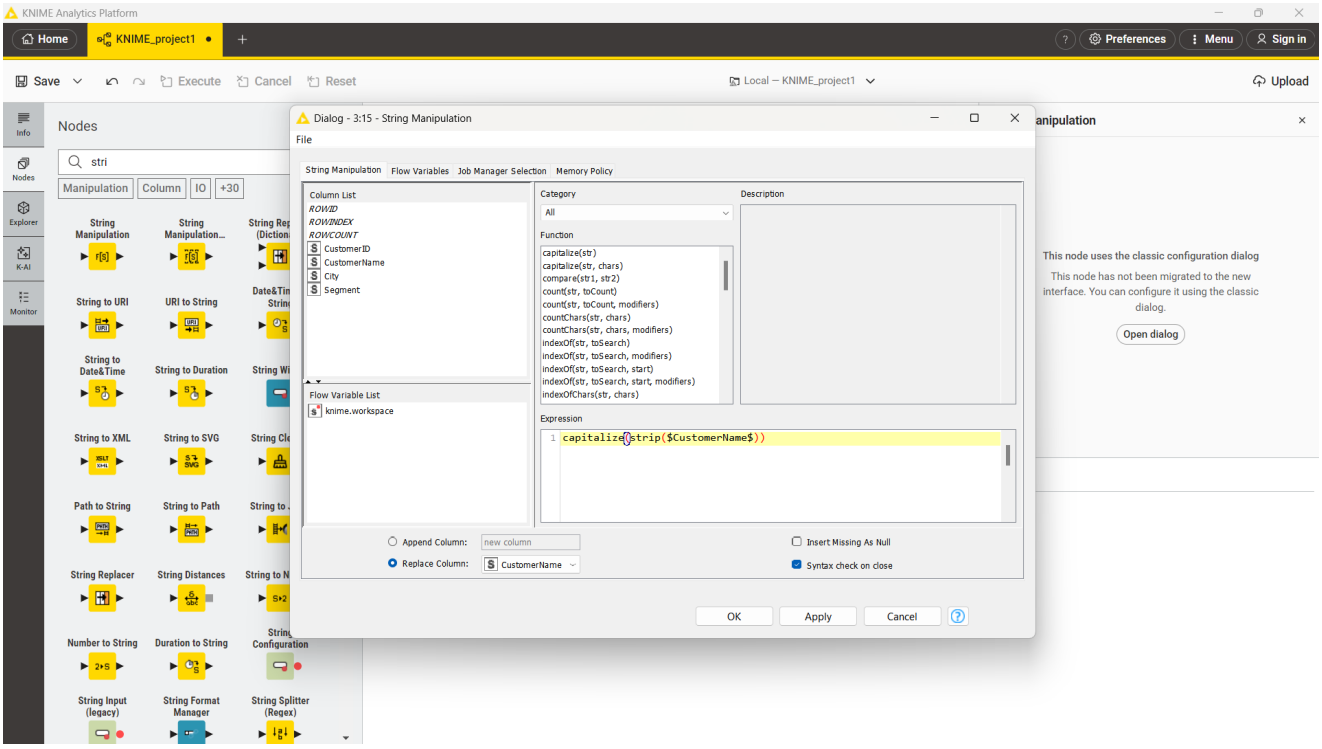


▶ 1: File Table Flow Variables

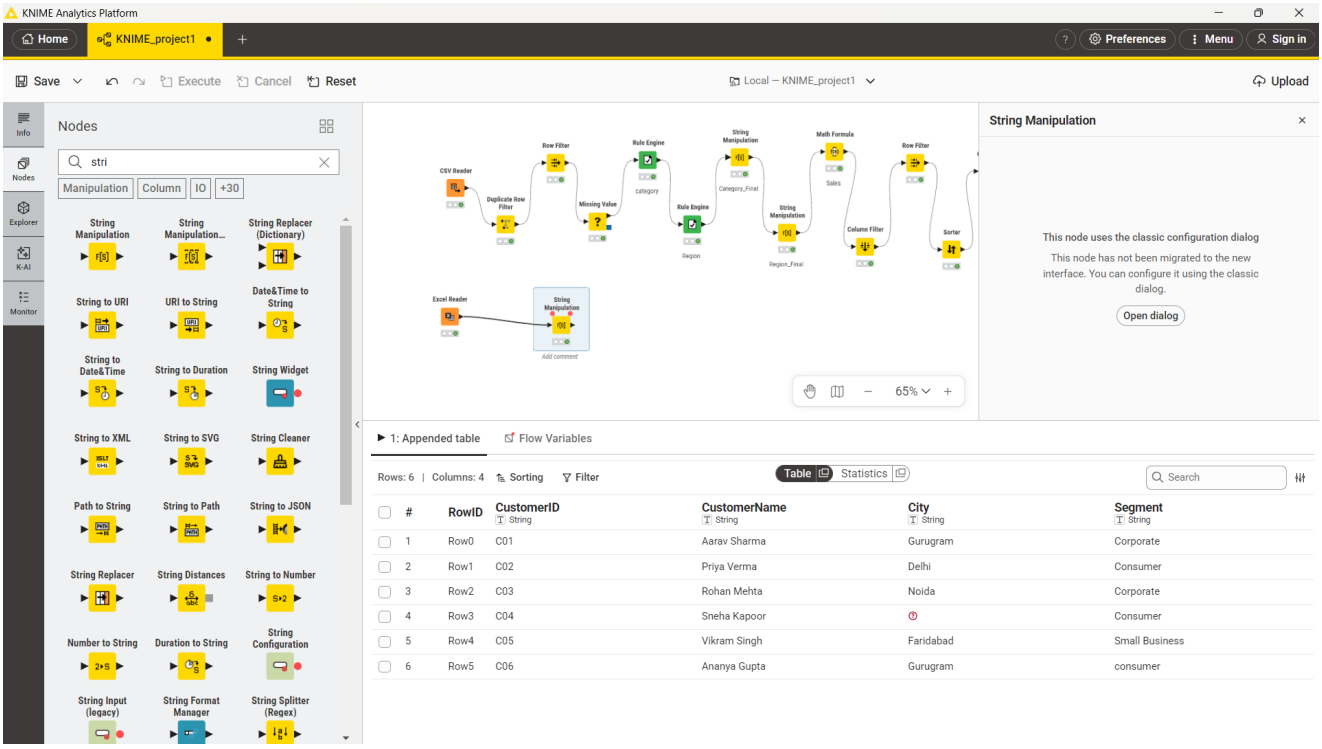
Rows: 6 | Columns: 4 Sorting Filter Table Statistics Search

#	RowID	CustomerID String	CustomerName String	City String	Segment String
1	Row0	C01	Aarav Sharma	Gurugram	Corporate
2	Row1	C02	priya verma	Delhi	Consumer
3	Row2	C03	Rohan Mehta	Noida	Corporate
4	Row3	C04	Sneha Kapoor		Consumer
5	Row4	C05	Vikram Singh	Faridabad	Small Business
6	Row5	C06	Ananya Gupta	Gurugram	consumer

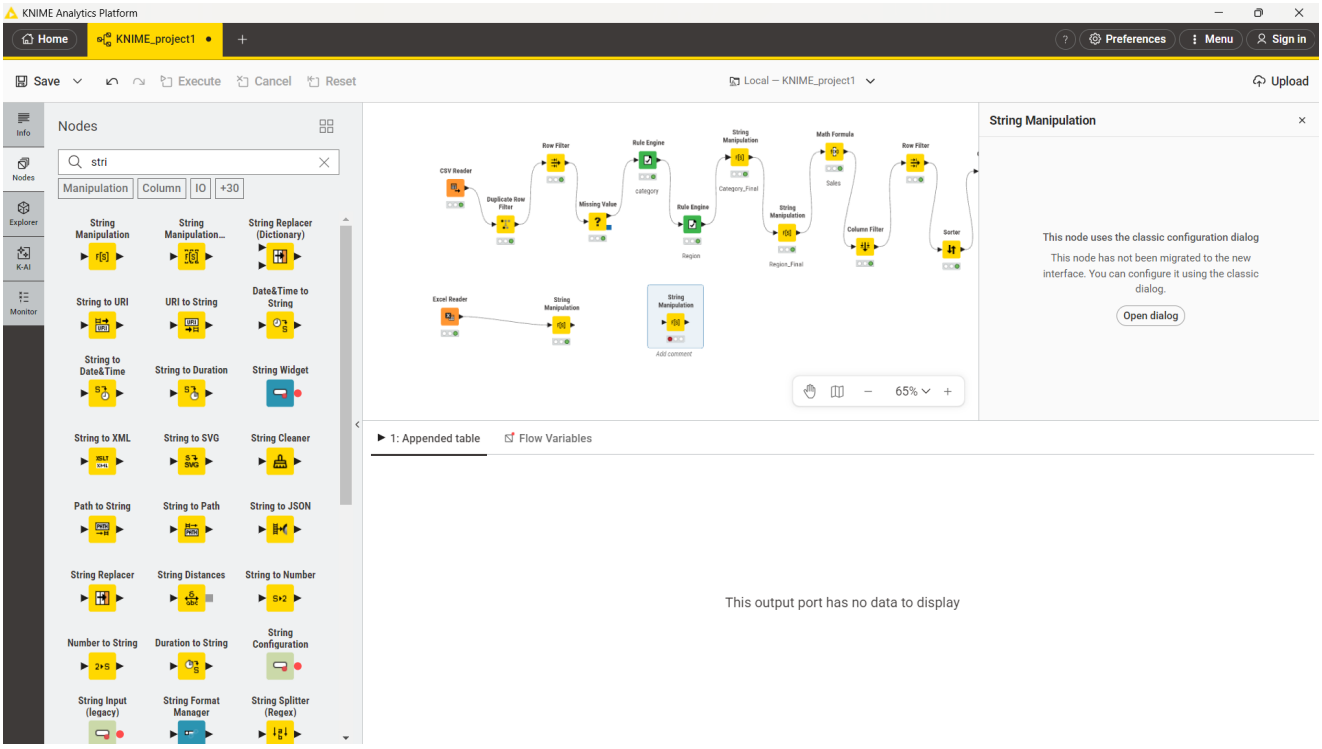
Step 48: Selected the matching key columns for the orders-to-customers join.



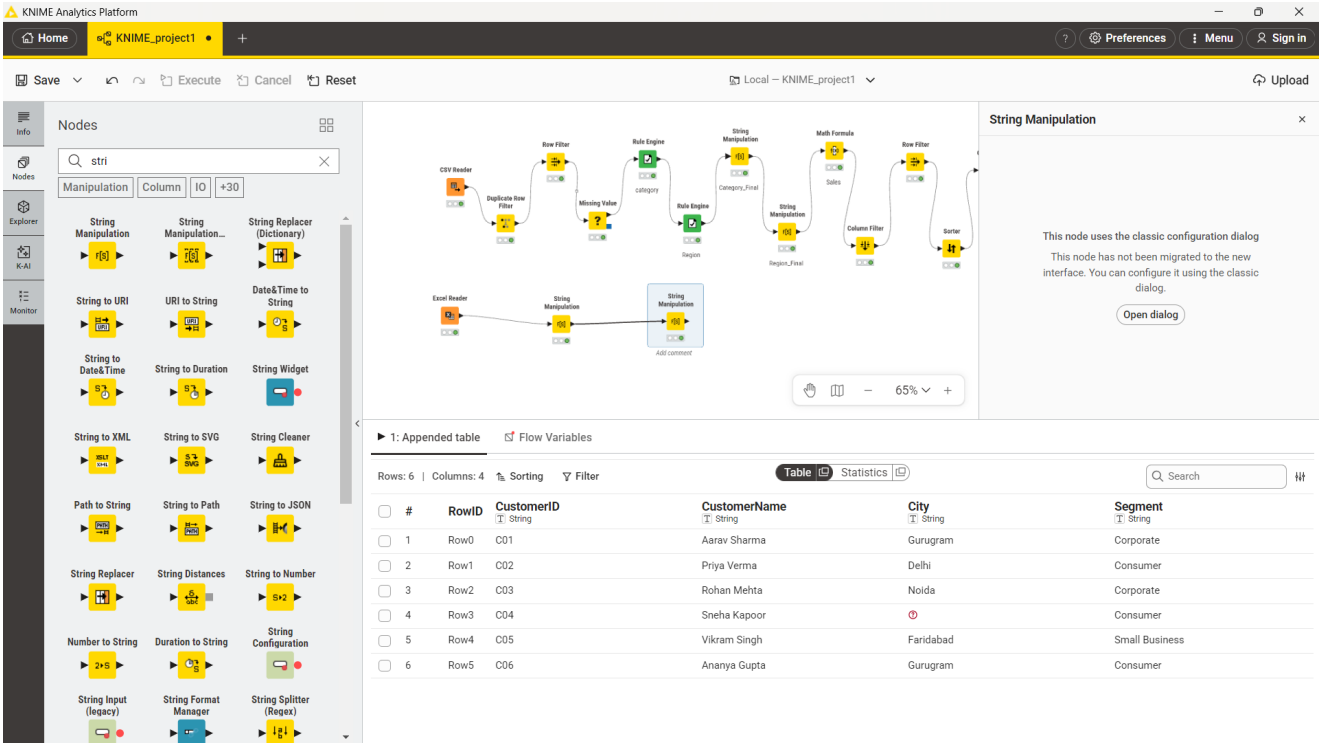
Step 49: Checked the Joiner configuration and prepared the merge for execution.



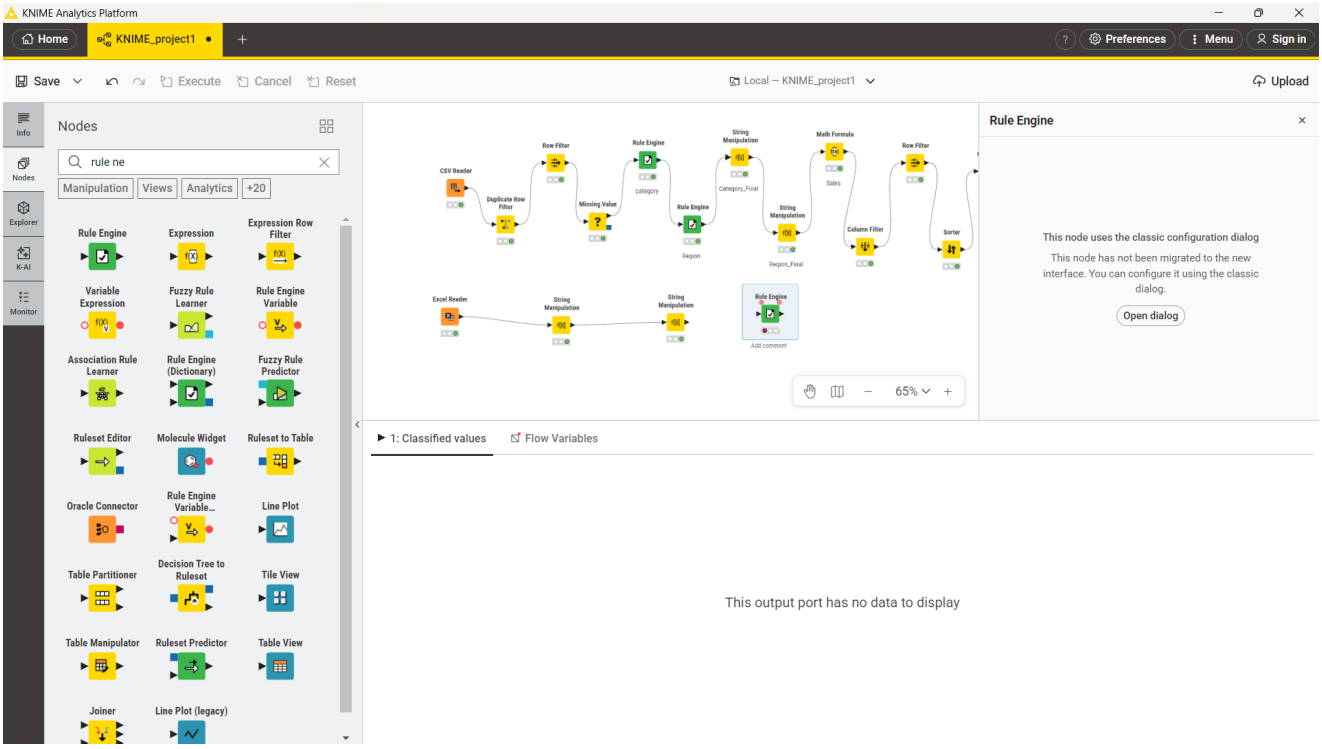
Step 50: Executed the Joiner and inspected the joined result.



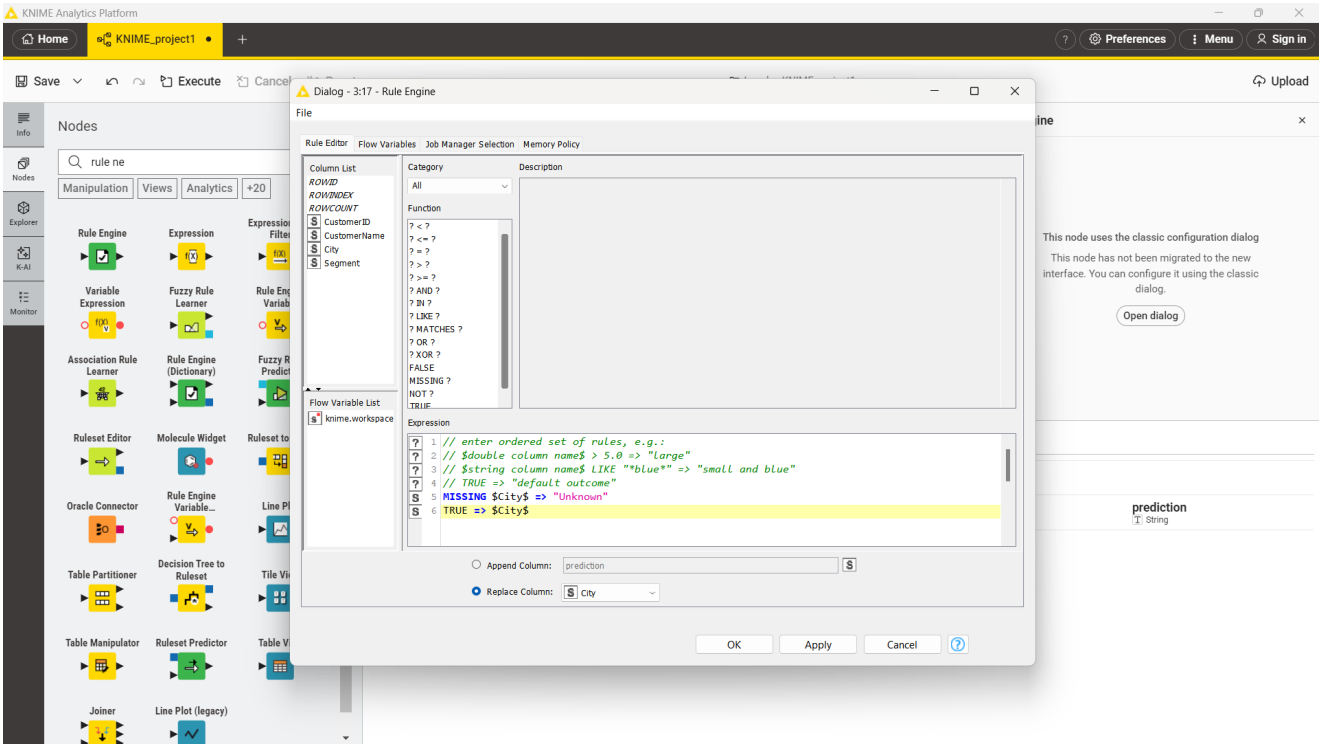
Step 51: Reviewed the merged table and verified that the required customer fields were brought into the orders data.



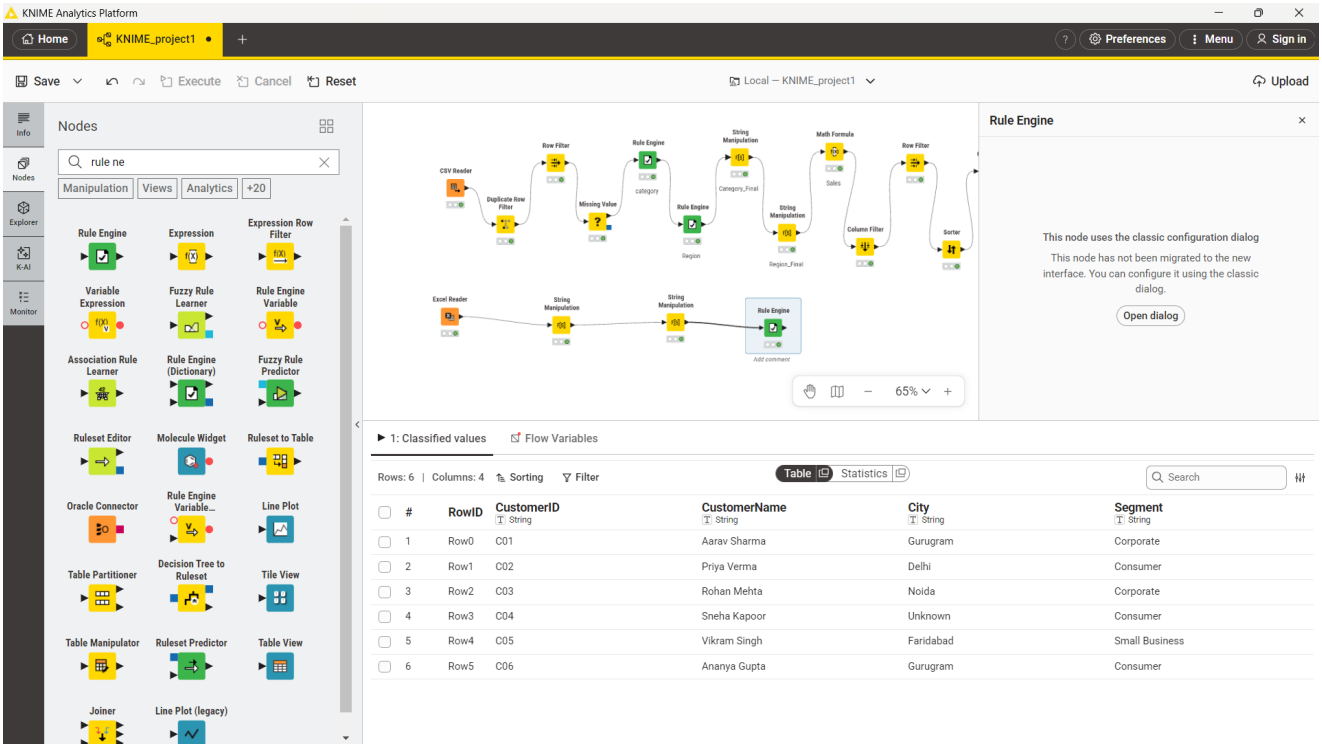
Step 52: Checked the complete merged workflow and its connected nodes.



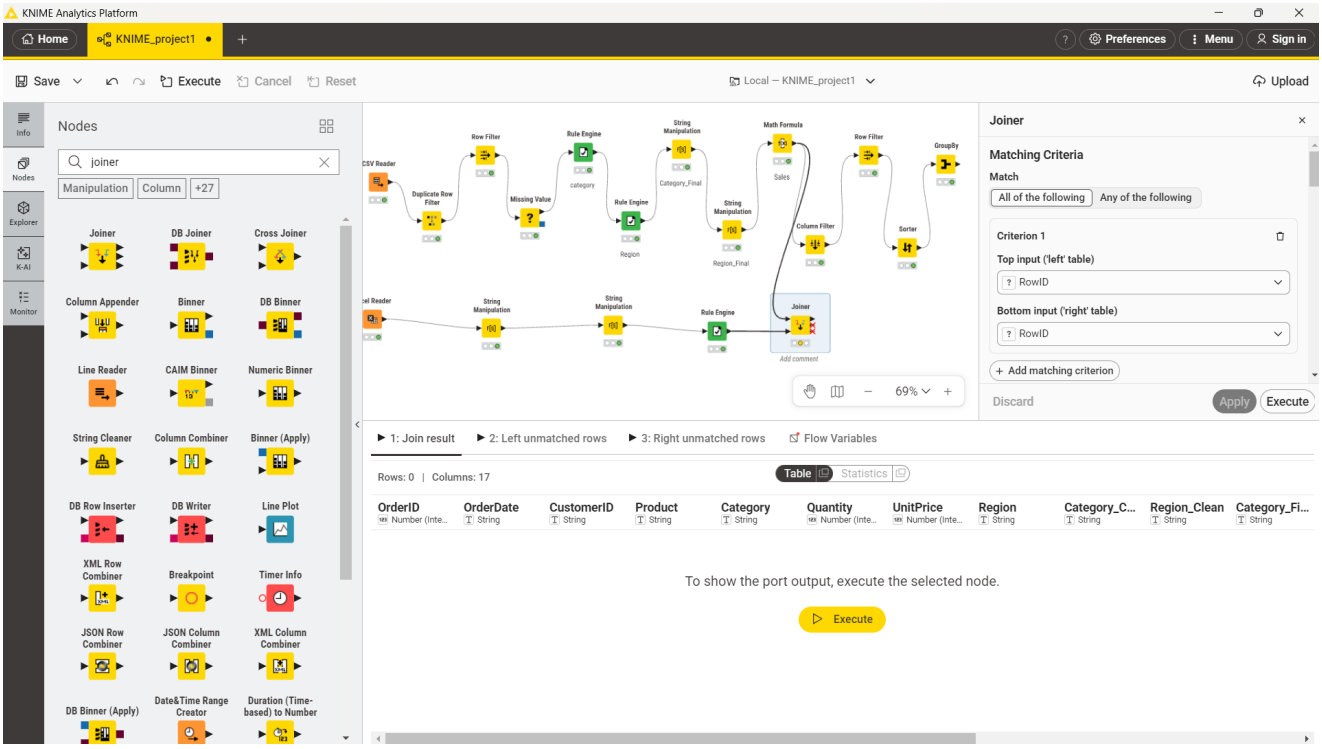
Step 53: Inspected the merged output table and verified the transformed fields.



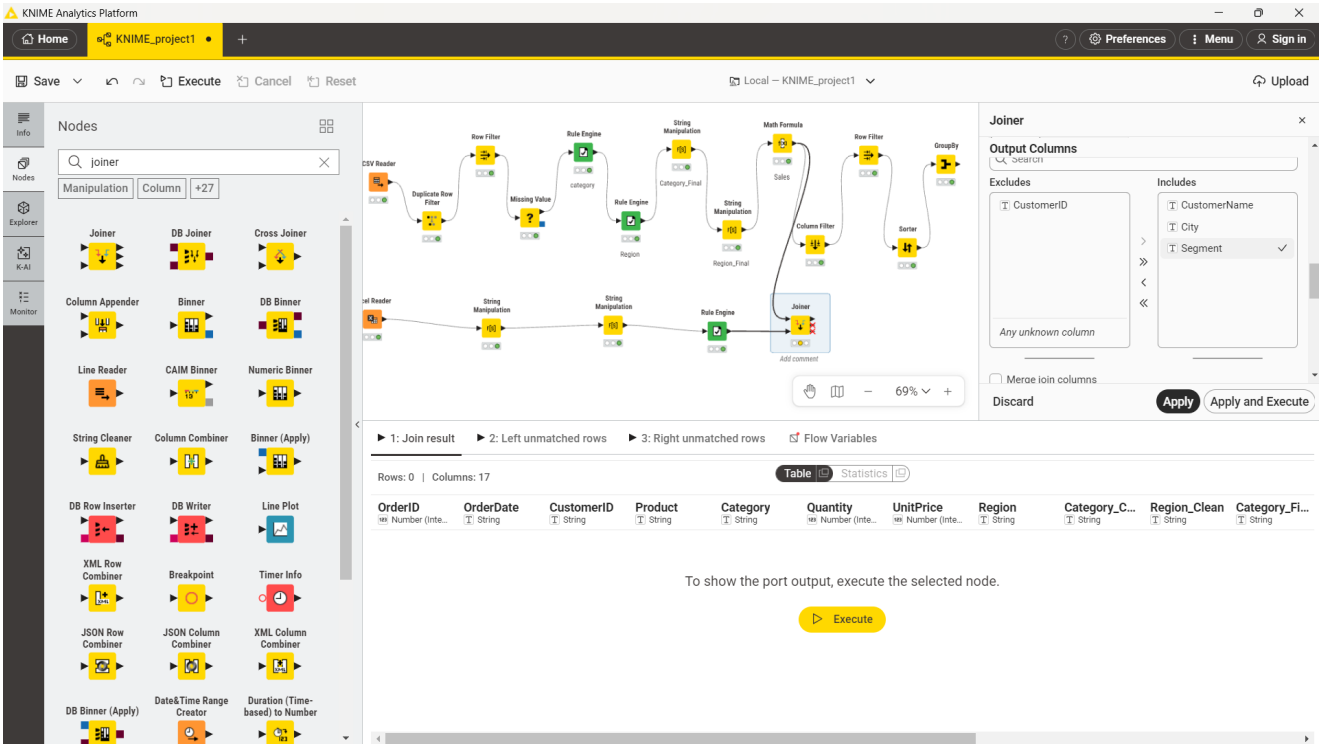
Step 54: Reviewed the workflow after the merge and confirmed the main data path.



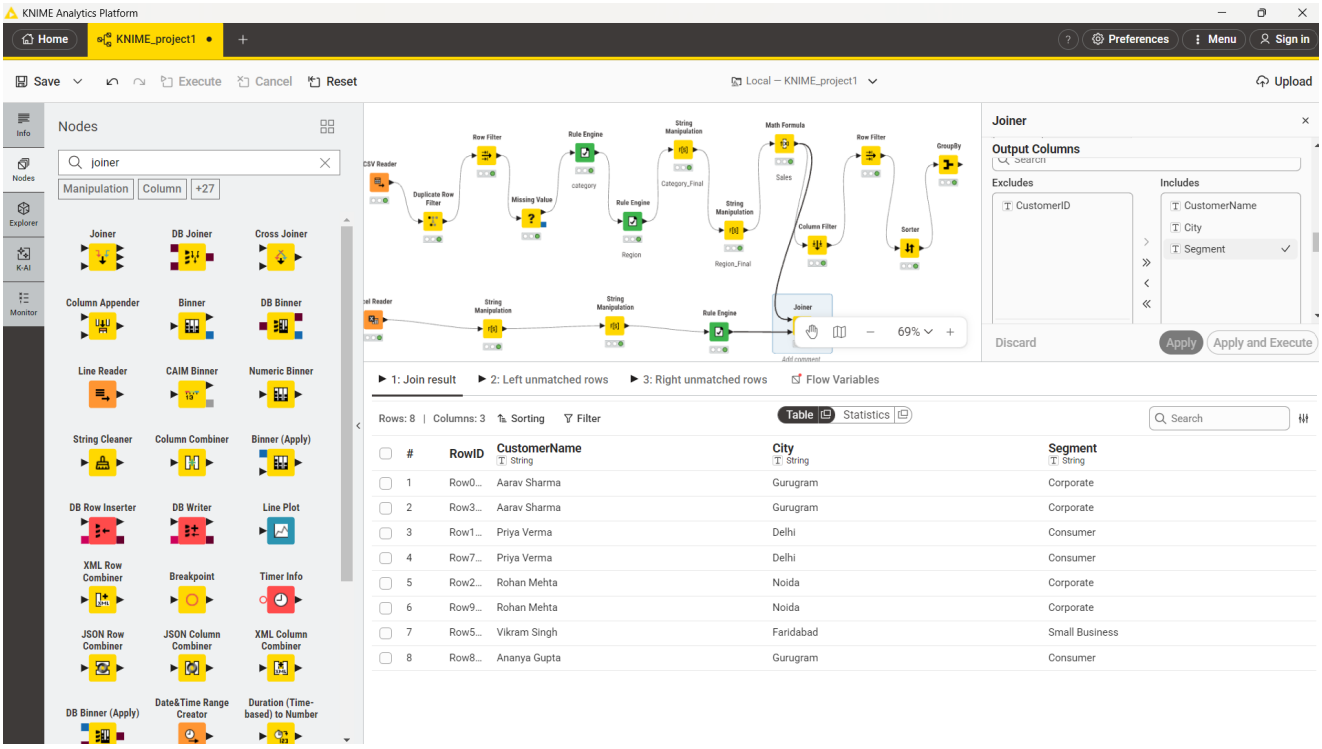
Step 55: Checked the customer-side output and its columns before final validation.



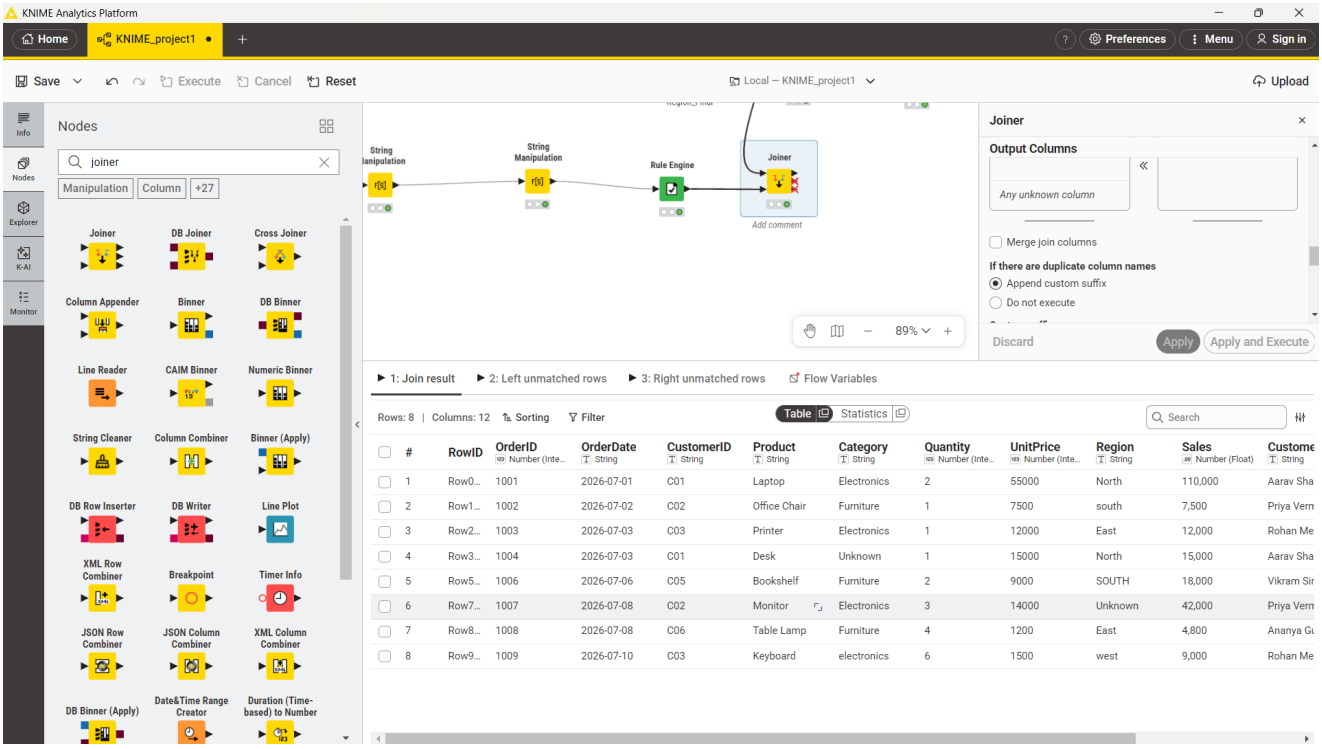
Step 56: Verified the merged result and its row/column structure.



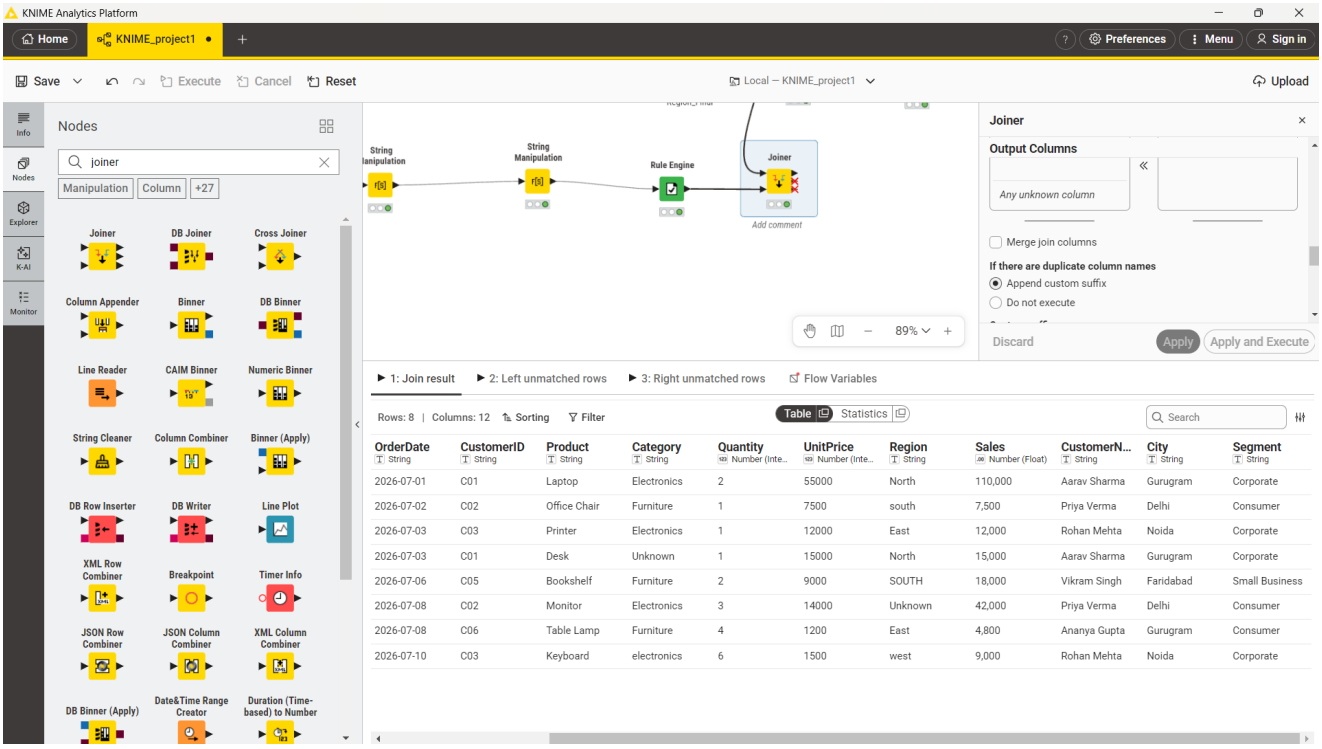
Step 57: Inspected the final merged data to confirm the cleaned 9-row result is preserved.



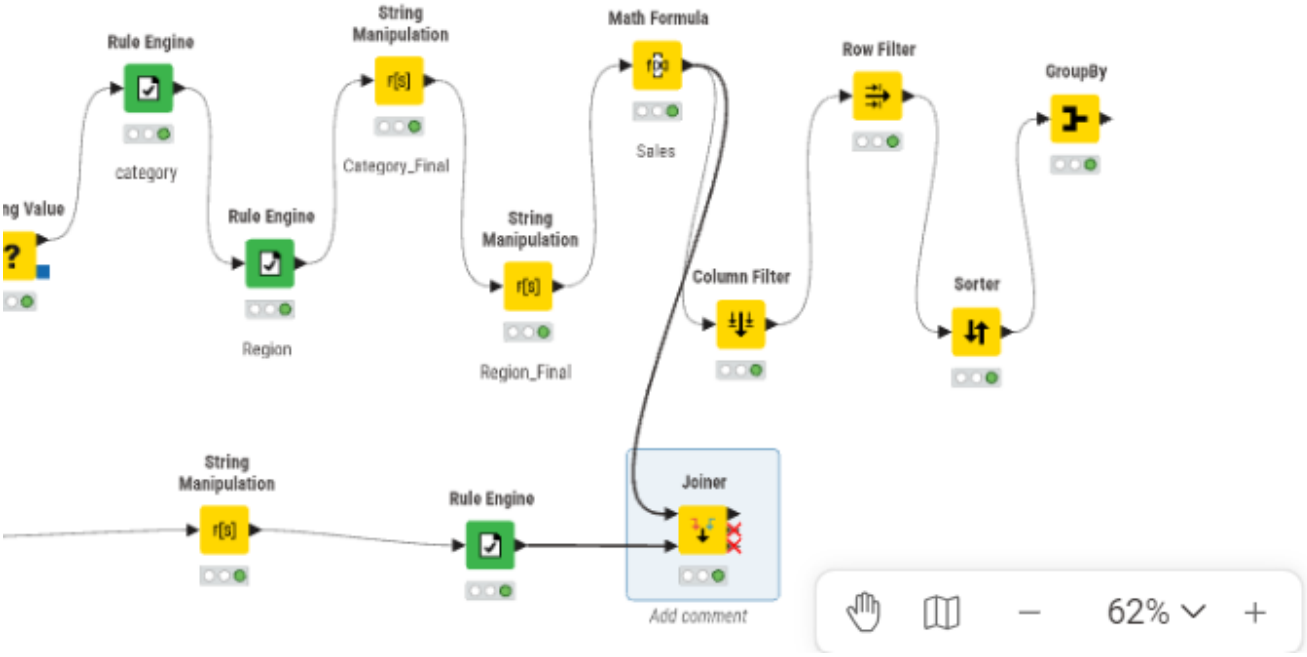
Step 58: Reviewed the final workflow layout and the output branch used for export.



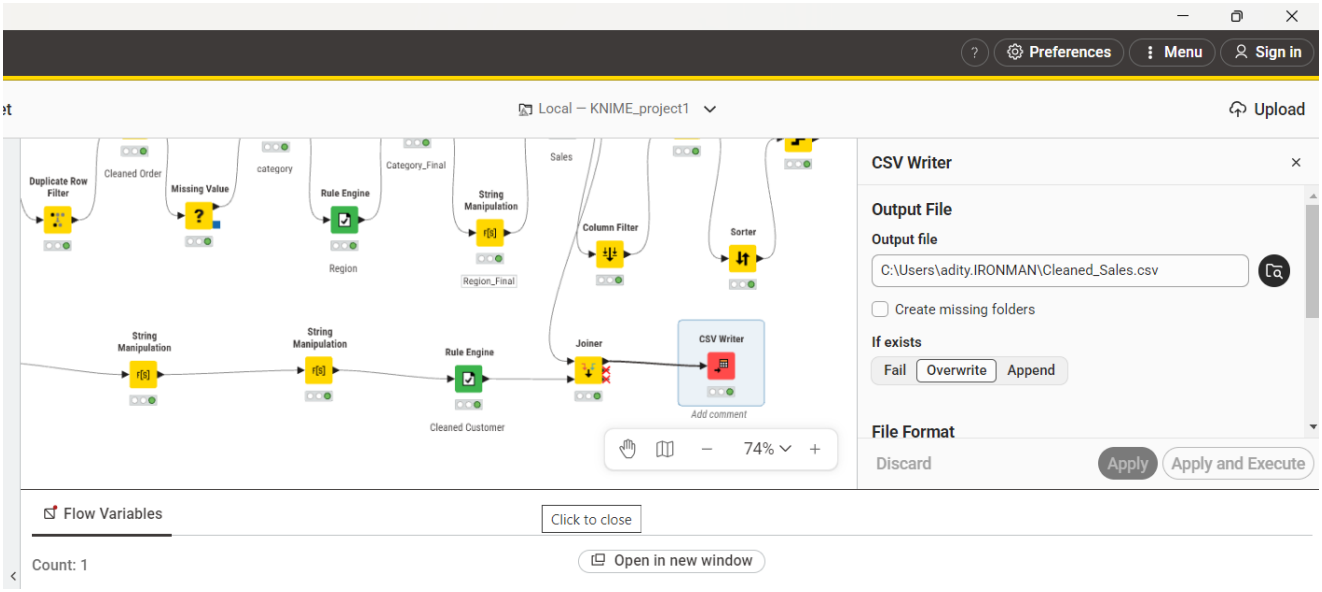
Step 59: Opened the CSV Writer configuration to export the cleaned merged dataset.



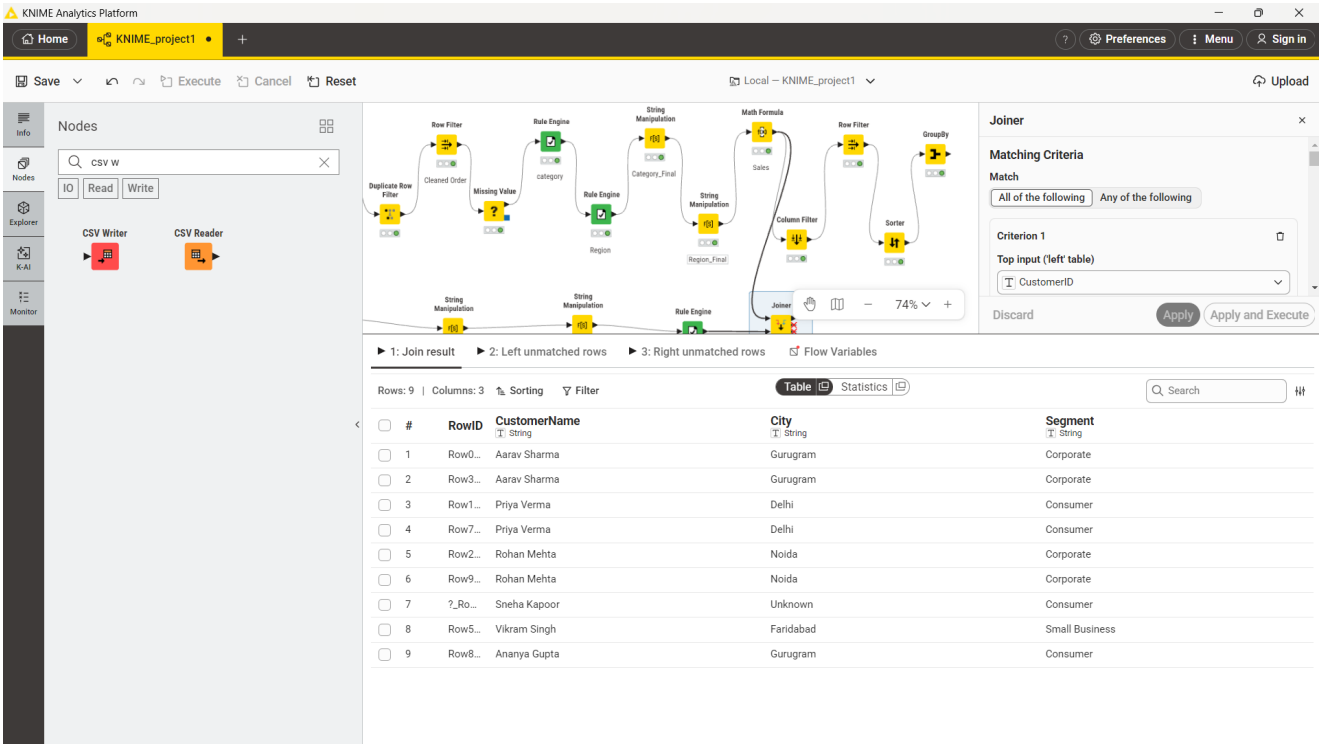
Step 60: Configured the output file and CSV writing options for the cleaned result.



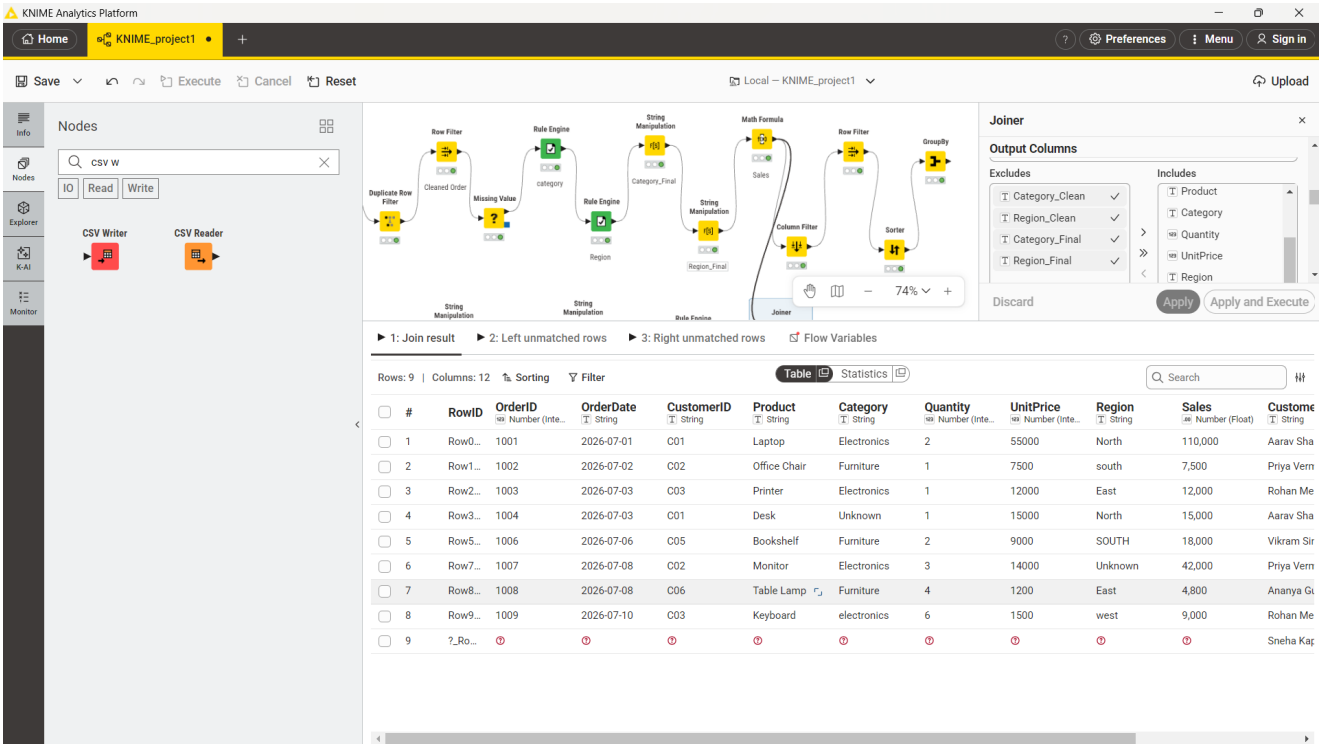
Step 61: Checked the CSV Writer settings before executing the export.



Step 62: Executed the final workflow/export step and reviewed the resulting output configuration.

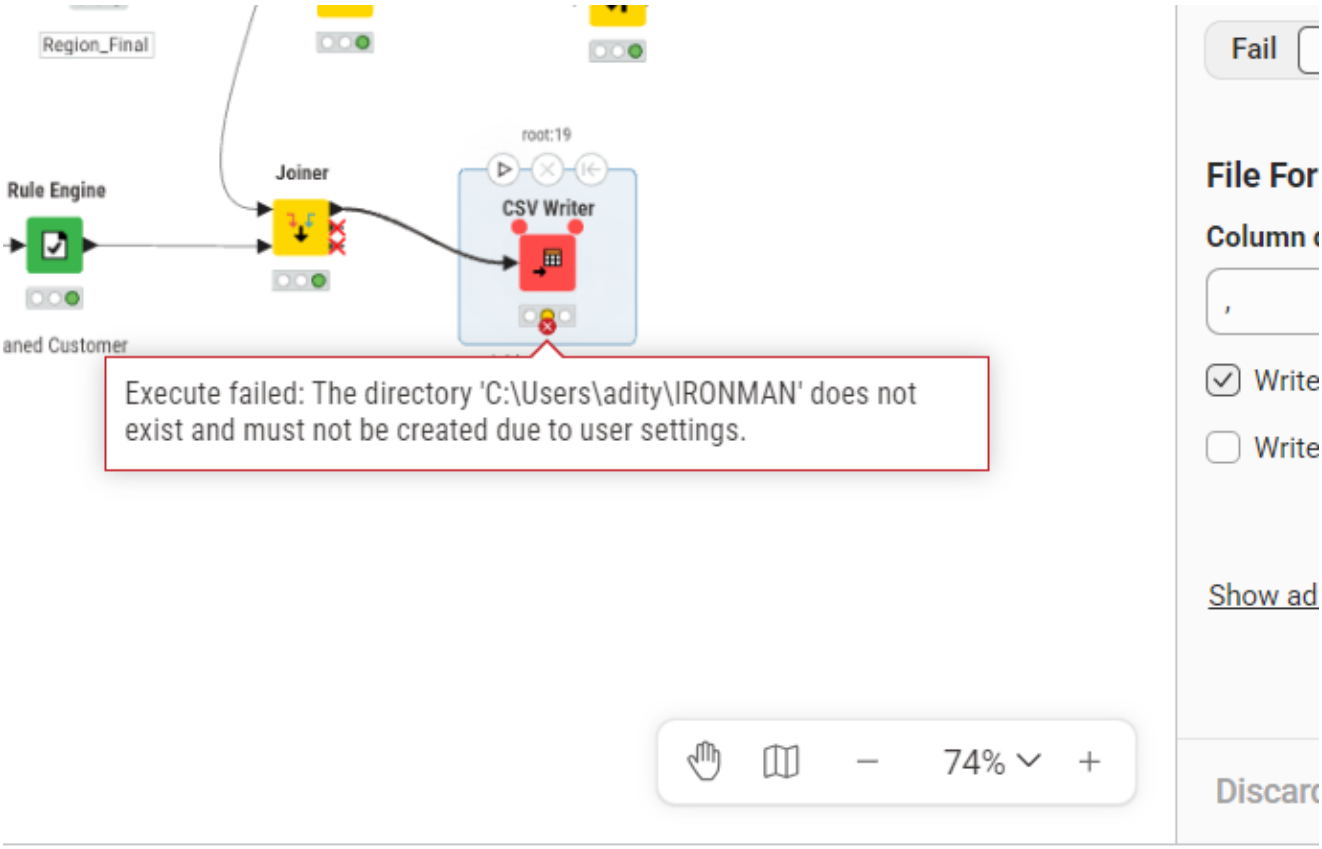


Step 63: Checked the completed workflow and final exported-data path.



#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Sales	Custome
1	Row0...	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	110,000	Aarav Sha
2	Row1...	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	7,500	Priya Verm
3	Row2...	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	12,000	Rohan Me
4	Row3...	1004	2026-07-03	C01	Desk	Unknown	1	15000	North	15,000	Aarav Sha
5	Row5...	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	18,000	Vikram Sir
6	Row7...	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	42,000	Priya Verm
7	Row8...	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	4,800	Ananya Gu
8	Row9...	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	9,000	Rohan Me
9	?_Ro...	?	?	?	?	?	?	?	?	?	Sneha Kap

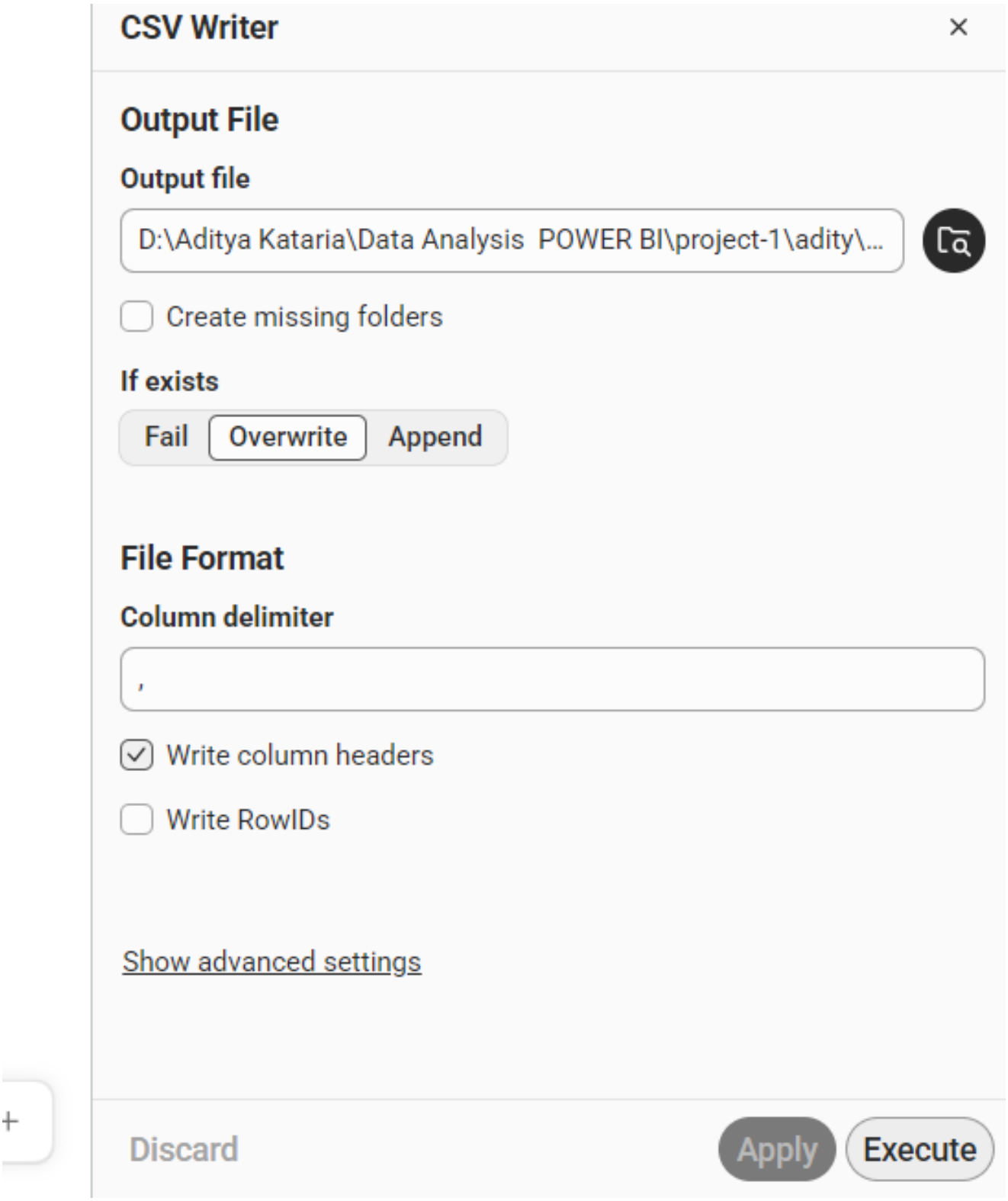
Step 64: Reviewed the final output table after export preparation.



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me Value

Step 65: Inspected the completed KNIME workflow and final data output.



Step 66: Final verification of the KNIME workflow and exported cleaned dataset.

